

Near and Middle East

Suheir Assady | Rawi Ramadan | Yaacov Frishberg

CHAPTER OUTLINE

ACUTE KIDNEY INJURY, 2518
CHRONIC KIDNEY DISEASE, 2523

MANAGEMENT OF END-STAGE KIDNEY
DISEASE IN THE MIDDLE EAST, 2537
SUMMARY, 2543

KEY POINTS

- Health care disparities do exist among countries of the Middle East (ME).
- Accurate and detailed epidemiologic data regarding acute and chronic kidney diseases in ME countries are still needed.
- Community-based screening programs for chronic kidney disease, risk factors, and intervention strategies are limited in most countries of the region.
- Diabetes mellitus is the major cause of end-stage kidney disease in most ME countries.
- Recessively inherited kidney diseases are prevalent in the ME, and targeted screening in populations at risk should be encouraged.
- Renal replacement therapies are available in all ME countries, and hemodialysis is mostly used.
- Hepatitis C infection is prevalent among dialysis patients in the ME. Implementation of infection control measures and introduction of new antiviral agents are of great importance to eradicate it.
- Legal kidney transplantation from both living and deceased donors should be expanded, and organ donation encouraged, in view of the positive ethical and religious authorizations.

This chapter is a discussion of the epidemiology, causes, predisposing factors, management, and prevention of kidney diseases. Future strategies for dealing with kidney diseases in the Near and Middle East are proposed. The term *Near and Middle East* is a historical, Eurocentric, and Western term that was used to describe a geographic region whose boundary is imprecise and whose internal borders are constantly changing because of political and historical evolution.¹ Therefore, the Near and Middle East, hereafter called the Middle East (ME), is defined for the purpose of this chapter as the region that encompasses the following 20 countries (in alphabetical order): Algeria, Bahrain, Egypt, Iran, Iraq, Israel, Jordan, Kuwait, Lebanon, Libya, Morocco, Oman, Palestinian Authority, Qatar, Saudi Arabia, Syria, Tunisia, Turkey, the United Arab Emirates (UAE), and Yemen (Fig. 78.1).

The ME has always been important for the following reasons: (1) its strategic location as a tricontinental hub that links Asia, Africa, and Europe; (2) its economic resources; and (3) its spiritual significance as the birthplace of the

world's three major monotheistic religions—Judaism, Christianity, and Islam. Of the adherents to the three religions, Muslims constitute the largest religious population in the ME; the sizes of the Christian and Jewish populations are smaller and vary from country to country.¹ Although the ME has several cultural, linguistic, and geographic associations, considerable disparity exists among the different countries in terms of economy, resources, political systems, health care systems, health expenditure, and disease incidence and prevalence.²

According to data from the World Bank, the estimated population in the 20 ME countries in 2016 was 514,854,000 (Table 78.1).³ The gross national income (GNI) per capita varies across ME countries. According to the classification system of the World Bank,^{3,4} most ME countries are considered “developing countries.” Egypt, Morocco, Syria, Palestinian Authority, and Yemen are considered lower middle-income countries because the GNI per capita of each is between \$1006 and \$3955 (USD) per annum, whereas Algeria, Iran,



Fig. 78.1 Map of Middle Eastern countries. Countries illustrated in *light green* are also known as “El Maghreb”; those in *dark green*, as “El Mashreq.”¹ (Modified from a public domain image available at http://en.wikipedia.org/wiki/Middle_East.)

Iraq, Jordan, Lebanon, Libya, Tunisia, and Turkey are considered upper middle-income countries because the GNI per capita of each is between \$3956 and \$12,235 per annum. The following countries are considered high-income countries because the GNI per capita of each is more than \$12,236 per annum: Bahrain, Israel, Kuwait, Oman, Qatar, Saudi Arabia, and UAE (Fig. 78.2). Inevitably, disease incidence, prevalence, course, and outcomes are affected by the different socioeconomic factors and health policies in each country (Fig. 78.2).^{5–7} For example, the infant mortality rate in each ME country is concordant with the socioeconomic, political, and health status of that country;⁸ and life expectancy for men is lowest in the low-income ME countries and in countries affected by man-made conflicts (see Table 78.1).⁶

The mortality rate for those younger than 5 years in Yemen, the poorest country in the ME, is at least five times higher (55/1000 live births) than that in industrialized, high-income ME countries (4–13/1000 live births; see Table 78.1).^{3,8} Health disparities also exist among minority populations (ethnic groups or expatriates) who live in industrialized ME countries, as well as among individuals of ME origin who live in countries that are not in the ME.^{9–19} Many developing

ME countries have been affected by various types of natural disasters (e.g., earthquakes, floods, droughts) and other disasters (e.g., military conflicts). Casualties, displacement, and migration are significant consequences of such disasters and adversely affect the socioeconomic strata and health status of a country, as in the recent devastating humanitarian crises in Yemen and Syria.^{2,20–23} Therefore, many ME countries and their communities need assistance to upgrade their existing health care structures to improve their ability to cope with any future disasters.^{2,20,22,24,25} Consequently, the World Health Organization (WHO) and national and international nephrology societies, such as the International Society of Nephrology (ISN), have pooled their resources to enhance the resilience of nations to the effects and consequences of disasters.^{9,22,23,26,28}

ACUTE KIDNEY INJURY

As in many countries worldwide, data on the incidence and prevalence of acute kidney injury (AKI) in many ME countries are scarce and imprecise because of an inconsistent definition

Table 78.1 Demographics, Health Indicators, and Human Development Indices in Middle Eastern and Western Industrialized Countries

Country	Total Population, 2016 ^a	Median Age (years)	Life Expectancy at Birth (years), 2015 ^b		Rate of Mortality Before Age 5 Years (/1000 Live Births, 2016) ^b	Country Ranking in Human Development Index (of 188 Countries) ^c
			Male	Female		
Middle Eastern Countries						
Algeria	40,606,050	28.1	74	78	25	83
Bahrain	1,425,170	32.3	76	78	8	47
Egypt	95,688,680	23.9	69	73	23	111
Iran	80,277,430	30.3	74	77	15	69
Iraq	37,202,570	20	66	72	31	121
Israel	8,547,100	29.9	81	84	4	19
Jordan	9,455,800	22.5	72	76	18	86
Kuwait	4,052,580	29.3	74	76	8	51
Lebanon	6,006,670	30.5	74	76	8	76
Libya	6,293,250	28.9	70	76	13	102
Morocco	35,276,790	29.3	73	75	27	123
Oman	4,424,760	25.6	75	79	11	52
Palestinian Authority	4,551,570	20	71	75	19	114
Qatar	2,569,800	33.2	77	80	9	33
Saudi Arabia	32,275,690	27.5	73	76	13	38
Syria	18,430,450	24.3	60	70	18	149
Tunisia	11,403,250	31.6	73	78	14	97
Turkey	79,512,430	30.9	73	79	13	71
United Arab Emirates	9,269,610	30.3	76	79	8	42
Yemen	27,584,210	19.5	64	67	55	168
Western Industrialized Countries						
Australia	24,127,160	38.7	81	85	4	2
Canada	36,286,430	42.2	80	84	5	10
France	66,896,110	41.4	79	85	4	21
Germany	82,667,680	47.1	79	83	4	4
Italy	60,600,590	45.5	80	85	3	26
Japan	126,994,510	47.3	80	87	3	17
United Kingdom	65,637,240	40.5	79	83	4	16
United States	323,127,510	38.1	77	82	7	10

^aData from World Bank: Indicators. <http://data.worldbank.org/indicator>.

^bSource data for country profiles from the World Health Organization: Countries. <http://www.who.int/countries/en>.

^cData from the 2016 Human Development Report, published for the United Nations Development Programme: http://hdr.undp.org/sites/default/files/2016_human_development_report.pdf.

of AKI in medical reports, underreporting, seasonal dependency on the occurrence of AKI, and the frequency and location of natural and human-induced disasters.^{29–32} There are large knowledge gaps about the age, number, natural history, and outcome of patients with AKI in both the community and hospitals and about the use of preventive measures in each ME country. Most published reports on AKI in ME countries are short-term studies done in tertiary-level hospitals or single-center studies and, of these studies, only a few were of the prospective type. Moreover, the definition of AKI is not consistent in these reports.^{33–39} Al-Homrany has reported that the incidence of AKI among hospitalized patients is 0.6% in a 2-year prospective study in one hospital in southern Saudi Arabia.³⁵ Of these cases of AKI, 62% were hospital acquired and 38% were community acquired. In a 1-year retrospective cohort study from a single center in northern

Israel, Shema and associates have reported that the annual incidence rate of AKI among hospitalized adult patients is 1% to 5.1%, depending on the AKI definition that was used.³⁶ In a case series study of the ME respiratory syndrome (MERS) coronavirus outbreak, AKI requiring renal replacement therapy (RRT) was reported in 58% of critically ill patients.⁴⁰ This percentage is high compared with that reported in equivalent critically ill patients who were treated during the severe acute respiratory syndrome (SARS) epidemic in Canada (5%).⁴¹ The Acute Kidney Injury–Epidemiologic Prospective Investigation (AKI-EPI) study was the first multinational and cross-sectional study to investigate the epidemiology of AKI in intensive care unit (ICU) patients worldwide (33 countries) prospectively using the Kidney Disease: Improving Global Outcomes (KDIGO) definition for AKI⁴² and a standardized data collection instrument.³¹ AKI was found in 57.3% of

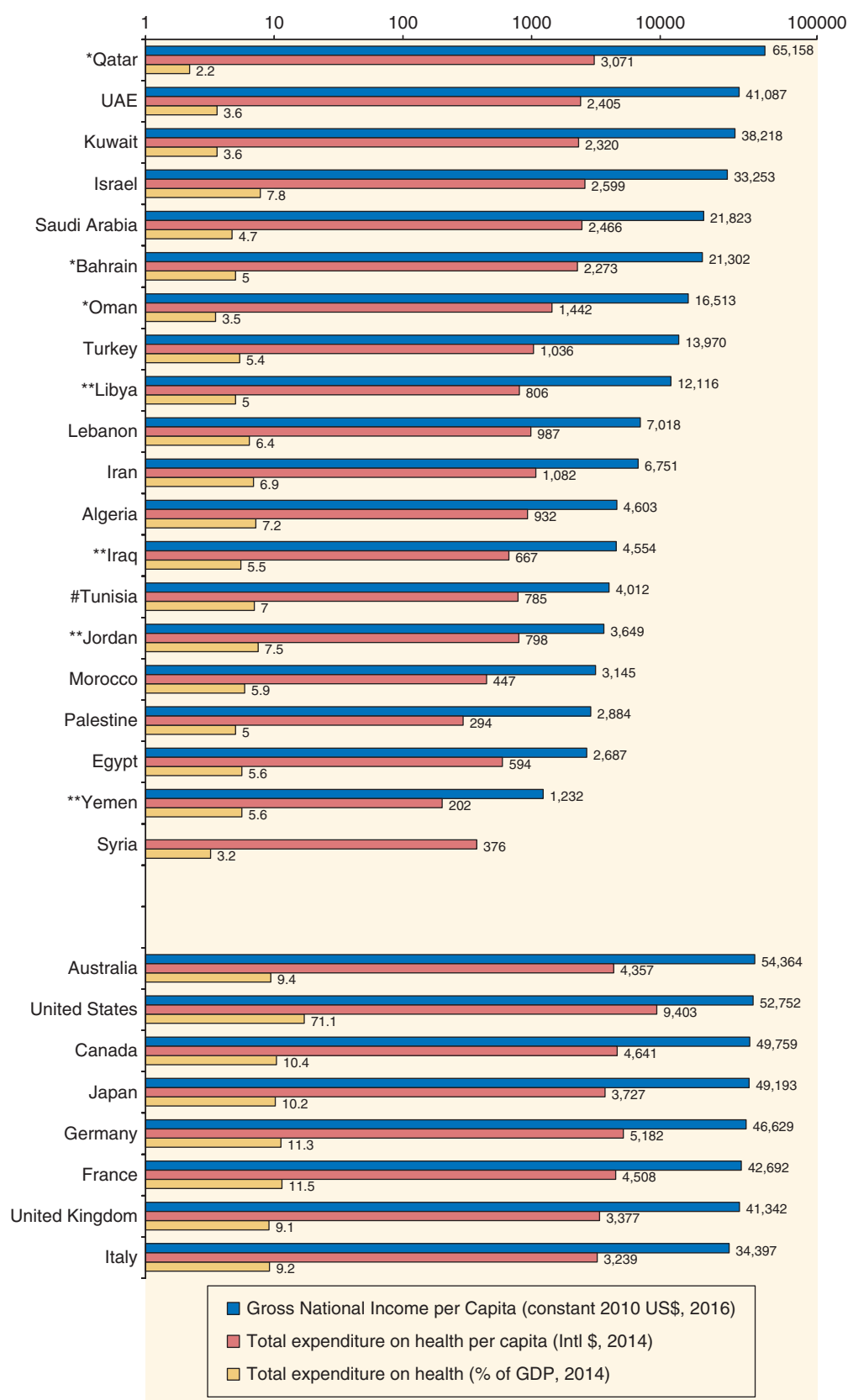


Fig. 78.2 Indices of wealth and health care expenditures in the 20 Middle Eastern and selected Western and industrialized countries in 2014. For each country, three indices are presented—gross national income per capita, on a purchasing power parity, expressed in constant 2010 US dollars (blue bars); total expenditure on health per capita, expressed in international dollars (violet bars); and total expenditure on health, expressed as percentages of gross domestic product (GDP; yellow bars). The upper seven countries are the wealthy high-income Middle East nations. Indices from *2015; *2013; **2010. UAE, United Arab Emirates; USA, United States of America. (Data from the World Bank: Countries and economies. <http://data.worldbank.org/country>; and from the World Health Organization: Countries. <http://www.who.int/countries/en>.)

subjects. Yet, only 5 of 97 participating centers were from the ME—three ICUs from Egypt, one from Tunisia, and one from Turkey. These are low numbers from which to draw firm conclusions regarding the whole ME region.

CAUSES

The causes of AKI have changed over time in ME countries. A nationwide prospective study, conducted by the Turkish Society of Pediatric Nephrology Acute Kidney Injury Study Group,⁴³ demonstrated that the cause of AKI in Turkish pediatric patients has greatly changed over the last 2 decades. Acute gastroenteritis and acute poststreptococcal glomerulonephritis have decreased significantly as causes of AKI, and prematurity, malignancy, and congenital heart disease have increased. In adults, obstructive uropathy, unspecified postsurgical complications, and crush injury were the most prevalent causes of AKI in Syria during the 1980s, according to Hadidy and colleagues.³³ In contrast, Said has reported findings similar to those from industrialized countries in 215 patients with AKI, aged between 12 and 90 years, in three Jordanian hospitals over an 18-month study period.⁴⁴ Renal parenchymal disease was the most common cause of AKI (58%), and acute tubular necrosis (ATN) and contrast-induced nephropathy were the two most prevalent causes of renal parenchymal disease. Prerenal causes (28%) and postrenal causes (14%) accounted for the remaining causes of AKI.⁴⁴ Of note, obstructive uropathy was a common cause of AKI as a result of the high prevalence of nephrolithiasis in the study patients who originally came from Yemen and Sudan. In another study from one center in Qatar, ATN was reported in 83% of all patients with AKI, who were referred mainly from ICUs.⁴⁵ Al-Homrany³⁵ has reported that ATN resulting from sepsis, ischemia, and rhabdomyolysis, as well as distinctive causes, such as malaria and snakebites (4.6% of all cases), were the major causes of AKI in tropical southern Saudi Arabia.

Malaria is rarely reported as a primary cause of AKI in most ME countries, with the exception of Yemen.⁴⁶ In the Hajjah and Sanaa regions of Yemen, malaria caused by *Plasmodium falciparum* was the cause of 29% of all cases of AKI. However, prerenal disorders, such as infectious diarrheal diseases, are still the predominant cause of AKI in Yemen because of the tropical climate and poor hygiene.⁴⁶ Malarial kidney injury is often a consequence of several hemodynamic, immune, and metabolic disturbances, which may also be accompanied by central nervous system sequelae and by fluid and electrolyte alterations.^{47–49} Malarial kidney disease can manifest as AKI in the form of the following: (1) ATN that accompanies or occurs as a complication of severe hemolysis, hemodynamic derangements, and tissue hypoxemia; (2) interstitial nephritis; or (3) glomerular mesangial proliferative lesions with immune complex deposits.⁴⁷ In areas where malaria is not common, a high index of suspicion is required.

Reporting the cause of AKI may, however, be biased by the type of referral hospital that documents the various causes. In a retrospective 18-month study from one cancer hospital in the UAE, sepsis and drug-induced nephrotoxicity were the leading causes of AKI because 30% of the patients were immunocompromised and were receiving chemotherapy.⁵⁰ Preexisting comorbid conditions, such as diabetes, hypertension, and chronic kidney disease (CKD), were documented

in approximately one-third of all patients with AKI in the UAE and Jordan^{34,39,50} and in as many as 87% of all such patients in Qatar.⁴⁵

AKI-associated mortality rates range between 12% and 77% in ME countries.^{33–36,39,44,45,50–52} The ISN's 0by25 AKI initiative aims to prevent all avoidable deaths from AKI by 2025, especially in low- and middle-income countries.⁵³ To implement such an initiative, a clear definition of a preventable death from AKI, and a strategy to regionalize any recommendations for AKI care in terms of health care infrastructure, socioeconomic conditions, and education and training were considered.^{53,54} The first global 10-week snapshot in 2014 included AKI data from 289 sites across 72 countries, mostly academic and referral centers, of which 32 centers represented ME countries—13 centers in Egypt, four in Iran, three in Saudi Arabia and three in UAE, two each in Morocco and Tunisia, and one center per country, including Israel, Kuwait, Oman, Palestinian Authority, and Syria.⁵² In this study, the median age of patients from ME countries was 60 years, with a male predominance. Two-thirds of AKI cases were community acquired and, as expected, mortality increased in hospital-acquired AKI and in patients requiring the ICU or dialysis. However, many areas of the ME were underrepresented in this study, and data from Egypt and Tunisia were combined with those from Africa; those from Israel were collected with Western Europe.

Devastating earthquakes have struck some ME countries and are a constant threat because their territory encompasses the Great African Rift Valley. Clinician investigators have described and analyzed the factors that have had major implications for kidney involvement and outcomes in survivors who sustained crush syndrome in catastrophic earthquakes in Turkey and Iran.^{55,56} Crush syndrome often causes profound hypovolemic shock that (1) is complicated and aggravated by gross disorders of acid-base balance and electrolytes, of which hyperkalemia is life-threatening, and (2) increases susceptibility to myoglobinuric AKI. These complications can occur within hours of the initial injury and can lead to early loss of limb or life.^{28,57} According to data collected after the 1999 Marmara earthquake in Turkey and the 2003 Bam earthquake in Iran, the number of casualties, incidence of crush injury, and incidence of myoglobinuric AKI are related to several variables, including the following^{55,56,58}:

1. The intensity of the earthquake and the magnitude of the aftermath.
2. The time of the day that the earthquake happened. The Turkish earthquake occurred during the night and was associated with more crush injuries than earthquakes that have occurred during the day because the victims were in the supine position. Erect subjects tend to be killed by head injuries from falling buildings.
3. The population density and type of residential area at the site of the earthquake. The population density in rural areas, where the buildings are single-storied and made from light construction materials, is less than that in urban areas, where the buildings may be multistoried and made of heavy construction materials, leading to more crush injury in urban areas.
4. The climate. Earthquake survivors suffer more volume depletion and dehydration, exacerbating AKI in hot weather compared with cold weather.

5. The time to rescue, because this reflects both the amount of time under the rubble and the magnitude of imposed pressure over a given time.
6. The extent of destruction of health care facilities at the site of the earthquake and the distance from referral hospitals.
7. The availability of medical support and the availability and efficiency of rescue teams.
8. The availability of dialysis.^{28,55,56,58,59}

An important issue in these disasters is the identification of patients with AKI who need early referral to specialized nephrology facilities.

The ISN Renal Disaster Relief Task Force, in cooperation with European Renal Best Practice, has published comprehensive clinical practice guidelines for the management of

crush victims in mass disasters.⁵⁸ These recommendations include medical and logistic principles and were based on lessons learned from the activities of the Renal Disaster Relief Task Force in disaster-affected areas during the past 2 decades.^{28,55,56,58} Of note, vigorous fluid resuscitation, which should begin while victims are still trapped under the rubble, and the use of mannitol with proper monitoring, are now proven therapies for reducing AKI and for saving limbs by preventing the development of compartment syndrome in crush injuries, the second leading cause of death after disasters (Fig. 78.3). These therapies can also reduce the need to perform fasciotomies, which are associated with severe bleeding, sepsis, and amputations.^{28,57,58,60}

In summary, it is still difficult to draw general conclusions about the epidemiology, causes, and outcomes of AKI in many ME countries because of regional variations and methodologic

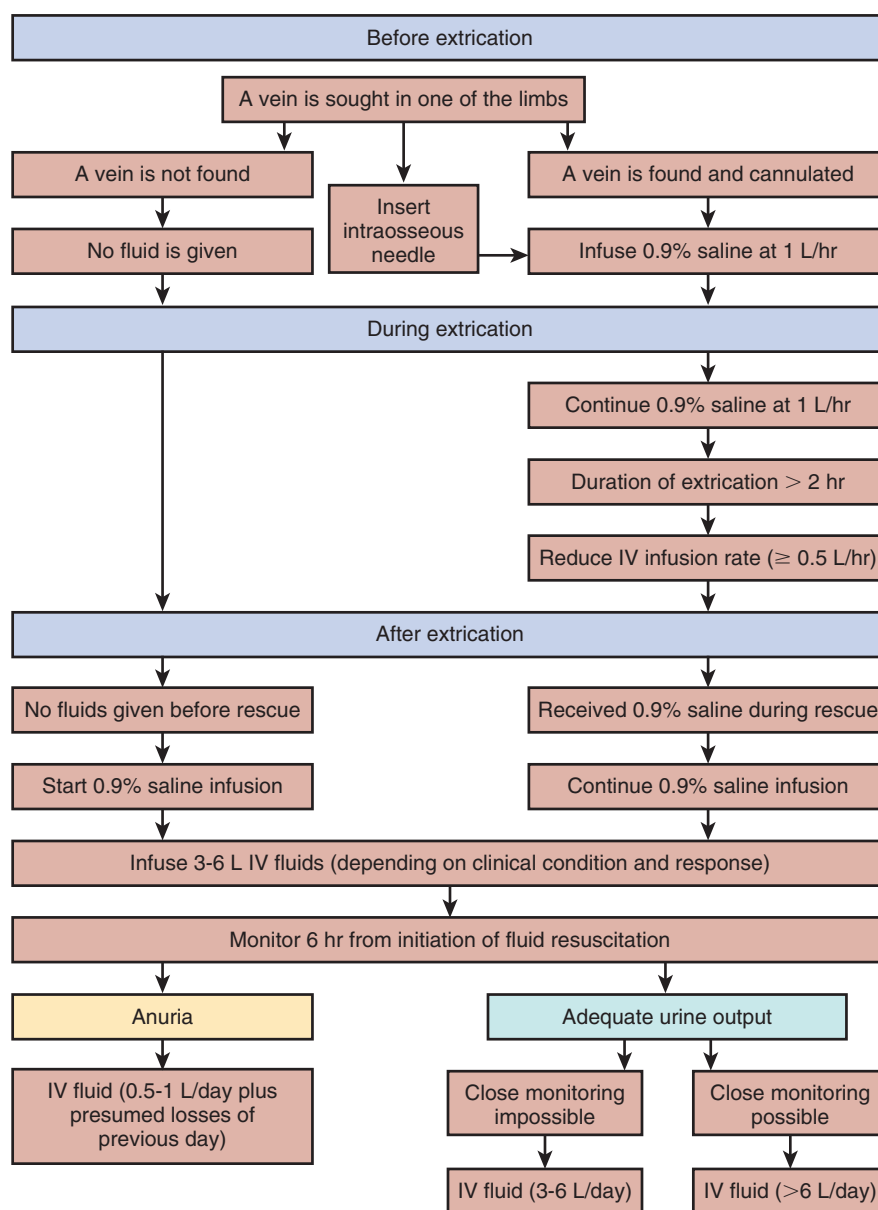


Fig. 78.3 Algorithm for fluid resuscitation in crush victims of mass disasters before, during, and after extrication. IV, Intravenous. (Modified from Sever MS, Vanholder R: Management of crush victims in mass disasters: highlights from recently published recommendations. *Clin J Am Soc Nephrol.* 2013;8:328–335; Gibney RT, Sever MS, Vanholder RC: Disaster nephrology: crush injury and beyond. *Kidney Int.* 2014;85:1049–057.)

differences in the few studies that have been performed in the region. Well-conducted regional studies in which the published definition of AKI⁴² is used are needed to clarify the actual occurrence of AKI. Accordingly, the results of these studies can then be used to identify the needed infrastructure and evaluate treatment strategies to prevent AKI and improve the clinical outcomes.

CHRONIC KIDNEY DISEASE

EPIDEMIOLOGY

The incidence and prevalence of noncommunicable diseases have been changing rapidly as a result of demographic transition, and the burden of disease has consequently shifted from infections in the pediatric population to chronic diseases of the adult population in many ME countries.^{13,61–64} This demographic shift in the burden of disease is exemplified by the emerging epidemic of diabetes mellitus that is occurring globally and affects Arab and Chaldean Americans. In 2017, the International Diabetes Federation ranked Saudi Arabia, Egypt, and UAE as having the highest age-adjusted (20–79 years) comparative diabetes prevalence in the ME (17.7%, 17.3%, and 17.3%, respectively) and worldwide after countries in the Western Pacific region.⁶⁵ The International Diabetes Federation has also predicted that the prevalence of diabetes mellitus in ME countries will increase by 2045. These alarming statistics for ME countries have been attributed to a combination of increasing urbanization, aging populations, increasing obesity, and falling levels of physical activity.^{7,13,63,64,66–77} Accordingly, the direct and indirect medical expenditures associated with diabetes mellitus will become a profound economic burden. As a result, ME countries with limited or scarce resources (see Fig. 78.2) will be unable to cope with the social, economic, and public health consequences of complications of diabetes mellitus, one of which is CKD.

In addition, the common practice of consanguineous marriages in ME countries has led to a high incidence of genetic disorders, some of which may lead to CKD and end-stage kidney disease (ESKD), especially in children, and has possibly altered the pattern of renal disease in these countries.^{74,78–84} Genetic renal diseases and their complications in the ME are discussed further later in this chapter.

It is difficult to estimate the incidence and prevalence of CKD in most ME countries. Obtaining the epidemiologic statistics concerning CKD, ESKD, and RRT is a work in progress that is hampered by the lack of well-conducted, cross-sectional, longitudinal cohort studies and a lack of reliable registries that incorporate data on comorbid conditions. Furthermore, additional research is required to establish the best and most cost-effective approach for screening for CKD.^{85,86}

Hence, the creation of a reliable and easily accessible medical registry is urgently needed in the ME countries where existing registries are lacking, unreliable, or difficult to access. The information in such registries would be essential for planning health policies and the allocation of resources. In long-term, nationwide, population-based, retrospective cohort studies, factors such as overweight, obesity, normal renal function with a history of any childhood kidney disease or persistent asymptomatic isolated hematuria detected in Israeli

adolescents and young adults were found to be strongly associated with an increased cumulative incidence of treated ESKD, with crude hazard ratios of 3, 6.89, 4.19, and 18.5, respectively.^{87–89} Therefore, although not yet proven, it is hoped that early detection of CKD and its risk factors will increase health awareness and permit early intervention, which in turn may modify the disease course, complexity, and costs of overall therapy.^{85,90–92}

Community-based screening programs for CKD and risk factors, such as diabetes mellitus, obesity, proteinuria, and hypertension, have been launched in developing countries worldwide under the auspices of the Research and Prevention Committee of the ISN.^{76,91} These programs are expected to detect CKD in populations in which the people are unaware of such chronic diseases. This initiative includes ongoing programs in Egypt and Morocco.⁹³ The Egypt Information, Prevention, and Treatment of Chronic Kidney Diseases (EGIPT-CKD) program was the first to report interim results. It investigated the prevalence of microalbuminuria among first-degree relatives of people with ESKD in the city of Damanhur and the surrounding towns in the Al-Buayrah governorate, in Lower Egypt.⁹⁴ In this study, microalbuminuria was detected in 10.6% of participants. In an adjusted logistic regression analysis, smoking and a personal history of cardiovascular disease were strongly associated with microalbuminuria. The well-conducted, cross-sectional *Maladie Rénale chronique au Maroc* (MAREMAR) study involved 10,524 individuals from two Moroccan towns, El-Jadida (coastal and industrial) and Khemisset (inland and rural).⁷⁷ It found that the adjusted prevalence of CKD in Morocco is among the lowest in the world (5.1%), including estimated glomerular filtration rate (eGFR) under 60 mL/min/1.73 m² (1.6%), confirmed proteinuria (1.9%), and hematuria (3.4%). The low rate of CKD in this study may reflect the fact that the diagnosis was made on repeat samples and not on purely cross-sectional data, suggesting possible overestimation of CKD in studies relying on a single GFR estimate. The MAREMAR-CKD cohort was also classified according to the KDIGO eGFR stages⁹⁵: G1, 17.8%; G2, 17.2%; G3A, 40.2%; G3B, 12.3%; G4, 4.4%; and G5, 7.2%. Patients with CKD G3A–G4 will be followed for 8 years. Furthermore in Morocco, population-adjusted prevalence rates for hypertension, obesity, and diabetes mellitus were reported to be 16.7%, 23.2%, and 13.8%, respectively.⁷⁷

According to the Chronic REnal Disease In Turkey (CREDIT) study,⁷⁵ the population-based estimated prevalence rates for CKD stages 1 to 5 among Turkish adults were 5.4%, 5.2%, 4.7%, 0.3%, and 0.2%, respectively, with an overall prevalence of 15.8%. Moreover, the frequency of concomitant cardiovascular risk factors such as diabetes, hypertension, dyslipidemia, and obesity was increased in patients with CKD, and odds ratios versus non-CKD population were 3.22, 2.86, 1.60, and 1.65, respectively. Based on analysis of one spot urine specimen per participant, 10.2% of the cohort subjects had microalbuminuria, and 2% had macroalbuminuria.⁷⁵ The causes of CKD were not detailed, but the study cohort continues to be followed longitudinally.

The actual or estimated number of patients with CKD at each eGFR stage of the KDIGO classification in other ME countries is unclear.⁹⁶ A small pilot study, which is part of the Global SEEK Project, demonstrated the feasibility of screening and early detection of CKD in a Saudi population.

Using standardized GFR prediction equations, the prevalence of CKD stages 1, 2, and 3 was relatively low—3.5%, 1.6%, and 0.6%, respectively—which may be attributed to the low mean age of the participants (37.4 ± 11.3 years).⁹⁷ In Kuwait, a 4-year prospective study from one referral center by El-Reshaid and associates⁹⁸ has reported a high incidence of CKD among Kuwaiti nationals. Specifically, they reported an average annual incidence of 366 per million population (pmp), with a higher incidence among the older population (≥ 60 years) of 913 pmp. Of patients with CKD who were admitted to the center, 6% presented with uremic syndrome, and 40% experienced acute deterioration of their kidney function that resulted mainly from drugs (mostly over-the-counter nonsteroidal antiinflammatory drugs [NSAIDs]), infection, and volume depletion. The authors did not report categories of eGFR or proteinuria in their patients.

In 2007, a study involving 31,999 taxi drivers in Tehran, mostly males, showed that 6.5% of drivers had an eGFR of less than 60 mL/min/1.73 m².⁹⁹ However, community-based studies revealed considerable differences in CKD prevalence between counties of Iran (10.2%–18.9%).^{73,100}

In a pioneering work, Tohidi and colleagues¹⁰¹ examined the incidence of CKD among a subgroup, 20 years or older,

in the Tehran Lipid and Glucose Study, which is a long-term community-based prospective study. During 10-year follow-up, the cumulative incidence of CKD stages 3 to 5 among women and men was 27.8% and 14.2%, respectively. Age, hypertension, diabetes, and current smoking were found to be independent predictors of CKD stages 3 to 5 in both genders. In addition, for males, high-normal blood pressure and, for females, being single, divorced, or widowed were associated with an increased risk of incident CKD.¹⁰¹ Interestingly, adherence to Mediterranean or Dietary Approaches to Stop Hypertension (DASH)-style diets had a positive impact on reducing the incidence and prevention of CKD,^{102–104} which may ultimately translate into favorable survival and cardiovascular outcomes.¹⁰⁵

Although the number of patients who underwent preemptive kidney transplantation might not be included, estimates of the prevalence and incidence of ESKD are more reliable in countries that have a national database in which each patient who has received RRT is registered (Tables 78.2 and 78.3). Nevertheless, patient numbers in a national registry in ME countries may be underestimated for two additional reasons: (1) the treatment of ESKD may be beyond the reach of the average citizen in low-income ME countries, such as

Table 78.2 Epidemiology of Renal Replacement Therapy in Middle Eastern Countries

Country	Dialysis				Kidney Transplantation		
	First Hemodialysis ^a	No. of Hemodialysis Centers	Incidence ^b	Prevalence ^{b,c}	Date of Country's First Transplantation	Incidence ^b	Type
Algeria	1962	281	104	482	1986	3.9	C
Bahrain	1971	5	62	410 (800)	1995	6.7	C
Egypt	1958	1050	NA	609.7	1976	14.8	L
Iran	1965	316	63.8	332.6	1967	32	C
Iraq	1967	27	NA	83	1973	12.2	L
Israel	1948	79	191	783.4	1964	41.5	C
Jordan	1968	78	117.3	754.74	1972	29.2	C
Kuwait	1976	9	100	400 (960)	1979	35.4	C
Lebanon	1960	66	NA	665.4	1972	19	C
Libya	1972	53	282	624	1989	8.6	L
Morocco	1980	197	42	405.8	1990	1.3	C
Oman	1983	21	60	365 (640)	1988	16.5	C
Palestinian Authority	1997	13	NA	240.3	2011	NA	L
Qatar	1981	6	75	320 (1230)	1986	22.1	C
Saudi Arabia	1971	187	136	513 (645)	1979	24.2	C
Syria	1974	NA	111	228	1976	16.2	L
Tunisia	1963	146	142	812	1986	13.2	C
Turkey	1965	860	125.1	754.2	1975	42.8	C
United Arab Emirates	1975	33	129	210 (760)	1985	2.3	L
Yemen	1982	NA	64	91	1998	1.4	L

^aFor acute kidney injury or end-stage kidney disease.

^bPer million population.

^cPrevalence not including expatriates—in parentheses.

C, Combined living and deceased kidney donors; L, only living kidney donors; NA, not available.

Adapted from Bello AK, et al: Global Kidney Health Atlas (GKHA)—country profiles; Bello AK, Levin A, Tonelli M, et al: Global kidney health atlas: A report by the International Society of Nephrology on the current state of organization and structures for kidney care across the globe. Brussels, Belgium, 2017, International Society of Nephrology; and personal communication with Dr. Bello.

Table 78.3 Registries, Awareness, and Strategies for Kidney Care—Nephrology Workforce Capacity in Middle Eastern Countries

Country	ESKD Registries ^a	Priority and Advocacy for Kidney Care		Presence of National Strategies For Kidney Care			Workforce
	Availability of Dialysis and Kidney Transplantation	CKD Recognized as Health Priority by the Government	Presence of CKD Advocacy Group	Nondialysis CKD	Dialysis	Kidney Transplantation	Nephrology Workforce (pmp)
Algeria							11.3
Bahrain							5.9
Egypt							21.7
Iran						NA	3.9
Iraq							1
Israel							9.7
Jordan							10.2
Kuwait							10.8
Lebanon							22.6
Libya							12.5
Morocco							10.5
Oman							30.4
Palestinian Authority						NA	2.2
Qatar						NA	11.4
Saudi Arabia							17.7
Syria							4.7
Tunisia							16.3
Turkey							6.3
UAE							8.7
Yemen					NA	NA	0.2

^aFew countries regularly report national registry data, such as Israel,¹²⁷ Turkey,¹²⁸ Jordan,¹²⁹ and Saudi Arabia.^{112,448}

Blue and red boxes indicate that the survey participants answered yes and no, respectively, to GKHA Questionnaire⁷⁶; yellow, only dialysis registry is available.

CKD, Chronic kidney disease; ESKD, end-stage kidney disease; pmp, per million population; NA, not available; UAE, United Arab Emirates. Data from Bello AK, et al: Global Kidney Health Atlas (GKHA)—country profiles; Bello AK, Levin A, Tonelli M, et al: Global kidney health atlas: A report by the International Society of Nephrology on the current state of organization and structures for kidney care across the globe. Brussels, Belgium, 2017, International Society of Nephrology; and personal communication with Dr. Bello.

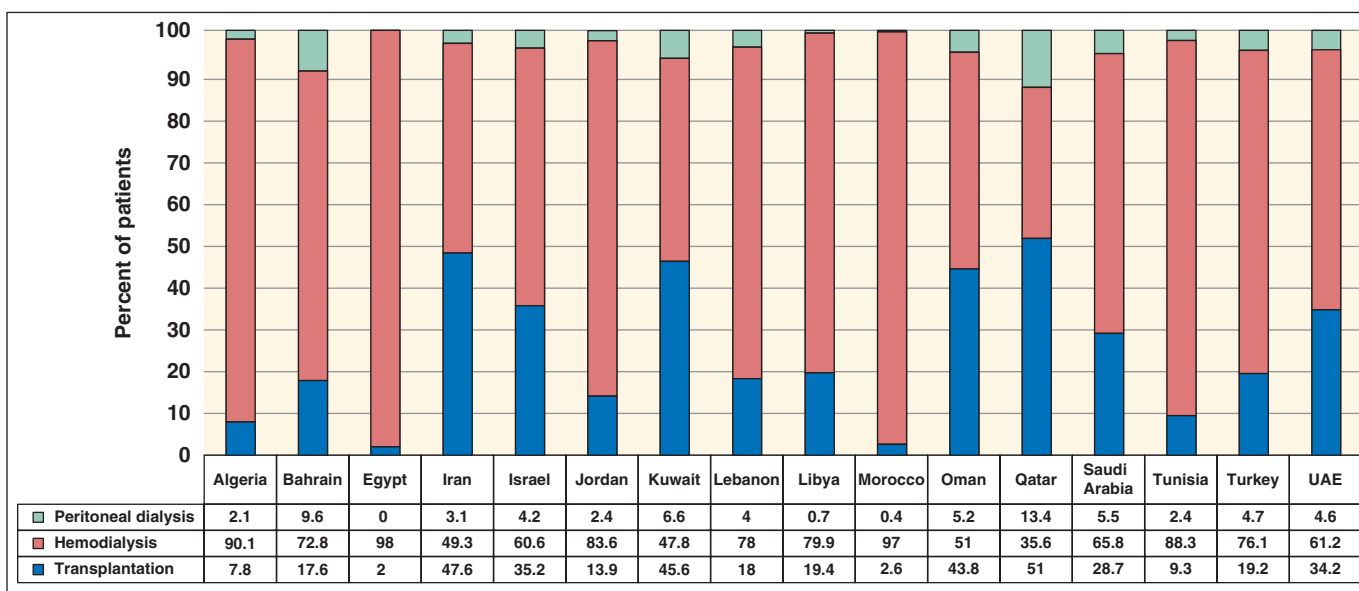


Fig. 78.4 Percentage distribution of renal replacement therapy modalities used for treating end-stage kidney disease in selected Middle Eastern countries. UAE, United Arab Emirates. (Adapted from US Renal Data System [USRDS]: Annual Data Report, Chapter 11: International comparisons. https://www.usrds.org/2017/download/v2_c11_IntComp_17.pdf; and Benghanem GM: Epidemiology of ESRD in the Maghreb [lecture]. Presented at the 5th Maghreb Congress of Nephrology, March 19–22, 2014, Djerba Island, Tunisia; data from references 107, 125, 128, and 130.)

Yemen; and (2) a high number of expatriates live in some ME countries that may not be included. Fig. 78.4 and Tables 78.2 and 78.3 summarize the available data on ESKD from published articles, registries, and abstracts of proceedings of international symposia in ME countries.^{66,76,106–132} Of note, most investigators relied mainly on the results of limited retrospective studies and answers to questionnaires that were sent to leading nephrologists in each country and not on accurate documented statistics, registries, or results of epidemiologic studies.

Israel and Turkey have reported their ESKD data to the European Renal Association–European Dialysis and Transplantation Association registry¹³¹ and the US Renal Data System.¹³² The registry in Turkey was established in 1990.^{116,133} In 2016, the prevalence of ESKD in Turkey plateaued at 933.1 pmp, and the incidence rate for new RRT patients declined to 139.9 pmp.^{128,132} Diabetic nephropathy and hypertension were the two main causes of ESKD in Turkey.^{128,131–134}

Data from the report of the Israeli Center for Disease Control (ICDC) have indicated that the incidence of ESKD increased by 73% during the past 25 years, from 113 to 196 pmp. The average annual increase was highest in the years 1996 to 1999, but it has stabilized since 2003. The increase in ESKD was observed mainly in the older population (≥ 65 years). Likewise, a 2.8-fold increase in the prevalence of ESKD was observed, from 416 pmp in 1990 to 1171 pmp in 2015. Of note, diabetes mellitus was the cause of ESKD in 11.5% of prevalent patients in 1990 and increased to 45.9% in 2015.^{127,135}

The Tunisian registry was started in 1990 and has also reported an increase in the incidence and prevalence of ESKD.¹³⁶ The incidence of ESKD increased from 81.6 pmp in the period 1992 to 1993 to 158.8 pmp in 2000 to 2001; the average annual increase was 9.6%. The incidence of ESKD in older persons, women, and individuals with diabetic

nephropathy has risen steeply. However, regional variations were noted among urban and rural districts.^{136,137}

A comprehensive 1-year observational study, conducted by Alashek and colleagues,^{66,138–141} shed light on the epidemiology of ESKD and RRT practices in Libya before the conflict. The prevalence and incidence of dialysis-treated ESKD from mid-2009 until August 2010 were 624 and 282 pmp, respectively.⁶⁶

The Saudi Center for Organ Transplantation (SCOT) has established an open-access RRT registry that provides annual information on the epidemiology and treatment of ESKD in Saudi Arabia. In 2016, the incidence and prevalence of RRT were 178 and 741 pmp, respectively, including non-Saudis, who comprise 15% of dialysis patients.¹⁴² Over the past 2 decades, the average annual increase in dialysis population was 6.2%.¹¹² In 2016, annual report of the Jordanian National Registry of End-Stage Renal Disease, established in 2007, did not include data on kidney transplantation.¹²⁹ The incidence and prevalence of dialysis were 191 and 783 pmp, respectively (excluding non-Jordanians).

In March 2011, Lebanon launched its national kidney registry, aiming to report reliable data on prevalence, incidence, patient management, practice patterns, clinical outcomes, and survival among CKD patients. In the first annual report of the registry in 2012, the incidence of ESKD and patient survival could not be accurately calculated because of incomplete data. However, the prevalence was estimated to be 855 pmp.¹³⁰ Since then, no other official report has been published. As of December 2015, 4496 dialysis patients were registered in the Lebanese registry.

In Iran, a treatment program for ESKD was introduced in 1975. The reported number of Iranians with ESKD has also increased, and this increase is mirrored in the growing number of dialysis centers and renal transplantation programs. The incidence and prevalence of ESKD were 49.9 and 238

pmp, respectively, in 2000 and increased to 63.8 and 357 pmp, respectively, in 2006.^{106,143,144} By the end of 2015, 50,223 patients were undergoing RRT in Iran, with a prevalence of 635.9 pmp.¹³²

In the ME countries for which detailed data are available, there is a male predominance among patients with ESKD, similar to that reported worldwide.*

The mean age of patients undergoing dialysis in each ME country varies between 42 and 68 years. Interestingly, the youngest patients with ESKD on dialysis are in the developing ME countries, and the oldest patients with ESKD on dialysis are in Israel.[†]

Of note, kidney diseases and risk factors or access to therapy in expatriates—who may comprise as much as 50% of the population, especially in the wealthy Gulf countries—have not been carefully investigated. This population needs to be distinguished from the resident population and may require special attention because of their different ethnic, socioeconomic, and environmental backgrounds.^{19,112,144,149,150}

CAUSES

The causes of CKD in ME countries are highly influenced by the bioecology of a particular region and the ethnic and socioeconomic background of its population. Accordingly, the various causes of CKD are ranked differently in ME countries. The populations of the five ME countries in North Africa—Morocco, Algeria, Tunisia, Libya, and Egypt—have similar ethnicity and socioeconomic backgrounds in that they are of African descent that has been intermixed with Berber, Arab, and Mediterranean population streams.^{7,151} In the 1990s, interstitial nephritis and glomerulonephritis each accounted for approximately 20% of all cases of CKD in these countries.¹⁵¹ The number of individuals with interstitial nephritis increased during the 2000s, possibly as a result of environmental pollution and abuse of over-the-counter drugs.¹⁵² Most cases of glomerulonephritis are of the proliferative type, whereas immunoglobulin A (IgA) nephropathy is rare. The high prevalence of proliferative glomerulonephritis in these ME countries reflects postinfectious glomerulonephritis caused by viruses, bacteria, and parasites. In Egypt, Libya, and southern Algeria, approximately 7% of patients with CKD suffered from obstructive uropathy as a result of urinary schistosomiasis caused by *Schistosoma haematobium* or *Schistosoma mansoni*. It is noteworthy that tuberculosis, other bacterial infections, and familial Mediterranean fever (FMF) are the main causes of renal (type AA) amyloidosis in many ME countries.^{151,153}

However, the frequency of causes of ESKD has changed. All ME countries except Algeria and Yemen have reported diabetes mellitus as the most frequent cause of ESKD (20%–48%) in incident patients undergoing RRT, followed by hypertension (11%–30%) and glomerulonephritis (11%–24%).^{7,62,66,107,111,127,129–132,142–144,149,154,155}

GENETIC DISORDERS

Genetic kidney diseases have received much attention in the fields of pediatric and adult nephrology because the

underlying molecular defects in many of these diseases have been elucidated as a result of advances in genetics and molecular biology. The ME population is ethnically and genetically diverse.¹⁵⁶ The primary demographic features of Arab Muslim and Druze communities in the ME include large families, rapid population growth, and high rates of consanguinity. Among Palestinian Arabs, more than 40% of marriages are between relatives and, of these, 50% are between first cousins.^{156,157} In Bedouin society, 40% of women of childbearing age are married to first cousins.¹⁵⁸ Nevertheless, a comprehensive recent study, conducted in a single Muslim village in Israel, has demonstrated significant sociodemographic changes during a 50-year time period.⁸⁴ A shift from the practice of marrying a first cousin to marrying a remote relative was noted. There was a significant reduction in the mean number of children born per woman and, in parallel, the mean age of first-time mothers progressively increased. The impact of these changes on the incidence of recessively inherited diseases among this population is of much interest.

In contrast, the consanguinity rate among Israeli Jews is reported to be 2.3% and, of these, first cousin marriages account for 0.8%. The highest consanguinity rate among Israeli Jews (7.1%) is found among Eastern (i.e., Asian, non-Sephardic) Jews.¹⁵⁹

High consanguinity rates have also been reported in the Saudi Arabian, Kuwaiti, Lebanese, and Moroccan populations.^{78,79,81,83,160–162}

Clinical Relevance

Inherited Kidney Disease

High rates of genetic kidney diseases occur in the ME because of high rates of consanguinity.

The results of an epidemiologic survey from Lebanon has revealed that 26% of patients undergoing chronic hemodialysis are the children of consanguineous parents.⁷⁹ In addition, Barbari and colleagues⁷⁹ have reported that the risk for a family history of kidney disease is particularly high among patients from consanguineous families who were undergoing hemodialysis. Populations with a high rate of consanguinity also have an increased prevalence of adulthood diseases associated with renal insufficiency, such as hypertension, metabolic syndrome, and diabetes mellitus.^{80,163} A study of kidney biopsies from three pathology centers in Lebanon has shown that mesangioproliferative glomerulonephritis is significantly more frequent among Muslims and offspring of consanguineous unions, whereas focal segmental glomerulosclerosis (FSGS) is most prevalent in Christians.⁸² It should be noted that mutation screening in genes known to be responsible for either of these histological diagnoses had not been performed.

A significant proportion of genetic kidney diseases are inherited in an autosomal recessive manner. It is therefore not surprising that these diseases occur most frequently in communities with high consanguinity rates.¹⁶⁴ The contribution of large families with many offspring, resulting in high carrier rates, to the augmented prevalence of recessively inherited disorders is also appreciated. Yet, the possibility of a higher mutation rate among consanguineous populations in the ME has thus far not been documented. Consanguinity

*References 111, 112, 128, 130, 132, 145, and 146.

†References 111–113, 127, 128, 130, 134, 144, 147, and 148.

Table 78.4 Genetic Kidney Diseases in Southern Israel, 1994 to 2010

Type of Disease	Disease	OMIM No.	No. of Patients	No. of Families	No. of Bedouin Patients	No. of Jewish Patients
Glomerular	Alport syndrome	301050	5	3	0	5
	Benign familial hematuria	141200	3	2	2	1
Tubular	Familial hyperkalemic hypertension	145260	1	1	0	1
	Cystinuria	220100	14	7	14	0
	Distal RTA	602722	4	3	2	2
	Nephrogenic diabetes insipidus	125800	18	11	17	1
	Type 2 Bartter syndrome	241200	10	6	3	7
	Type 4 Bartter syndrome	602522	20	13	20	0
	Unclassified Bartter's syndrome		3	3	1	2
	Familial hypomagnesemia	602014	21	15	21	0
	Gitelman syndrome	263800	3	3	1	2
	Hypophosphatemic rickets	307800	4	2	3	1
Cystic/NPHP	ADPKD	173900	14	13	0	14
	ARPKD	263200	16	12	13	3
	Bardet-Biedl syndrome	209900	5	4	5	0
	Juvenile nephronophthisis	256100	3	3	2	1
	Infantile nephronophthisis	602088	5	4	5	0
	Fanconi-Bickel syndrome	227810	2	2	2	0
Metabolic	Xanthinuria	278300	5	2	5	0
	Lowe syndrome	309000	1	1	0	1
	Cystinosis	219800	3	3	0	3
	Atypical HUS	134370	8	4	8	0
Other	Renal tubular dysgenesis	267430	5	5	4	1
Total			173	122	128	45

ADPKD, Autosomal dominant polycystic kidney disease; ARPKD, autosomal recessive polycystic kidney disease; HUS, hemolytic-uremic syndrome; NPHP, nephronophthisis; OMIM, Online Mendelian Inheritance in Man catalogue (<http://www.ncbi.nlm.nih.gov/omim>); RTA, renal tubular acidosis.

Modified from Landau D, Shalev H: Childhood genetic renal diseases in southern Israel. *Harefuah*. 2010;149:180–185.

may enhance allelic and locus heterogeneity.^{80,163–165} In Palestinian Arabs, 71 different autosomal recessive kidney diseases have been reported¹⁵⁷; these include primary renal diseases (congenital nephrotic syndrome, nephronophthisis), metabolic and tubular defects (cystinuria, Bartter syndrome, renal tubular acidosis, primary hyperoxaluria type 1), and FMF (discussed in detail later in this chapter). The influence of genetic factors is very evident in pediatric patients with kidney diseases, particularly in Saudi Arabia and Syria, where increased numbers of congenital and hereditary kidney and urologic diseases are now being reported.^{156,166} The results of an extensive study of Jewish and Bedouin populations in southern Israel have shown that genetic kidney diseases are overrepresented in the pediatric population (see Table 78.4).⁸¹

By combining whole-exome sequencing with a high-resolution metabolomics profiling for a highly consanguineous Arab cohort in Qatar, 21 common variant and 12 functional rare variant metabolomics quantitative trait loci (mQTLs) were discovered, of which 45% were novel altogether.¹⁶⁷ This project may have wide implications for precision medicine in the ME in various areas, including kidney diseases.

Cystic Kidney Diseases

Autosomal dominant polycystic kidney disease is one of the most common genetic renal diseases in adults and occurs in 1.2% to 10% of ESKD patients in several ME communities.^{127,155,168–170} Autosomal recessive polycystic disease (ARPKD; Online Mendelian Inheritance in Man [OMIM] catalogue

number 263200) occurs with a proposed incidence of 1/20,000 to 1/40,000. Mortality during the first month of life is particularly high among those born with severe oligohydramnios leading to pulmonary hypoplasia. Its principal manifestations include fusiform dilation of the renal collecting ducts and distal tubules and dysgenesis of the hepatic portal triad. ARPKD is associated with mutations in the *PKHD1* gene on chromosome 6p21.1-p12, encoding fibrocystin, a membrane protein located on the primary cilium.¹⁷¹ Over 700 mutations in *PKHD1* gene have been reported.¹⁷² Several mutations were reported in Turkish and Israeli children who participated in a large international study.^{173,174}

Meckel-Gruber syndrome (MKS) is a lethal, recessively inherited ciliopathy characterized by multiple congenital anomalies, including polycystic kidneys, occipital encephalocele, and polydactyly. At least 13 genes have been reported to date to underlie MKS. The incidence of this relatively rare disorder in Qatar has been reported as 1 in 500 live births for a population with over 40% consanguineous marriages.¹⁷⁵ The underlying genetic cause was identified in 11 of 12 Arab families residing in a distinct region in Israel, comprising mutations in seven different genes and revealing high genetic heterogeneity.¹⁷⁶ Likewise, other ciliopathies, including various forms of nephronophthisis (including the infantile type) with mutations in *NPHP1*, *NPHP2* (*INVS*), and *NPHP3*, encoding nephrocystin 1, inversin, and nephrocystin 3, respectively, have been reported in several ME countries^{177–179} (Table 78.5). Cases of Joubert syndrome, characterized by cerebellar vermis

Table 78.5 Rare Genetic Diseases With Renal Involvement Reported in Middle Eastern Communities

Country, Community	Disease	OMIM No.	Phenotype	Defect, Mutation	References
Saudi Arabs, Turks	Alström syndrome ^a	203800	Retinal degeneration, obesity, cardiomyopathy, sensorineural hearing loss, insulin resistance, renal impairment	<i>ALMS1</i> gene, ubiquitously expressed, encodes a protein of unknown function	220, 247
Israeli Jews of Iraqi origin	Renal hypouricemia type I	220150	Hypouricemia, hyperuricosuria, nephrolithiasis, exercise-induced acute kidney injury	<i>SLC22A12</i> gene, encodes uric acid transporter URAT1	242
Israeli Arabs	Renal hypouricemia type II	612076	Increased renal clearance of uric acid, hypouricemia, nephrolithiasis, exercise-induced acute kidney injury	<i>SLC2A9</i> gene, encodes glucose transporter 9 (GLUT9)	233, 237, 243
Israeli Jews, Turks	Dent disease ^a	300008	Low-molecular-weight proteinuria, hypercalciuria, nephrocalcinosis, nephrolithiasis, rickets, renal failure, hypokalemic metabolic alkalosis, focal glomerulosclerosis	<i>CLCN5</i> , encodes the chloride/proton ClC-5 antiporter	234, 236, 244
Egyptian and Saudi Arabs, Turks, Iranians, Jordanians	Cystinosis	219750 (adult) 219900 (juvenile) 219800 (infantile)	Failure to thrive, polydipsia, polyuria, Fanconi-like syndrome; corneal, conjunctival, and retinal deposition	<i>CTNS</i> , encodes the lysosomal cystine transporter	232, 240, 248, 249, 251
Israeli Druze	Autosomal recessive proximal tubulopathy with hypercalciuria	138160	Proximal tubulopathy, hypercalciuria, normal or slightly elevated urinary phosphate excretion	<i>SLC2A2</i> (<i>GLUT2</i>), encodes the glucose transporter 2 ^b	246
Israeli Arabs	Autosomal recessive Fanconi syndrome and hypophosphatemic rickets	613388	Proximal tubulopathy, renal phosphate wasting, normocalciuria, bone mineral deficiency, decreased glomerular filtration rate	<i>SLC34A1</i> , encodes the renal sodium phosphate cotransporter IIa	245
Israeli Arabs	Familial renal glycosuria and aminoaciduria ^c	233100	Glycosuria, aminoaciduria	<i>SLC5A2</i> , encodes the kidney-specific Na ⁺ /glucose cotransporter	239
Israeli Jews	Proximal renal tubular acidosis and glaucoma	604278 and 603345	Short stature, deformed teeth, bilateral glaucoma, blindness, metabolic acidosis	<i>SLC4A4</i> , encodes the sodium bicarbonate cotransporter (NBCe1)	235
Israeli Arabs and Jews	Familial autosomal recessive renal tubular acidosis	267300	Distal renal tubular acidosis, deafness	<i>ATP6V1B1</i> , encodes the B1-subunit of H ⁺ -ATPase	250
Israeli Arabs	Familial autosomal recessive renal tubular acidosis	259730	Proximal and distal renal tubular acidosis, osteopetrosis, mental retardation	<i>CA2</i> , encoding carbonic anhydrase	250
Egyptian Arabs	Familial hypomagnesemia with hypercalciuria and nephrocalcinosis	248250	Hypocalcemia, hypomagnesemia, hypercalciuria, nephrocalcinosis, congenital cataracts	<i>CLDN16</i> , the claudin-16 gene	241
Iranians	Familial lecithin-cholesterol acyltransferase deficiency ^d	245900	Lower extremity edema, proteinuria, corneal opacities, hypercholesterolemia, hemolytic anemia	Lecithin-cholesterol acyltransferase (<i>LCAT</i>) gene	238

^aX-linked.^bHomozygous mutations in *GLUT2* also cause Fanconi-Bickel syndrome.^cAutosomal recessive.^dDiagnosis made on the basis of family history and electron microscopic findings on kidney biopsy.OMIM, Online Mendelian Inheritance in Man catalogue (<http://www.ncbi.nlm.nih.gov/omim>).

hypoplasia, developmental delay, seizures, retinal dystrophy, and kidney involvement, have been reported in the ME. There is genetic heterogeneity with pathogenic mutations in 33 distinct genes responsible for this disorder. Founder mutations have been identified in Iranian families and in Ashkenazi Jews.¹⁸⁰

Genetic Glomerular Diseases

A substantial number of patients in the ME with renal disease have familial glomerular diseases whose spectrum includes familial hematuria, Alport syndrome, IgA nephropathy, and familial steroid-resistant nephrotic syndrome (SRNS) with histologic findings consistent with FSGS.^{81,166,181} A recessive form of SRNS has been shown to be associated with mutations in the *NPHS2* gene, which encodes the glomerular barrier protein podocin.¹⁸² The most common phenotype is of a nephrotic syndrome that is resistant to immunosuppressive treatment in early childhood, and most patients will develop ESKD by the end of their second decade of life. In a group of children from two consanguineous families of Israeli Arab descent, mutation analysis of the *NPHS2* gene revealed homozygosity for the C412T nonsense mutation (R138X).¹⁸³ The same mutation was subsequently found in additional children who presented with SRNS from the same ethnic background, which points to the possibility of a founder effect.¹⁸³ Interestingly, cardiac anomalies, especially left ventricular hypertrophy, pulmonary stenosis, and discrete subaortic stenosis, were detected in a high proportion of the children with *NPHS2* mutations.¹⁸⁴ Frishberg and associates¹⁸⁴ have speculated that podocin may have a role in normal cardiac development because podocin messenger RNA is reported to be expressed in the human fetal heart.

At least 53 *NPHS2* mutations have also been found in Turkish children with familial and “sporadic” SRNS.^{185,186} In this context, the term *sporadic* is a misnomer. As a recessively inherited disease, both parents are obligatory carriers of pathogenic mutations, and the lack of positive family history may merely represent the first case in a given nuclear family. Among Turkish patients with *NPHS2* mutations, the proportion of patients with CKD or ESKD was significantly higher (19 of 73) than that of patients without these mutations (28 of 222). Furthermore, the mean time for progression to ESKD was significantly shorter in patients with mutations compared with those without mutations.^{185,186} The prevalence of *NPHS2* mutations among 49 children with SRNS and 50 children with steroid-sensitive nephrotic syndrome from Iran was 31% and 4%, respectively.¹⁸⁷ Of these children, 33% experienced recurrence of primary disease after kidney transplantation, none of whom had an *NPHS2* mutation.

The *NPHS1* gene encodes nephrin, an essential protein for maintaining the normal structure and function of the slit diaphragm of the visceral glomerular epithelial cell. Three mutations in the *NPHS1* gene were reported in 12 children with congenital nephrotic syndrome of the Finnish type from a large consanguineous Israeli Arab family.¹⁶⁵ Steroid-sensitive nephrotic syndrome is rarely reported to have a familial pattern. However, a familial pattern associated with this condition has been reported in Israeli Bedouin families with a high rate of consanguinity, and the authors of the report proposed that the increased incidence of steroid-sensitive nephrotic syndrome resulted from selective enrichment of susceptibility genes in this population.¹⁸⁸

A study has evaluated causative mutations related to childhood nephrotic syndrome in 49 families from Saudi Arabia¹⁸⁹; 62 patients were screened for mutations in genes associated with congenital, infantile, or childhood nephrotic syndrome, such as *NPHS1*, *NPHS2*, *LAMB2* (laminin- β_2), *PLCE1* (phospholipase C_{ϵ_1}), *CD2AP* (CD2-associated protein), *MYO1E* (myosin 1E), *WT1* (Wilms' tumor suppressor gene 1), *PTPRO* (protein tyrosine phosphatase receptor type O), and *NEIL1* (Nai endonuclease VII-like 1).¹⁸⁹ A homozygous mutation in the *NPHS2* gene was found in 11 families (22%) and was the most common cause of nephrotic syndrome. Other mutations were found in *NPHS1* (12%), *PLCE1* (8%), and *MYO1E* (6%) genes. FSGS was the most common pattern of histopathologic injury, and a significant proportion of patients developed ESKD requiring RRT.¹⁸⁹

The Podonet Consortium, an international registry for congenital nephrotic syndrome and childhood SRNS, has analyzed the data retrieved from 1655 patients, 9.2% of whom were from the ME (Syria, Lebanon, Iran, and the UAE). Pathogenic mutations were detected in 23.6% of the screened patients.¹⁹⁰ A cohort of children with SRNS from a single center in Saudi Arabia was tested for causing mutations in *NPHS1*, *NPHS2*, and *WT1*. All 44 children were older than 1 year, and presumably pathogenic mutations were identified in five children (11.4%), three in *NPHS2* and the remaining two in *NPHS1*.¹⁹¹ None of those with an identified genetic cause showed any response to immunosuppressive treatment.

In Turkey, 25 children with SRNS underwent mutation screening in *TRPC6*, a dominantly inherited form of SRNS. Positive family history was found in nine children.¹⁹² The results showed a variant in one patient, intronic nucleotide substitution in six patients, and previously described missense and synonymous amino acid variants in nine patients. Although the pathogenicity of all these genetic variants remains to be proven, the authors concluded that *TRPC6* mutations may play a significant role in childhood SRNS.

Galloway-Mowat syndrome (GAMOS) is an autosomal recessive disease characterized by the combination of early-onset nephrotic syndrome (SRNS) and microcephaly with brain anomalies. Recessive mutations in *OSGEP*, *TP53RK*, *TPRKB*, and X-linked mutations in *LAGE3* genes encoding the four subunits of the KEOPS complex have been recently detected in 37 individuals from 32 families with GAMOS, 6 of whom are from the ME (Iran, Turkey, Jordan, and Egypt).¹⁹³

Coenzyme Q10 (CoQ10), also known as ubiquinone, is a lipid-soluble component of virtually all cell membranes and plays a critical role in transporting electrons along the respiratory chain of the mitochondrial inner membrane. *ADCK4* appears to have a role in CoQ10 biosynthesis; mutations in *ADCK4* may cause SRNS and/or CKD of presumably unknown cause.¹⁹⁴ A total of 146 index patients aged 10 to 18 years from Turkey, with newly diagnosed nonnephrotic proteinuria, nephrotic syndrome, or CKD of varying severity, were screened for *ADCK4* mutations.¹⁹⁵ In this study, 28 individuals with a biallelic mutation were identified, and CoQ10 supplementation appeared efficacious in reducing proteinuria in a subgroup of this cohort.

The Rat sarcoma homolog gene family of small guanosine triphosphate (GTP)-ases (Rho GTPases) is associated with actin remodeling and podocyte migratory ability. Alterations

in Rho GTPases signaling interfere with podocyte mobility and cause proteinuria.¹⁹⁶ Rho guanosine diphosphate (GDP) dissociation inhibitor 1, a regulator of Rho GTPases, is a protein that is encoded in humans by the *ARHGDI* gene and is also expressed in podocytes. *ARHGDI* mutations, associated with SRNS progressing to ESKD during the first decade of life, have been reported in three siblings from a family of Ashkenazi Jews and in a Moroccan infant.¹⁹⁷

Genetic Metabolic Diseases and Inherited Tubular Disorders

The primary hyperoxalurias (PHs) are a group of rare autosomal recessive inborn errors of glyoxylate metabolism that result in the excessive production of oxalate.^{198,199} Currently, three types of PH have been characterized genetically and phenotypically.²⁰⁰ The relatively more frequent PH type 1 (OMIM number 259900) is caused by the absence, deficiency, or mistargeting to the mitochondria of the liver-specific peroxisomal enzyme alanine-glyoxylate aminotransferase (AGT), which catalyzes the conversion of glyoxylate to glycine. Accumulating glyoxylate diffuses from the peroxisome into the cytosol, where it is oxidized to oxalate, a reaction catalyzed by lactate dehydrogenase (LDH). Excessive oxalate synthesis and massive urinary excretion of the insoluble calcium oxalate crystals results in nephrocalcinosis, kidney stones, and declining kidney function. As the glomerular filtration rate (GFR) declines, the oxalate load, produced by the liver, can no longer be cleared by the kidneys, the plasma oxalate level increases, and calcium oxalate crystals are deposited in almost all tissues in a devastating process termed “systemic oxalosis.” PH type 2 (OMIM number 260000) is caused by a deficiency of the mitochondrial enzyme glyoxylate reductase/hydroxypyruvate reductase (GRHPR), which catalyzes the reduction of glyoxylate to glycolate. In 2010, PH type 3 (OMIM number 613616) was described in a cohort of patients with calcium oxalate nephrolithiasis due to hyperoxaluria in whom PH1 and PH2 were excluded.²⁰¹ In affected members of nine unrelated families, Belostotsky and colleagues found biallelic loss-of-function mutations in *HOGA1*, formerly *DHDPSL*, which encodes a mitochondrial 4-hydroxy-2-oxoglutarate aldolase that catalyzes the fourth step in the hydroxyproline pathway.^{201,202} A founder mutation in *HOGA1* among Ashkenazi-Jews was identified.

PH types 1 to 3 are relatively prevalent in the ME. Cases of PH1 and isolated cases of type 2 have been reported in Israeli families.^{203–205} In 22 Israeli Arab families with type 1 hyperoxaluria, at least 15 different mutations in the AGT-encoding *AGXT* gene have been detected.^{203,205} Marked intrafamilial phenotypic variability, with no definite genotype-phenotype correlation, was noted in these families, and the prevalent phenotype was of early onset of disease with progression to ESKD in the first decade of life.²⁰³ The diagnosis of PH1 was confirmed in 15 of 19 children tested in a single tertiary center in Egypt.²⁰⁶ Two-thirds of this cohort reached ESKD, with one-third during infancy. A large European study of 155 patients from 129 families with type 1 hyperoxaluria, including a large proportion of ME individuals, has shown that the most common mutation is *p.Gly170Arg* (allelic frequency, 21.5%). This mutation results in mistargeting of AGT into the mitochondria while the catalytic activity of the enzyme remains intact. A subgroup of these patients will respond to vitamin B₆ treatment, a product of pyridoxal

phosphate, which is a known cofactor in all aminotransferase reactions and therefore may have a better long-term prognosis.²⁰⁷

Hyperoxaluria manifesting as kidney stone disease and/or progressive CKD has been described in the following reports from the ME. Although, systematic genetic analysis has not been performed, it is likely that PH accounts for the disease in a significant number of individuals. For example, among 260 adult patients with recurrent nephrolithiasis from a tertiary center in Israel, hyperoxaluria was detected in 24.2%.²⁰⁸ Hyperoxaluria was found among 10.7% of 84 patients with CKD, treated conservatively or by hemodialysis, in Jenin district in the Palestinian Authority.¹⁵⁵ PH1 is the probable cause of hyperoxaluria in most of these cases. In a retrospective report from a single center in Amman, Jordan, 70 children with suspected PH were identified.²⁰⁹ At the time of initial presentation, 15.7% were in ESKD and an additional 25% had impaired kidney function. Although not tested at the time of this report, PH1 may be responsible for hyperoxaluria in many of these children, particularly in those with compromised kidney function. Similarly, a high prevalence of hyperoxaluria was demonstrated among Turkish children with kidney stone disease.²¹⁰ Hyperoxaluria causing nephrolithiasis, nephrocalcinosis, and kidney failure was reported in children from other ME countries in western North Africa, Saudi Arabia, and Kuwait.^{151,211,212} Primary hyperoxaluria is probably underdiagnosed because, by the time of diagnosis, advanced CKD or early graft dysfunction in kidney transplant recipients may have already developed.¹⁹⁹

Cystinuria is a recessively inherited kidney stone disease, prevalent in various ethnic groups in ME countries.^{151,211,213} Cystinuria is a different disorder from cystinosis, which is a recessively inherited lysosomal transport disease leading to the accumulation of cystine, presenting with renal Fanconi syndrome, and resulting in CKD, with various extrarenal manifestations (see Table 78.5). Cystinuria is caused by biallelic mutations in either one or both of two genes:

1. The *SLC3A1* gene on chromosome 2p16.3-p21 encodes rBAT, a protein required for normal trafficking of the amino acid transporter to its appropriate location in the apical membrane of affected epithelia. In its mutated form, it leads to impairment of cystine and dibasic amino acid (e.g., ornithine, lysine, arginine) transport.
2. The *SLC7A9* gene located on chromosome 19q12-13.1 encodes an amino acid transporter, b⁰+AT. When this protein is defective, its appearance at the apical membrane, assuming that rBAT is normal, does not lead to normal amino acid reabsorption.

Because parents of patients with cystinuria due to *SLC7A9* mutations have increased levels of urinary cystine excretion, they may sometimes form stones if the urine volume is low or if animal protein intake is high. Otherwise, disease manifestations caused by either mutation are similar.^{214–216} Some patients may bear heterozygous digenic mutations, and they do not tend to form cystine stones unless there are mutations in two copies of a single gene. Mutations in *SLC3A1* have been detected in Turks, Muslim Arabs, Druze, and Ashkenazi and Sephardic Jews of Persian and Yemenite origin. The disease is also common among Libyan Jews, in whom the estimated prevalence is 1/2500 and the carrier rate is 1/25.

In this population, the disease is caused by a single founder mutation, V170M, in the *SLC7A9* gene.^{214–220}

Fabry disease is a rare X-linked inborn error of the glycosphingolipid metabolic pathway caused by deficiency of α -galactosidase A. A mutation in the gene encoding this enzyme results in insufficient breakdown of lipids, which accumulate to harmful levels in the eyes, kidneys, autonomic nervous system, and cardiovascular system. In untreated patients, the accumulation of globotriaosylceramide (Gb3) in lysosomes may result in multiple organ damage that includes the development of ESKD between the fourth and fifth decades of life. Histologic evidence of Fabry disease has been detected in graft biopsy samples many years after successful kidney transplantation.^{221,222} Although most disease features have been reported in adults, a pediatric disease phenotype that includes acroparesthesias, skin manifestations, and glomerular alterations has been described.^{222,223} Fabry disease has been diagnosed in families in Israel and Turkey.^{221–225} The disease may be underdiagnosed in other ME countries because of phenotypic variability of the disease, with often nonspecific signs and symptoms or insufficient awareness. However, screening of adult patients with ESKD undergoing dialysis in Turkey and in Lebanon has yielded an extremely low prevalence rate of Fabry disease.^{226,227}

Bartter syndrome and Gitelman syndrome belong to a group of inherited salt-losing tubulopathies with distinct phenotypes; they are caused by inherited defects in ion transporters in the loop of Henle and distal convoluted tubule, respectively. Most cases of the variants of Bartter syndrome in the ME have been reported in Israeli Arabs, in large Bedouin communities living in southern and northern Israel, and in Kuwaiti children.^{81,228,229}

Congenital anomalies of the kidney and urinary tract (CAKUT) comprise the most common cause of CKD in childhood. To understand the underlying pathogenetic mechanisms better, Vivante and associates have identified 20 Israeli pedigrees with isolated nonsyndromic renal hypodysplasia and screened for mutations in genes known to be involved in kidney development.²³⁰ Two brothers were found to have a heterozygous *PAX2* nonsense mutation, responsible for renal coloboma syndrome, which includes kidney dysplasia, eye coloboma, and hearing impairment. Nine affected subjects from two unrelated families were found to harbor heterozygous *HNF1B* mutations, encoding the hepatocyte nuclear factor-1 β . These mutations are associated with variable renal phenotypes, maturity-onset diabetes of the young (MODY5), hypomagnesemia, and hyperuricemia. In one family, two affected brothers were heterozygous for a missense mutation in *WNT4*. Functional analysis of this variant in different cell lines revealed both agonistic and antagonistic canonical WNT stimuli. In primary cultures of human fetal kidney cells, this mutation causes loss of function, resulting in diminished canonical WNT/ β -catenin signaling. These findings were interpreted as suggestive of a role for heterozygous *WNT4* variants in renal hypodysplasia.²³⁰ A three-generation Yemenite Jewish family with seven individuals who had CAKUT was investigated by whole-exome sequencing. Renal hypoplasia-dysplasia was the predominant phenotype (six individuals), together with vesicoureteral reflux (VUR; four individuals) and/or ectopia (two individuals).²³¹ A heterozygous truncating mutation of the nuclear receptor interacting protein 1 gene (*NR1P1*) was identified in all seven affected members. *NR1P1*

encodes a nuclear receptor transcriptional cofactor that directly interacts with the retinoic acid receptors (RARs) to modulate retinoic acid transcriptional activity.

Some rare genetic kidney diseases that have been reported in ME communities are listed in Table 78.5.^{232–251} Other nonrenal genetic diseases may cause glomerular and tubulointerstitial complications; an example is sickle cell anemia, a hemoglobinopathy that is prevalent in the ME countries in western North Africa and the Arabian Peninsula.^{151,156,215,252,253} Inherited diseases of the kidney are discussed further in Chapters 43, 44, and 72.

Familial Mediterranean Fever

FMF is the most common of the hereditary periodic fever syndromes and is a prevalent recessively inherited disease in the ME. FMF is an autoinflammatory disease that is characterized by recurrent attacks of fever, serositis, arthritis, and erysipelas-like skin lesions. The disease affects several ethnic groups in the ME, including Sephardic Jews, Armenians, Turks, and Arabs.²⁵⁴ The most significant complication of FMF is renal amyloidosis that progresses to nephrotic syndrome and ESKD. Renal amyloidosis is occasionally diagnosed in patients without a typical history of FMF attacks.

In the 1990s, two groups identified the *MEFV* gene by positional cloning as the underlying genetic cause of FMF.^{255,256} At least 327 mutations in the *MEFV* gene have been reported, and more than 195 of these mutations have been found to be associated with FMF.²⁵⁷ The *MEFV* gene encodes the protein pyrin (or marenostin). The gene is located on the short arm of chromosome 16 (16p13.3) and includes 10 exons that encode 781 amino acids. Pyrin is expressed predominantly in polymorphonuclear cells (PMNCs), eosinophils, and monocytes, but not in lymphocytes. It is also expressed in dendritic cells and fibroblasts from the synovium, peritoneum, and skin.^{258,259} Pyrin has an important role in the innate immune system, which constitutes a primary defense against external pathogens and other noxious agents.²⁶⁰ Its exact role and mechanism of action appear to be in sensing changes in the Rho GTPases.²⁶¹ Rho proteins are involved in controlling GTPase activity, which in turn are important in the regulation of actin and tubulin dynamics. Actin-tubulin interactions contribute to neutrophil motility and chemotaxis. Bacterial toxin (e.g., *Clostridium difficile* toxin) may modify the effect of Rho proteins on GTPases, thereby inhibiting actin-tubulin activity and neutrophil chemotaxis. Pyrin does not directly recognize Rho modification but probably senses an event downstream of Rho modification in the actin cytoskeleton pathway and forms a caspase-1-activating inflammasome.^{262,263} Therefore, pyrin-mediated innate immunity is unique in that it senses bacterial virulence rather than microbial molecules.²⁶²

In patients with FMF, the mutations in the *MEFV* gene lead to the production of pyrin protein, which is capable of constructing the inflammasome, even in the absence of a known external trigger such as a toxin or infectious agent. This possibility suggests that the mutations in *MEFV* gene cause gain of function of pyrin protein so that it acts without a known provocation. The outcome of this process is the secretion of interleukin (IL)-1, IL-18, and other mediators of inflammation, which enhance chemotaxis and neutrophilia and induce an attack of FMF.

The carrier frequency of mutant alleles is high, as much as one-fifth to one-third, in certain populations of the ME

countries (Armenians, Jews, and Turks). The most common reported mutations are *M694V*, *V726A*, *M680I*, and *M694I* in exon 10 and *E148Q* in exon 2.^{264,265} The first four mutations are believed to be pathogenic, with the *M694V* homozygotes having the most severe phenotype. Although the *E148Q* homozygotes are asymptomatic in 50% of cases, this variant may be associated with other systemic inflammatory diseases, including Behçet disease, vasculitis, ulcerative colitis, rheumatoid arthritis, and multiple sclerosis.²⁶⁶ The most frequent *MEFV* mutations in the ME populations are listed in Table 78.6.^{266–294} Data from five databases encompassing whole-genome and whole-exome datasets for 2115 individuals from multiple subpopulations in the ME were available for analysis.²⁹⁵ A compendium for *MEFV* genetic variants was also created. The carrier frequency for genetic variants in *MEFV* was found to be 8%, with differences in allele frequency among subpopulations. Also, three pathogenic variants were added to the four popular studied variants. The former include c.1261-11T>G, Y688X, and E148V. Mutations in the *MEFV* gene have also been reported in Spanish, Italian, Greek, Portuguese, Indian, Chinese, and Japanese populations.^{296,297} Most *MEFV* mutations are single amino acid substitutions (missense), and some patients with FMF have a single *MEFV* mutation. The latter may imply a dominantly inherited trait, which may go along with a gain of function of the gene product. Alternatively, additional genes or immune factors may modulate the innate immune response in FMF.

The clinical spectrum of FMF and the elucidation of the molecular biology, structure, and regulation of pyrin and its role during inflammation have been reviewed.^{263,298–301} Pyrin belongs to the large family of proteins sharing a conserved domain structure with the tripartite motif (TRIM) consisting of an N-terminal RING domain, B-box domain(s), and C-terminal coiled coil domain. In pyrin, the RING domain with the ubiquitin ligase activity is replaced by the PYD domain, which belongs to the death domain. It also carries an additional approximately 200–amino acid C-terminal B30.2/rfp/PRY/SPRY domain. Most of the disease-associated mutations are located in the C-terminal B30.2 domain. The B30.2 domain is present in primate and human pyrin, but not in rodent pyrin.³⁰¹ It includes a SPRY domain that is located in front of the C-terminal region and a SPRY extension at the N terminus. The SPRY domain is a protein interaction module, which is implicated in several biologic pathways, including those that regulate innate and adaptive immunity.³⁰²

During the last decade, the functions of pyrin have been the subject of intensive research. Advances in our understanding of its functionality are the result of identifying its interaction with different proteins and oligomers.²⁶³ In particular, the PYD domain, via cognate pyrin domain association, interacts with an adapter protein termed *apoptosis-associated specklike protein with a caspase recruitment domain* (ASC)³⁰³ and thus participates in the regulation of apoptosis, inflammation, and IL-1 β processing.³⁰⁴ The B30.2/rfp/PRY/SPRY domain-mediated protein interactions indicate the role of this domain as an adaptor module to assemble macromolecular complexes.³⁰⁵ As a part of the pyrin inflammasome, pyrin also acts as a pattern recognition receptor, sensing pathogen modification and inactivation of Rho GTPases (see earlier).²⁶¹

Paradoxically, pyrin also acts as a component directing the inflammasome components NLRP1, NLRP3, and procaspase 1 to selective autophagic degradation.³⁰⁶ On its own,

the dimerization of B30.2 domains by means of the CHS domain appears to be crucial for the recognition of higher order oligomers.³⁰⁷ Inflammasomes contain members of the nucleotide-binding oligomerization domain, leucine-rich repeat, and PYRIN domain-containing subfamily of proteins, now known as NLRP proteins. Mutations in the gene expression of the NLRP protein, *NLRP3* (cryopyrin), are associated with monogenic autoinflammatory diseases—cryopyrin-associated periodic syndromes.³⁰⁸

Clinical Spectrum and Renal Disease. The wide clinical spectrum and possible genotype-phenotype correlations in FMF are influenced by genotype (extent and position of the *MEFV* mutation), ethnicity, and environmental factors. Although mutations are found throughout the entire *MEFV* gene in patients with FMF, the mutations with the most severe forms of disease are clustered in exon 10, which encodes a motif known as the B30.2/SPRY domain at the C terminus of the protein. Genetic variants found in exons 2 (e.g., *E148Q*) and 3 are usually associated with less severe clinical presentations of FMF or even just mild nonspecific inflammatory manifestations. Overall, *M694V* homozygotes have a severe phenotype and are more likely to have arthritis, renal amyloidosis, erysipelas-like skin lesions, high fever, splenomegaly, and more frequent attacks as compared with individuals with other *MEFV* mutations.³⁰⁹ In addition, patients with these mutations require higher doses of colchicine to prevent attacks as compared with patients with other genotypes. Among Arab patients with FMF, the worst disease severity is associated with alleles carrying the mutations *M694V/M694V* and *M694V/M726A*, whereas the *M694I/M694I* mutation is associated with a mild form of the disease.³¹⁰ In Turkish patients with FMF, the common *M694V* mutation is associated with more severe disease but not with amyloidosis.^{293,300} In contrast, the homozygous *M694V* mutation was reported to be associated with amyloidosis in Jews from North Africa, in Arabs, and in Turks and also with the protracted febrile myalgia syndrome, which is a type of myalgia that is associated with FMF in Arabs.*

FMF has traditionally been considered an autosomal recessive genetic disease, but as many as 25% of patients with clinical FMF have only one *MEFV* mutation.^{292,293,299} This finding could explain the vertical transmission of the disease in some families. In patients with FMF who have a single *MEFV* mutation, an additional but less common mutation may account for the disease, but this has not yet been confirmed in carefully performed studies. A dominant segregation of a heterozygous *MEFV* mutation was reported in some families. Therefore, it was suggested that complex alleles can lead to a more severe form of disease.³¹¹ One study has reported that heterozygous mutations at amino acid position 577 of pyrin (T577) can induce autosomal dominant autoinflammatory disease. The T577 is located in front of the C-terminal B30.2/SPRY domain and is crucial for pyrin function.^{309,312} The T577 mutations were found in a family of Turkish descent with autosomal dominant FMF phenotype and also in European patients with other periodic syndromes. The impact of *MEFV* mutations has been shown in other inflammatory diseases, such as Henoch-Schönlein purpura,

*References 276, 278, 285, 287, 293, 294, and 310.

Table 78.6 Genotype-Phenotype Correlations in Several Patient Populations With Familial Mediterranean Fever

Population, Community	Frequent Mutations	Phenotype	Amyloidosis (% of Patients)	Associated Syndromes and Diseases	Reference
Israeli Jews and Arabs	<i>M680I</i> , <i>M694V</i> , <i>M694I</i> , <i>V726A</i>	Arthritis, fever, serositis, vasculitis	1.4% (associated with mutation in <i>M694V</i>)	NR	Brik et al ²⁷⁸
Israeli Jews and Arabs	<i>M694V</i> (very common in Jews), <i>M694I</i> (exclusive in Arabs), <i>M680I</i> , <i>V726A</i> , <i>E148Q</i>	Arthritis, fever, serositis, vasculitis	NR	NR	Ben-Chetrit et al ²⁷⁶
Israeli Arabs	<i>M694V</i> (associated with severe disease), <i>V726A</i> (most common)	Arthritis, fever, serositis, vasculitis	NR	NR	Shinawi et al ²⁹¹
Israeli Jews of North African origin and Arabs ^a	<i>M694I</i> , <i>M694V</i> (very common in North African Jews), <i>E148Q</i>	Synovitis, pleuritis, abdominal pain, skin rash	95% ^a	Focal glomerulosclerosis	Ben-Chetrit and Backenroth ²⁷⁵
Israeli Jews of North African origin, Ashkenazi Jews, Jews of Iraqi origin, Israeli Arabs, and Druze	<i>E148Q</i> , <i>M694V</i> (very common in North African Jews), <i>V726A</i>	FMF criteria ²⁸⁴	4.6% (most common in <i>M694V</i> homozygous)	NR	Zaks et al ²⁹⁴
Israeli Jews (Ashkenazi and non-Ashkenazi), Arabs, and Druze	<i>M694V</i> (very common in Jews), <i>E148Q</i> , <i>M694V</i> , <i>V726A</i> (equally common in Arabs), <i>E148Q</i> (most common in Druze)	Fever, serositis	NR	NR	Sharkia et al ²⁹⁰
Turks	<i>M680I</i> , <i>M694V</i> , <i>M694I</i> , <i>V726A</i> , <i>E148Q</i>	Abdominal pain, fever, arthralgia, chest pain, skin rash	3% (mostly associated with <i>M694V</i>)	NR	Solak et al ²⁹²
Turks	<i>M680I</i> , <i>M694V</i> , <i>V726A</i>	Abdominal pain, fever, arthralgia, pleuritis, muscle pain, skin rash	12.9%; 0.9% as the main disease manifestation (phenotype II, associated with <i>M694V</i>)	Nonamyloid renal disease, Henoch-Schönlein purpura, polyarteritis nodosa, Behçet syndrome, rheumatic fever, uveitis, inflammatory bowel disease	Tunca et al ²⁹³
Turks ^b	<i>M680I</i> , <i>M694V</i> (most common), <i>V726A</i> , <i>E148Q</i>	Abdominal pain, fever, arthralgia, chest pain, erysipelas-like lesion, vomiting, family history of renal failure	0%	NR	Caglayan et al ²⁷⁹
Turks	<i>M680I</i> , <i>M694V</i> (most common), <i>M694I</i> , <i>V726A</i> , <i>E148Q</i> (common)	Abdominal pain, fever, arthritis, pleuritis, erysipelas-like erythema, peritonitis	NR	NR	Ozdemir et al ²⁷⁰
Turks	<i>M680I</i> , <i>M694V</i> (most common), <i>V726A</i> , <i>E148Q</i>	Fever, arthritis, pleuritis, erysipelas-like erythema, peritonitis, vasculitis	8.6% (most common in <i>M694V</i> homozygous)	NR	Kasifoglu et al ²⁶⁸
Azeri Turks	<i>M680I</i> , <i>M694V</i> (most common), <i>M694I</i> , <i>V726A</i> , <i>E148Q</i>	Fever, serositis, synovitis, kidney failure	NR	NR	Mohammadnejad and Farajnia ²⁶⁹
Jordanian Arabs Palestinian Arabs	<i>M680I</i> , <i>M694V</i> , <i>M694I</i> , <i>V726A</i> , <i>E148Q</i>	Abdominal pain, fever, arthralgia, myalgia, skin rash	1% (associated with <i>M694V</i>)	Protracted fever, myalgia syndrome; 42% homozygous for <i>M694V</i>	Langevitz et al ²⁸³ ; Majeed et al ²⁸⁵

Table 78.6 Genotype-Phenotype Correlations in Several Patient Populations With Familial Mediterranean Fever (Cont'd)

Population, Community	Frequent Mutations	Phenotype	Amyloidosis (% of Patients)	Associated Syndromes and Diseases	Reference
Jordanian Arabs	<i>M680I</i> , <i>M694V</i> , <i>V726A</i> , <i>E148Q</i>	Abdominal pain, fever, arthralgia	9% (<i>M694V</i> in one patient, <i>V726A/M680I</i> in two patients)	Celiac disease, folliculitis	Medlej-Hashim et al ²⁸⁷
Palestinian Arabs	<i>M680I</i> , <i>M694V</i> , <i>V726A</i> , <i>E148Q</i>	NR	NR	NR	Ayesh et al ²⁷³
Arabs from Jordan, Egypt, Syria, Iraq, and Saudi Arabia	<i>M694V</i> , <i>V726A</i> , <i>E148Q</i>	NR	NR	NR	Al-Alami et al ²⁷²
Iranian Azeris who live in Turkey	<i>M680I</i> , <i>M694V</i> , <i>M694I</i> , <i>V726A</i> , <i>E148Q</i>	Abdominal pain, fever, arthralgia, pleuritis, skin rash	7%	NR	Esmaeili et al ²⁸²
Iranians (72% Azeri)	<i>M680I</i> (most common), <i>M694V</i> , <i>V726A</i>	Fever, peritonitis, arthralgia, pleuritis, skin rash	5.6%	NR	Bidari et al ²⁷⁷
Egyptian Arabs	<i>M680I</i> , <i>M694V</i> , <i>V726A</i>	Abdominal pain, chest pain, fever, arthritis, myalgia	NR	NR	Settin et al ²⁸⁹
Egyptian Arabs	<i>M680I</i> , <i>M694I</i> (most common), <i>M694V</i> , <i>V726A</i> , <i>E148Q</i>	Fever, serositis	Mostly associated with <i>M694V</i>	NR	el Gezery et al ²⁸⁰
Egyptian Arabs	<i>V726A</i> (most common), <i>M694V</i> (common), <i>M680I</i> , <i>M694I</i> , <i>E148Q</i>	Abdominal pain, fever, arthralgia, pleuritis, skin rash, myalgia	2.9% (associated with <i>M694V</i>)	NR	el Garf et al ²⁶⁷
Syrian Arabs	<i>M680I</i> , <i>M694V</i> , <i>M694I</i> , <i>V726A</i> , <i>E148Q</i> , <i>A744S</i> , <i>R761H</i>	Serositis, fever, arthritis, pleuritis	5%	NR	Mattit et al ²⁸⁶
Lebanese Arabs	<i>M680I</i> , <i>M694V</i> (very frequent), <i>M694I</i> , <i>V726A</i> , <i>E148Q</i> (very frequent); minor alleles also detected	Serositis, fever, arthritis, chest pain	NR	NR	Sabbagh et al ²⁸⁸
Algerian, Moroccan, and Tunisian Arabs	<i>M694V</i> , <i>M694I</i> (most common), <i>M680I</i> , <i>M680I</i> , <i>A744S</i> , <i>V726A</i> , <i>E148Q</i>	Serositis, fever, arthritis, chest pain	NR	NR	Belmahi et al ²⁷⁴
Algerians	<i>M694I</i> (most common), <i>M694V</i> , <i>E148Q</i> , <i>A744S</i> , <i>M680I</i>	Abdominal pain, fever, arthritis, chest pain, erythema	8%	NR	Ait-Idir et al ²⁷¹
Tunisian Arabs	<i>M680I</i> (most common), <i>M694V</i> , <i>M694I</i> , <i>V726A</i> , <i>E148Q</i> , <i>A744S</i> , <i>R761H</i> , <i>1692del</i>	FMF criteria ²⁸⁴	3.5%	NR	Chaabouni et al ²⁸¹

^aStudy performed in patients with end-stage kidney disease.
^bStudy performed in compound heterozygous patients.
 FMF, Familial Mediterranean fever; NR, not reported.

polyarteritis nodosa, Behçet disease, rheumatic heart disease, and rheumatoid arthritis (reviewed by Guz and associates).³⁰⁰ Of interest, individuals who carry a single mutated allele in the *MEFV* gene can also suffer from the syndrome of periodic fever, aphthosis, pharyngitis, and adenitis (PFAPA), ankylosing spondylitis, and Crohn disease.³¹³

Secondary or reactive AA amyloidosis is the most severe complication of FMF. Before the advent of colchicine

treatment, amyloidosis was reported to occur in approximately 75% of patients with FMF who were older than 40 years.³¹⁴ The disease is caused by the extracellular deposition of amyloid A fibrils. These fibrils consist of β -pleated sheet polymers of the N-terminal fragments, the products of incomplete proteolytic digestion of the acute-phase precursor serum amyloid A (SAA) protein, whose production is markedly increased in chronic inflammatory processes.³¹⁵ Amyloidosis

is frequently found in Jews of North African descent, in Turks, and in Armenians. As mentioned earlier, a positive association of amyloidosis with the *M694V* mutation has been reported in patients from several ethnic groups.*

The *M694V* mutation has also been found in patients with FMF phenotype II, in which amyloidosis is detected before the onset of clinical symptoms of FMF.²⁹³ Phenotype III is defined as the presence of two *MEFV* mutations (homozygous or compound heterozygote state) without clinical manifestation of FMF or amyloidosis. This condition, however, like phenotype II, could predispose to the development of amyloidosis in families in which one sibling is afflicted with FMF.³¹⁷ A screening study of patients with FMF from 14 countries worldwide—which included Israel, Turkey, Qatar, Jordan, and Lebanon—has revealed that amyloid nephropathy is present in 11.4% of patients with FMF.³¹⁸ In this study, however, the country of recruitment, rather than *MEFV* genotype, was found to be the leading risk factor for amyloidosis.³¹⁸ These findings underscore the relative contribution of environmental factors to the genetic background in defining the phenotypic variation and severity of the disease.³¹⁹

Male gender, a positive family history of amyloidosis, pain flares in joints, and the *SAA1* α/α genotype have been suggested as additional risk factors for amyloidosis.³¹¹ Albuminuria is usually the first manifestation of renal involvement in patients with FMF and amyloidosis, and in untreated patients it may appear early in the course of FMF.³²⁰ In one study, the serum cystatin C level was found to be significantly increased in patients with FMF and secondary amyloidosis. It was suggested that the cystatin C serum level may be an early marker of renal impairment, even before the onset of albuminuria.³²¹ Proteinuria, mostly represented by albumin, can reach the nephrotic range, with urinary protein excretion rates higher than 20 g/day. If untreated, AA amyloidosis progresses to ESKD, which ultimately necessitates RRT. Renal disease due to amyloidosis accounts for 35% and 60% of deaths in men and women, respectively, with FMF.³²² Extrarenal deposits of AA amyloid can be found in the liver, spleen, lung, thyroid, heart, adrenal glands, stomach, and testes. These deposits can be clinically significant and are associated with intestinal malabsorption and adrenal insufficiency, which are often diagnosed after the initiation of chronic hemodialysis or after kidney transplantation.^{320,323,324}

The diagnosis of amyloidosis can be confirmed by tissue biopsy; the sensitivities of the rectal or bone marrow biopsy and renal biopsy are 79.5% and 88%, respectively, whereas the sensitivity of biopsy of abdominal fat is lower.^{325–327} The presence of amyloid deposits is demonstrated histologically with Congo red staining of the biopsy tissue sample. Congo red–stained amyloid has an orange appearance when viewed under a light microscope and an apple green birefringence when viewed under polarized light. Another amyloid-specific stain, thioflavin T, is used less frequently than Congo red. Amyloidosis is diagnosed definitively on electron microscopy by the demonstration of characteristic amyloid fibrils. The AA protein in tissue can also be detected with antibodies against AA protein.³²⁵

In addition to AA amyloidosis, other kidney diseases have been diagnosed in patients with FMF. It is worth mentioning

that many of the reports of systemic and inflammatory diseases in patients with FMF contain descriptions of kidney involvement.²⁹³ In fact, the frequency of the reported nonamyloid glomerular lesions reflects the number of kidney biopsies performed in patients with abnormal findings on urinalysis or on kidney function tests. Recurrent focal and proliferative glomerulonephritis and polyarteritis nodosa were described in an early report of the disease.³²⁸ Rapidly progressive glomerulonephritis, mesangial proliferative glomerulonephritis, IgA nephropathy, and membranous nephropathy have also been reported.^{293,299,329–332}

Treatment of Familial Mediterranean Fever. Daily oral colchicine is the most effective therapy for patients with FMF, and its use since the 1980s has dramatically changed the course of the disease. Colchicine prevents both acute attacks of FMF and SAA amyloidosis.^{333,334} The mechanism of colchicine action is not completely understood. The pathophysiology of FMF involves the recruitment and activation of neutrophils at serosal surfaces. Colchicine affects the motility of neutrophils by reducing their deformability and elasticity. These properties are crucial for their extravasation in response to inflammatory stimuli. On a cellular level, it is believed that one of its major actions is to interact with cytoskeletal structures, such as microtubules. It has been shown that colchicine accumulates in PMNCs, where it depolymerizes the microtubules and suppresses microtubule dynamics. Pypin is also expressed in PMNCs and associates with microtubules.³³⁵ Because colchicine also has an inhibitory effect on chemotaxis and reduces serum levels of IL-6, IL-8, and tumor necrosis factor- α (TNF- α),³³⁶ its antiinflammatory action is thought to be related to its ability to suppress nuclear factor-kappa B (NF- κ B) activation by attenuating calpain-mediated I κ B kinase α degradation, which is enhanced by N-terminal cleaved pypin.³³⁷ Suppression of caspase-1 expression in PMNCs was also suggested to be a potential mechanism of the action of colchicine.³³⁸

In vitro, colchicine has been shown to have a specific effect on pypin and pypin-interacting protein, such as proline-serine-threonine phosphatase interacting protein 1 (PSTPIP1) and ASC. In THP-1 cells, colchicine reorganized cytoskeleton and downregulated *MEFV* expression. This effect, which may result in a reduction of the level of a proinflammatory mutant pypin, may explain the suppressive mechanism of colchicine on FMF attacks.³³⁹

The most common adverse effects of colchicine are gastrointestinal, especially abdominal pain and diarrhea. Colchicine toxicity from overdosing is associated with hepatic, renal, muscle, and cerebral effects.³⁴⁰ Approximately 5% to 10% of FMF patients are nonresponders, and 2% to 5% do not tolerate the drug, mainly due to gastrointestinal side effects. In *M694V* homozygotes, a significant portion of patients show a partial or minimal response to colchicine, despite being treated with appropriate doses.³⁴¹ In contrast, prolonged colchicine-free remission has been noted in some individuals, none of whom was homozygous for this mutation.³⁴²

One study has revealed the existence of an association between polymorphism of the adenosine triphosphate-binding cassette, subfamily B, member 1 (*ABCB1*) gene and the response to colchicine in Turkish patients with FMF. The *ABCB1* gene encodes p170—also known as multidrug resistance 1 (MDR1)—a glycoprotein that functions as a drug

*References 275, 278, 285, 287, 292, 293, and 316.

transport pump, which can cause a variety of drugs, including colchicine, to become extruded from cells. In this study, patients with FMF who had the TT genotype for the c.3435C>T variant of *ABCB1* responded better to colchicine in terms of treatment efficacy and lower dose requirements in comparison with patients who had the CT and CC genotypes.³⁴³ These results warrant validation in further studies in patients with FMF from different ethnic populations.

Thalidomide, NSAIDs, corticosteroids, azathioprine, prazosin, and eprodisate disodium, an inhibitor of fibril formation, had occasionally been used to treat patients with FMF whose disease was refractory to colchicine.^{325,344–346} Interferon- α (IFN- α) was one of the first drugs used to treat patients with FMF who were refractory to colchicine.³⁴⁷ Two studies of patients with FMF who were treated with IFN- α have shown that the duration and pain intensity of most of the attacks were reduced by more than 50%.^{348,349} In addition, clinical improvement has been reported in patients with FMF who were treated with the TNF- α antagonists etanercept or infliximab.^{345,350,351}

Because most inflammatory manifestations of FMF are believed to be associated with the induction of IL-1 β production by mutations in the C-terminal B30.2 domain of pyrin, it has been proposed that IL-1 β antagonists may be an effective therapy for FMF. In fact, more studies have suggested that IL-1 inhibition seems to be a safe and effective alternative for patients with FMF who do not respond to or cannot tolerate colchicine. The IL-1 β receptor antagonist anakinra has been shown to have a beneficial effect in patients with FMF and with cryopyrin-associated periodic syndromes.^{344,346} Indeed, significant improvement and resolution of FMF symptoms have been reported in patients with colchicine-resistant FMF (crFMF) who were treated with anakinra.^{346,352,353} Anakinra elicited a beneficial therapeutic response in individuals with FMF and chronic kidney disease, and even in a patient on maintenance hemodialysis and after kidney transplantation.^{353,354} The main limitation of anakinra treatment in patients with FMF is probably the need for daily subcutaneous injections. The administration of canakinumab, a human monoclonal antibody against IL-1 β , every 8 weeks has been associated with rapid remission of symptoms in FMF and cryopyrin-associated periodic syndromes.³⁵⁵ In a systematic review of eight studies that included 40 patients with crFMF, rates of complete and partial response to canakinumab were 68% and 32%, respectively.³⁵⁶

Rilonacept is a dimeric fusion protein consisting of the extracellular portions of the human IL-1 receptor and the Fc region of human IgG1 that binds and neutralizes IL-1. A relatively small, randomized controlled study has shown a decrease in the attack rate, but the duration of attacks did not differ between placebo and rilonacept.³⁰² Similarly, a beneficial effect has been noted with the use of rilonacept in cryopyrin-associated periodic syndromes.³⁵⁷

Familial Mediterranean Fever and End-Stage Kidney Disease. In contrast to the poor prognosis of patients with primary (AL) amyloidosis who require RRT, most patients with FMF and secondary (AA) amyloidosis do relatively well with chronic dialysis.³⁵⁸ However, the issue of which dialysis modality is ideal for the treatment of FMF-associated ESKD is still unresolved,⁸³ and prospective studies that compare the different modalities are still needed. Of the dialysis

modalities, hemodialysis is more widely used in patients with FMF. Continuous ambulatory peritoneal dialysis (CAPD), however, has been performed successfully in selected patients. The effects of CAPD on azotemia and overall survival are similar in patients with nonamyloid diseases and patients with FMF. However, serum albumin levels are reported to be lower, and the rate of peritonitis and requirements for erythropoietin-stimulating agents (ESAs) higher, in patients with FMF who underwent CAPD than in patients with FMF who underwent hemodialysis.^{359,360}

A considerable number of patients with FMF and ESKD have undergone kidney transplantation in the last few decades. The rates of survival of patients and allografts are reported to be worse or similar to those in the general population of kidney transplant recipients.^{83,293,361–363} Analysis of the data from a single center in Israel has shown that graft survival is comparable between patients with FMF and controls.³⁶⁴ However, FMF was associated with more than a twofold increased risk for death following transplantation. Infections and cardiovascular disease were the leading causes of death. Maintenance colchicine therapy is obligatory after kidney transplantation because it may prevent the recurrence both of FMF symptoms and of amyloidosis. However, recurrence of AA amyloidosis has been frequently reported 8 to 10 years after kidney transplantation,³⁶⁵ and the rate of recurrence may be as high as 71%.³⁶⁶ Adverse drug effects are seen in kidney transplant recipients with FMF who are taking cyclosporine. In addition, increased gastrointestinal complications can be observed in patients who are treated with a combination of mycophenolate mofetil and colchicine.³⁶⁷

MANAGEMENT OF END-STAGE KIDNEY DISEASE IN THE MIDDLE EAST

Dialysis and kidney transplantation are available in all ME countries. In all industrialized and many developing ME countries, RRT is accessible to every patient with ESKD, regardless of the patient's socioeconomic status or whether the patient has health insurance (see Table 78.3 and Fig. 78.4).*

On the other hand, the number of patients offered RRT in other developing ME countries may be affected by late diagnosis and referral, the presence of comorbid conditions, the country's health system, reimbursement, and the availability of dialysis facilities.^{2,151}

DIALYSIS

PERITONEAL DIALYSIS

Throughout the ME, hemodialysis is the preferred treatment modality for ESKD (see Fig. 78.4). Peritoneal dialysis is an underused treatment modality for ESKD, although its use has been increasing in Tunisia, Kuwait, Iran, Saudi Arabia, Qatar, and Turkey.[†]

Several medical and nonmedical factors play an important role in inhibiting the widespread use of peritoneal dialysis throughout the ME, including the following: (1) health care providers generally offer low or no reimbursement for

*References 19, 109, 111–113, 118, 119, 127, 128, 136, 144, 151, and 368–373.

†References 112, 121, 125, 127, 132, 374, and 375.

peritoneal dialysis; (2) the numbers of peritoneal dialysis training programs, qualified nephrologists, and skilled dialysis nurses are limited, and the salaries of attending physicians for peritoneal dialysis are low; (3) many patients have poor education and poor hygiene practices; and (4) patients and caregivers have concerns about the high rates of peritonitis.

At the end of 2006, 38,824 patients were treated with dialysis (87.4% with hemodialysis, 12.6% with peritoneal dialysis) in Turkey, and 99.7% of all patients undergoing dialysis were covered by the social security system.³⁷⁰ The Turkish experience in establishing a peritoneal dialysis program has been outstanding and could be used as an example by other ME countries. The Turkish Multicenter Peritoneal Dialysis Study Group (TULIP) has established a platform for organizing peritoneal dialysis facilities and units. The group helped standardize patient records and various peritoneal dialysis treatments and maximized the number of patients with ESKD who can benefit from this treatment modality. TULIP has published clinical research that has affected peritoneal dialysis practice locally and worldwide.³⁷⁰ The Turkish peritoneal dialysis program has good rates of survival of patients and technique efficacy, which are comparable with those in Western industrialized countries.¹⁵⁴ The peritoneal dialysis dropout rate was 21%, and the incidence of peritonitis was one episode per 35.5 patient-months in 2007. Dyslipidemia was the most common noninfectious complication¹³⁴; cardiovascular diseases (42.3%), followed by infections (19.9%) and cerebrovascular events (13.6%), were the major causes of death among Turkish patients undergoing peritoneal dialysis.^{134,154} Moreover, peritoneal dialysis has been suggested as a means of reducing the seroconversion rate for hepatitis C virus (HCV) in populations with ESKD and a high prevalence of infection³⁷⁶; such treatment thereby confers an advantage to kidney transplantation candidates.³⁷⁷

Peritoneal dialysis was started in Iran in 1978 and in Saudi Arabia and Kuwait during the 1980s with imported peritoneal dialysis solutions. The costs of this treatment are low in Turkey and Iran because peritoneal dialysis solutions began to be produced locally in Turkey in 1994 and in Iran in 1995.¹²⁵ Establishment of a peritoneal dialysis registry and a multidisciplinary approach has dramatically increased the numbers of patients undergoing peritoneal dialysis in Iran since 2001.^{107,125} In 2010, the 1-, 3-, and 5-year patient survival rates in Iran were reported as 89%, 64%, and 49%, respectively, whereas death- and transplantation-censored technique survival rates were 90%, 73%, and 58%.¹⁰⁷ The main causes of death among Iranian patients undergoing peritoneal dialysis were cardiac events (46%), cerebral stroke (10%), and infection (8%). More recently, the rate of peritonitis has improved from one episode in 15.2 to 19.4 patient-months to one episode in 25 patient-months (in 2010).^{107,125}

The use of peritoneal dialysis has been growing in Qatar since its introduction in 1997, and peritoneal dialysis is offered to approximately 23% of ESKD patients. Most of these patients are non-Qatari male expatriates, a group that favors a mode of dialysis that can be self-administered, secures the ability to work or travel, requires less dietary restrictions, and potentially decreases the hospital and medical costs.³⁷⁴

These advantages of peritoneal dialysis, which provides flexibility and ease of adjustment for lifestyle and daily schedule, has encouraged some clinically stable Muslim

patients on peritoneal dialysis to fast during the holy month of Ramadan.³⁷⁸ However, patients who intend to fast need to be comprehensively educated and followed up by their peritoneal dialysis units.

HEMODIALYSIS

There are limited detailed published data on hemodialysis in the ME with regard to practice patterns and outcomes, such as the quality of dialysis, survival of patients per technique, type of vascular access, standard dialysate solutions and dialyzers, medical complications and their management, comorbid conditions, and other measures of quality assurance.*

In general, quality is ensured according to best medical practice guidelines in industrialized ME countries with more resources, easily accessible health care systems, and availability of advanced medical supplies.^{111,112,128,142,386-390} In ME countries that lack resources, the quality of dialysis is dictated by nonmedical financial considerations. In these countries, dialysis may be offered only once or twice weekly because of recent conflicts, lack of skilled personnel, distance of the dialysis centers from the patient's residence, and inability of the patient to pay for more frequent treatment.^{22,86,128,138,373} The number of patients who withdraw from hemodialysis therapy because of financial problems has not been documented, however. A small number of patients do seek pre-ESKD care in most ME countries.^{128,381,391,392}

Practice guidelines for hemodialysis have been developed in Egypt. These included five main domains: personnel, patient care practices, infection prevention and control, facility, and documentation or records. Before distribution of the guidelines, one study had disclosed flawed compliance and variability in adoption of evidence-based hemodialysis guidelines among 16 Egyptian facilities affiliated with the Ministry of Health and Population.³⁷⁹ Therefore, it is mandatory to distribute these guidelines to all hemodialysis facilities in Egypt, along with the organization of workshops to educate staff and follow-up to ensure implementation of the guidelines. In Libya, the provision of dialysis was adequate in 2010. However, several areas for improvement were identified by Alashek and associates,¹³⁸ such as lack of national dialysis practice guidelines, and few clinical practitioners were familiar with KDOQI guidelines. Because of continuous drought, many dialysis centers were obliged to dig their own wells to maintain the water supply. However, testing of water quality was not regularly practiced. In addition, wide discrepancy in methods applied to manage and monitor patients was observed among centers.^{138,140}

According to the 2016 SCOT report,¹¹² 16,315 patients undergoing hemodialysis were treated in 243 centers in Saudi Arabia. Of these patients, approximately 17% were on active waiting lists for kidney transplantation. Bicarbonate dialysis solutions were used in all centers, and only a few centers (6.3%) still use acetate dialysis solutions.³⁹³ The predominant type of vascular access for hemodialysis varies among the Gulf Cooperation Council (GCC) countries.³⁸⁹ In Saudi Arabia, arteriovenous fistulas (71.2% of all patients) are mostly used, followed by jugular catheters (12.6%) and arteriovenous grafts (7.6%),³⁹³ whereas in Kuwait 56% of patients undergo dialysis via catheters.³⁸⁹ In Tehran, arteriovenous fistula is the type of

*References 109, 111, 112, 124, 128, 129, 138, 140, 369, and 379-390.

vascular access used most in patients receiving hemodialysis (91%); only 3% of such patients have arteriovenous grafts, and 4% have permanent catheters.¹⁴⁴ Arteriovenous fistula was also preferred in Turkey. In 2016, it was used in 79.1% of Turkish patients undergoing routine hemodialysis, whereas permanent tunneled catheters and temporary catheters were used in 19.4% of patients.¹²⁸

Reuse of dialyzers is not practiced in Saudi Arabia, Libya, and Iran^{138,394} and is prohibited by law in Egypt.³⁸¹ Approximately 87% of Saudi patients undergoing hemodialysis are treated with ESAs and, of these patients, approximately 23% have hemoglobin levels lower than 10 g/dL. In Turkey, ESAs are administered to 61.8% of patients undergoing hemodialysis. In comparison, 48.2% of patients undergoing hemodialysis in Tehran have anemia as a result of low-dose treatment with ESAs, and dialysis may be inadequate because the mean measured treatment adequacy for these patients was 0.97 ± 0.25 .¹⁴⁴ Thus, common clinical practice for hemodialysis in Iran does not meet all the targets recommended by international guidelines.^{144,382}

Limited studies have been conducted to describe the patterns, prevalence, or therapy of CKD-associated mineral bone disorders (CKD-MBDs) in patients undergoing dialysis in ME countries.^{128,387} In developing ME countries, the use of aluminum-based phosphate binders, presence of high amounts of strontium in the soil, and use of dialysis acetate concentrates that are contaminated with strontium have been implicated as causes of osteomalacia in adult patients undergoing hemodialysis and of rickets in pediatric patients undergoing hemodialysis.^{395,396} Hyperphosphatemia and increased serum levels of calcium-phosphorus product are serious problems in dialysis recipients in developing ME countries, and their occurrence is correlated with poor patient knowledge of an appropriate diet. In a large cross-sectional multicenter study that included 1005 patients in Egypt who were undergoing hemodialysis, two-thirds of the patients had hyperphosphatemia and one-third had an elevated serum level of calcium-phosphorus product. The patients used mainly calcium-based phosphate binders.³⁹⁷ In most developing ME countries, economic considerations hinder the use of the newer non-calcium-based phosphate binders, calcium-sensing receptor agonists (calcimimetics), or vitamin D receptor agonists. Therefore, prolonged or additional dialysis sessions are frequently prescribed to control hyperphosphatemia in patients undergoing hemodialysis.^{128,144,387,397}

The results of one survey have suggested that practicing Saudi nephrologists have an adequate perception of the morbidity in patients with CKD-MBDs.³⁹⁸ However, the results of this survey also identified a need for creating national guidelines because the physicians' assessment of the prevalence, patterns, and results of therapeutic interventions in patients with CKD-MBDs was relatively inadequate. Approximately 25% of their patients had hyperphosphatemia (serum phosphorus levels >6 mg/dL), and 20% had hypocalcemia (serum calcium levels <8.4 mg/dL), although vitamin D was administered orally to most patients. In view of the current status of CKD-MBDs in the ME and the associated economic and logistic constraints, a committee of experienced nephrologists from the region was created. Its members examined the KDIGO guidelines and formulated recommendations that can be implemented practically for the management of CKD-MBD in the ME.³⁹⁹

MANAGEMENT OF END-STAGE KIDNEY DISEASE WITH VIRAL HEPATITIS

Available global data have indicated that the prevalence of chronic infection with hepatitis B virus (HBV) and HCV is high in African and ME populations.^{400–405} These two viral infections have considerable effects on morbidity and mortality among patients with ESKD, as well as on graft survival in kidney transplant recipients.^{406–410} However, surveys are still needed to estimate the incidence and prevalence correctly of these infections in patients with CKD and ESKD in each ME country. In addition, screening will identify patients who would benefit from a number of highly efficacious, direct-acting, antiviral oral therapies.^{410–412} The HCV epidemic is particularly devastating in Egypt, where its prevalence in the general population is estimated to be more than 18%. The start of the HCV epidemic in Egypt is attributed to the use of unsterilized needles and syringes during the mass anti-schistosomiasis treatment programs that were conducted during the 1960s and 1970s.⁴¹³ Several studies have reported a dismal prevalence of anti-HCV serologic findings among Egyptian patients undergoing hemodialysis and kidney transplant recipients that ranged between 46% and 100%.^{404,410,413–415} Implementation of an infection control program by 60 Egyptian dialysis facilities has 3 years later resulted in improvements in infection control practices among health care workers and a subsequent decrease in the annual incidence of HCV infection among patients undergoing dialysis from 28% to 6%.⁴¹⁴

Among Saudi patients undergoing hemodialysis, according to the 2016 SCOT report,¹¹² the prevalence of HBV positivity is 4%, whereas HCV infection continued to be problematic (12%). Nonetheless, it has decreased compared with findings in previous reports.³⁹³ However, there is considerable variability in the prevalence of HCV infection between the various regions of Saudi Arabia (9%–16%).¹¹² This variability is important to consider when patients who are undergoing hemodialysis travel to other units and emphasizes the importance of screening them for seroconversion after their return.

Clinical Relevance

Hepatitis C in Dialysis

Owing to high prevalence of HCV infection in some countries in the ME, dialysis patients who travel to the ME and receive dialysis should be monitored for seroconversion after their return.

The annual rate of HCV seroconversion in Saudi patients undergoing hemodialysis is between 7% and 9%. The most common variant of the virus is HCV genotype 4, which is also prevalent in North African countries in the ME, followed by HCV genotypes 1a and 1b.^{406,408,416} Implementation of strict measures of infection control and isolation of HCV-positive patients and dialysis machines has resulted in a significant drop in the annual incidence of HCV infections from 2.4% to 0.2% in one Saudi hemodialysis center.⁴¹⁷ A similar high prevalence of HCV was observed in neighboring GCC countries, which was associated with a higher risk of death, hospitalization, and worse quality of life scores.^{111,418}

In Iran, where genotypes 3a and 1a are the common variants, impressive drops in the rates of HCV and hepatitis B surface antigen (HBsAg) positivity have been reported in patients undergoing hemodialysis. These decreases (14.4% in 1999 to 4.5% in 2006 for HCV and 3.8% in 1999 to 2.6% in 2006 for HBV) have been attributed to the introduction of several measures, such as strict isolation policies, no reuse of dialyzers, compulsory HBV vaccination in patients, and early kidney transplantation.³⁹⁴ Implementation of routine virologic testing, strict isolation measures, HBV vaccination, and ESA administration with few blood product transfusion requirements have resulted in a low prevalence of both HBV (1.7%–2%) and HCV (3.7%–5.7%) infection among Israeli patients undergoing dialysis between 2005 and 2010, according to the 2012 ICDC report.¹⁴⁶

Using second-generation immunoassays to detect HCV infection in 30 Turkish dialysis centers, Köhler has found that the prevalence of HCV was 49.9% in 1995.⁴⁰⁸ Sayiner and colleagues have reported that the duration of dialysis, kidney transplantation history, and history of blood product transfusion were all related to HCV transmission and high prevalence of HCV among Turkish patients undergoing hemodialysis.⁴¹⁹ Turkey is also endemic for HBV infection; 10% of patients undergoing hemodialysis and 2.9% of blood donors are HBsAg-positive. Of note is that the 2008 and 2016 registry of the Turkish Society of Nephrology^{128,420} has reported a progressive decline in the prevalence of HBV infection (4.5%–3.88%), HCV infection (12.7%–5.2%), and both infections (0.9%–0.3%) in Turkish patients undergoing hemodialysis. A similar trend was observed among hemodialysis patients in Lebanon, who have the lowest prevalence of HBV and HCV (1.6% and 4.7%, respectively).⁴²¹ The results of a survey conducted among Jordanian patients undergoing hemodialysis during 2003 revealed HBV positivity in 4% and HCV positivity in 21%, with annual seroconversion rates of 0.34% for HBV and 2.6% for HCV.¹¹³ In 2007, high seropositivities for HBV (8%) and HCV (22%) were also detected in Palestinian patients undergoing hemodialysis who were residing in the Gaza Strip.⁴²²

Alashek and colleagues have found a remarkably high prevalence of seropositivity for HCV among Libyan patients undergoing hemodialysis (31.1%), which is 25-fold higher than the general population.¹³⁹ The overall incidence of seroconversion during this 1-year prospective study was 7.1%, with wide variation between hemodialysis units. Conversely, the prevalence of HBsAg positivity was low (2.6%), with a seroconversion incidence of 0.6%. Factors associated with seroconversions, such as duration of dialysis, history of receiving dialysis in another center in Libya, and prior kidney transplantation, possibly suggest nosocomial transmission.¹³⁹ In Jordan, 1.2% and 7.8% of prevalent dialysis patients are seropositive to HBV and HCV, respectively.¹²⁹ In comparison, the adjusted prevalence of HBV is 4.6% among patients in Germany who are undergoing hemodialysis, 4.3% in Italy, 3.7% in France, 2.1% in Spain, 2.1% in Japan, and 2.4% in the United States, and adjusted seroconversion rates range from 0.4 to 1.8/100 patient-years.⁴²³ The adjusted prevalence of HCV also varies among these countries—22.9% in Spain, 20.5% in Italy, 10.4% in France, 3.8% in Germany, 2.6% in the United Kingdom, 14.8% in Japan, and 14% in the United States. The adjusted seroconversion rates range from 1.2 to 3.9/100 patient-years and are highest in Italy and Spain.⁴²⁴

DIALYSIS-RELATED OUTCOMES

With regard to other dialysis-related outcomes, reports from Egypt have revealed significant center-specific effects (which were not defined) on the survival and quality of life of patients undergoing dialysis; these effects were dependent largely on funding.³⁸¹ In Syria, the 3-year survival rate among patients undergoing hemodialysis was unsatisfactory, ranging from 26% to 64% in different hemodialysis centers.¹¹⁹ Regrettably, during the recent conflicts, death due to lack of dialysis or underdialysis has been common in Syria and Yemen, although the exact figures are difficult to extract from published literature.²² In Jordan, the 1-year mortality rate among patients undergoing hemodialysis was approximately 20%,¹¹³ although it decreased to 7% in 2016.¹²⁹ In Libya, the 1-year mortality was similar to many other countries (21.2%).¹⁴¹ However, this is a poor outcome when taking into consideration the young mean age of patients undergoing hemodialysis. Although not reported, it is expected that the survival of Libyan ESKD patients was also aggravated by the conflicts in their country.¹⁴¹ According to the 2018 ICDC report,¹²⁷ the unadjusted 1-, 2-, 3-, and 10-year survival rates among Israeli patients undergoing dialysis were 86.6%, 73%, 61.3%, and 15%, respectively, from 1990 to 2010. Of note, there appears to be a survival advantage for Arab patients over Jewish patients on maintenance dialysis in Israel, in contrast to the life expectancy of Arabs in the general population, which is 3 to 4 years lower than that of the Jewish population.¹³⁵ The survival of patients undergoing hemodialysis was high in Qatar, with 1-year and 5-year survival rates of 84% and 53%, respectively,¹²⁴ as well as in other GCC countries (annual mortality, 8%–14.65%).¹¹¹

It is of interest that a health disparity exists for both legal and illegal immigrants from ME countries with CKD who live and work in Western industrialized countries and may need RRT. The ethnic and educational backgrounds of these patients are different from those of the local population. Accordingly, they face potential problems with communication, health insurance, and care.^{9,27,425} Fogazzi and Castelnovo have reported their experience with these patients who were treated in an Italian dialysis unit.⁴²⁶ Although the number of patients was small, they reported that those undergoing hemodialysis who came from developing countries, which included some ME countries, were younger at the initiation of dialysis (38.2 ± 7.9 years vs. 63 ± 12.6 years), were referred at a later stage of their disease, and had more infections, such as tuberculosis and viral hepatitis, than did patients from the local population who were undergoing hemodialysis. Furthermore, they also reported that a visit to their native countries was usually associated with medical complications, such as worsening of anemia.

KIDNEY TRANSPLANTATION

In ME countries, as in all other countries, kidney transplantation is recognized as the treatment of choice for ESKD. The beneficial effects of kidney transplantation on life expectancy, quality of life, and medical expenses are greater than those associated with maintenance dialysis. However, legislative obstacles, an underdeveloped and poorly funded health care infrastructure, poor public awareness of the importance of organ donation, lack of effective kidney transplantation

programs, cultural and religious barriers, and lack of trained multidisciplinary medical teams are some of the obstacles that prevent the promotion of kidney transplantation in some ME countries.^{86,122,126,427–433}

Most ME countries are members of the Middle East Society for Organ Transplantation (MESOT). MESOT includes Iran, Turkey, and all the Arab countries in the ME, as well as Pakistan, Cyprus, and some countries of central Asia.^{126,427} There are very few organ procurement centers in the MESOT member countries for overseeing the activities of organ donation, sharing, and transplantation at a national level. Israel is not a member of MESOT; it has its own national center for organ transplantation. The first successful living- and deceased-donor kidney transplants in the ME were performed in Israel (1964 and 1966, respectively)¹¹⁷ and were reported to have been performed in Iran in 1967 and 2003, respectively.^{126,434} The first successful kidney transplantation in an Arab country in the ME was performed with a kidney from a deceased donor with no heartbeat in Jordan in 1972. Most other ME countries, such as Lebanon and Turkey, started their kidney transplantation programs during the early 1970s, and Egypt and Saudi Arabia commenced their programs in 1976 and 1979, respectively. Since then, the remaining ME countries have established their own kidney transplantation programs; Libya did so in 2004. Currently, kidney transplantation programs are active in most ME countries but are relatively limited in Algeria, Yemen, the UAE, and Bahrain (see [Tables 78.2 and 78.3](#)).^{86,121–123,126,435–438}

Most kidneys for transplantation are obtained from living donors. An important milestone that paved the way for organ donation for transplantation from deceased donors in Arab countries in the ME was the Amman declaration in 1986. In this declaration, Islamic theologians recognized that brain death was irreversible and could be used to declare a person legally dead, thereby making it permissible to disconnect that person from mechanical life support systems. This declaration was preceded in 1982 by a resolution of the Islamic Council in Saudi Arabia that permitted the use of organs from both living and deceased donors for transplantation.^{121,439,440} Similar declarations have since been made by religious authorities in Egypt, Turkey, and Iran.^{121,439} As a result of these declarations, most Arab countries in the ME, including Egypt, have now enacted laws for regulated organ donation for transplantation from both living and cadaveric donors, and transplantation programs for kidneys and other organs are beginning to expand.^{108,119,441} Programs for donation after cardiac death are limited in many Arab countries because of religious, legal, ethical, and social issues.⁴⁴² Such a program has recently begun in Israel.⁴⁴³

In his code of Jewish law, the Mishneh Torah (Laws of Sanhedrin, 12: 3), the 12th century philosopher and physician Maimonides interpreted the Talmud as saying that someone who saves the life of one person is considered to have saved the entire world. This is similar to what is written in the Koran, Chapter 5, verse 32: “... if any one saved a life, it would be as if he saved the life of all mankind.” Therefore, organ donation that saves a life is considered in Islam and Judaism to be a good deed. With regard to living related donors, Bulka has argued that organ donation is permissible because the danger to the donor is minimal, but it is not obligatory.⁴⁴⁴ Likewise, Christians believe that organ donation is an act of love and nobility (e.g., as interpreted from

Corinthians and from the parable of the Good Samaritan in Luke, Chapter 10, verses 25 to 37).⁴⁴⁵

Active deceased-donor kidney transplantation programs exist in Iran, Israel, Turkey, and 10 Arab countries (Algeria, Bahrain, Jordan, Kuwait, Lebanon, Morocco, Oman, Qatar, Saudi Arabia, and Tunisia; see [Table 78.2](#)).^{86,122,446} However, deceased-donor kidney transplantation is inadequate to address the current need for allografts, and the number of patients on kidney transplantation waiting lists in these ME countries has been progressively increasing. The use of kidneys from living related donors is increasing, but the gap between demand and supply is continuously widening in all ME countries. To meet the growing demand for kidneys and the shortfall in donors, some countries have initiated kidney transplantation programs involving organs from living unrelated donors. Using emotionally related donors extends donor eligibility to include individuals who are not genetically related to recipients.

Although kidney transplantation involving organs from deceased donors has been increasing in the ME, it accounts for less than 21% of kidney transplantations in most ME countries, with large national variations.^{126,447} In 2016, 3416 kidney transplantations were performed in Turkey, of which 22.8% were from deceased donors.¹²⁸ In Saudi Arabia, kidney transplantation from living unrelated donors is forbidden. SCOT reported that 798 kidney transplantations were performed in 2016, and 16% of the kidneys (125) were obtained from deceased donors⁴⁴⁸; the rest were obtained from living related donors. In fact, in 2006, Saudi Arabia had the highest reported rate of living-donor kidney transplantations worldwide, at 32 procedures pmp, followed by Jordan (29 procedures pmp), Iceland (26 procedures pmp), Iran (23 procedures pmp), and the United States (21 procedures pmp).³⁷¹ However, the quoted Saudi statistic should be interpreted cautiously because the rate of living related donor kidney transplantations is 10.1 pmp. According to Noël,⁴⁴⁹ the report by Horvat and colleagues³⁷¹ probably included “transplant tourism” activity because it incorporated data on kidney transplantation in Saudi patients from living unrelated donors that was performed in other countries.

In Egypt, approximately 7% of Egyptian patients with ESKD are offered kidney transplantation in the year of diagnosis, and no kidneys are obtained from deceased donors—80% of the kidneys are obtained from living unrelated donors, and 20% are obtained from living related donors. In 2013, 264 kidney transplantations were performed in Israel; of the transplanted kidneys, 43.5% were obtained from deceased donors and 54.5% from living related and living unrelated donors. The ratio of deceased donors to living donors is not constant—according to the annual reports of the Israeli National Transplant Center—yet it remains higher than in most ME countries (e.g., 38.2% in 2017). This could be attributed to a high consent rate (about 60%) for organ donation after brain death.

Kidney transplantation in Syria commenced in 1976 with exclusive reliance on kidneys from living related donors. Since 2003, transplantation of kidneys from deceased donors has been allowed. As of 2010, 13.2 kidney transplantations pmp were performed, of which 22% were from deceased donors.^{119,122} Unfortunately, the war in Syria destroyed the hospitals’ infrastructure and decreased the kidney transplant

programs from eight to four centers, along with difficulties in providing immunosuppressive therapies.⁴⁵⁰

In Iran, the living unrelated donor transplantation program operates under the close supervision and scrutiny of the Ministry of Health and Medical Education, Iranian Scientific Society of Organ Transplantation, Foundation for Patients with Special Diseases, and Dialysis and Transplant Patients Association (DATPA).⁴⁵¹ In the Iranian model, patients who need kidney transplantation are referred to the DATPA, a charity founded in 1978 by Iranian patients with ESKD, which acts as a liaison agency between patients and potential donors. The altruistic volunteers are also registered by the DATPA and undergo evaluation in the foundation's clinics. Several important features characterize the Iranian model of living unrelated donors for kidney transplantation⁴⁵⁶:

- No coercion is allowed.
- Written consent is obtained from the donor and the donor's parents or spouse.
- Donors are rewarded with gifts from the government.
- No commercialism is allowed.
- The medical teams receive no financial benefit.
- Recipients and donors must be Iranian citizens.
- Time on the waiting list is minimal.

Consequently, the annual number of kidney transplantations has substantially increased in Iran, from 1421 in 2000 to 2285 in 2010.^{120,436} However, the 86% share of living unrelated kidney donation in 2000 (20.1 pmp) decreased to 75% in 2006 (23 pmp) and to 69% in 2010 (21.8 pmp). This change was mainly due to a substantial parallel increase in brain-dead kidney donation (2.2% in 2000, equivalent to 0.4 pmp, increasing to 26% and 7.9 pmp in 2010), after the Organ Transplantation and Brain Death Act in 2000, which legalized brain-dead organ donation.¹²⁰ Of note, 10 years after the introduction of the controlled living unrelated donor kidney transplantation program in Iran, the national waiting list for renal transplants was eliminated.^{452,453}

Simforoosh and associates have reported that patient and graft survival rates of 2155 Iranian recipients of kidneys from living unrelated donors after 15 years were 76.4%, and 53.2%, respectively.⁴⁵⁴ These results are comparable with survival data in the 2003 annual report of the US Renal Data System on living donor kidney transplantations.⁴⁵⁵ However, the Iranian model has been criticized for ethical reasons because 85% of the vendors are very poor, a fact that may increase the risk that potential donors will withhold relevant medical information.⁴⁵⁶ A study from the Shiraz Transplant Center has revealed that Iranian paid unrelated donors have a lower quality of life and a higher incidence of microalbuminuria compared with related donors.⁴⁵⁷ Hence, the lack of long-term donor follow-up and the direct financial connection between donor and recipient are major weak points that should be ethically reviewed.^{120,458–460}

COMMERCIAL KIDNEY TRANSPLANTATION

The lack of deceased donors for kidney allograft transplantation, combined with other medical factors (prolonged graft survival and improved surgical techniques, including laparoscopic nephrectomy) and nonmedical factors (economic and cultural), have motivated many patients with ESKD to seek kidney transplants from living unrelated donors outside

their home countries. This practice is often called commercial kidney transplantation or organ tourism⁴⁶¹ because the donor sells his or her kidney for a certain amount of money. These kidney transplantations are performed in many countries worldwide, such as the Philippines, Russia, India, China, Pakistan, and South Africa, as well as some ME countries, such as Turkey, Iraq, and Egypt.^{428,462–464} The surgery is commonly performed in substandard conditions under the cover of secrecy. Many transplant recipients return to their homeland 1 to 2 weeks after transplantation without crucial information (e.g., a complete medical report, important information about the donor, HLA typing, and details of the medical treatment, surgery, and postoperative care). In addition, the recipients are exposed to potential risk for infection, such as HBV, HCV, or human immunodeficiency virus infection, as well as tuberculosis from inadequately evaluated donors. Thus the results of such commercial kidney transplantations are reported to be substandard.^{461,463,464}

To undermine the practice of global commercial kidney transplantation, the Transplantation Society and the ISN convened an international summit meeting in Istanbul, Turkey, in April 2008. The outcome of this meeting was the “Declaration of Istanbul on Organ Trafficking and Transplant Tourism,” which suggested that strategies to increase the donor pool and encourage legitimate, lifesaving transplantation programs be developed by countries to prevent organ trafficking, transplant commercialism, and transplant tourism. In addition, strategies should be aimed at stopping and prohibiting unethical activities, as well as encouraging safe and accountable practices that both meet the needs of transplant recipients and protect donors.^{462,465,466} Many ME countries need to develop national self-sufficiency in organ donation to combat organ tourism and achieve global justice in transplantation.^{462,465}

In 2008, two laws were approved by the Israeli parliament, the Brain-Respiratory Death Law and the Organ Transplantation Law. As a result, health insurance reimbursement for transplantation performed abroad that contravenes local Israeli laws was prohibited, and criminal penalties for brokering organ sales were introduced. Consequently, Israel has both increased organ donations from living and deceased sources and has reduced the number of transplant candidates seeking transplantation abroad (from 150 in 2006 to 44 in 2017).^{446,467} In Egypt, however, in spite of the April 2010 organ transplantation law, which prohibits and penalizes organ trafficking and permits deceased donation in accord with the WHO Guiding Principles and the Declaration of Istanbul, little enforcement has been evident, and the problem of a black market in organs from living donors has increased.^{428,441,462,468}

PHARMACOLOGIC TREATMENT OF KIDNEY TRANSPLANT RECIPIENTS

In general, all approved medications that are used worldwide to treat kidney transplant recipients are also used in the ME.^{120,435} In all ME countries with kidney transplantation programs, induction therapy with methylprednisolone, lymphocyte-depleting agents, or IL-2 receptor antagonists is widely used. In most of the commercial transplantation programs, induction therapy with lymphocyte-depleting agents is routinely given to reduce the frequency of acute graft rejection so that the recipient can be discharged early after the

transplantation surgery. No reports have yet been published on the long-term complications of this routine treatment, such as bone marrow suppression, cytomegalovirus infections, and posttransplantation lymphoproliferative disorder.

Maintenance treatment after kidney transplantation in most ME countries consists of triple therapy with corticosteroids, azathioprine, or mycophenolate mofetil and calcineurin inhibitors. A small number of kidney transplantation centers have started prescribing rapamycin, a drug that inhibits a serine-threonine kinase called the mammalian target of rapamycin (mTOR), to prevent transplant rejection.^{120,439} Treatment of acute rejection in most ME countries consists mainly of methylprednisolone (Solu-Medrol) and rabbit antithymocyte globulin (Thymoglobulin). Hyperimmune globulins and plasmapheresis are used by some centers when indicated.

POSTTRANSPLANTATION COMPLICATIONS

There have been only a few reports in the literature on the outcomes, as well as patient and graft survival rates, after kidney transplantation in ME countries. In general, these reports claim that outcomes in Saudi Arabia, Kuwait, Egypt, and North African countries in the ME are in line with international standards, although infections remain the main reason for morbidity and mortality.^{128,149,151,432,454, 464}

Distinctive infections are of special concern in some ME countries. Tuberculosis is prevalent among dialysis recipients in some countries such as Saudi Arabia, Yemen, Turkey, and the North African countries in the ME.^{431,469–473} Patients in these countries are at increased risk for developing active tuberculosis after kidney transplantation, and approximately 30% have symptoms of tuberculosis in extrapulmonary organs, particularly the lymph nodes, gastrointestinal tract, and peritoneal cavity. The Mantoux test often yields negative results in kidney transplant recipients, possibly because of the suppression of cellular immunity. Tuberculosis in the graft kidney tends to manifest as granulomatous interstitial nephritis. Urinalysis results are often negative for bacilli. The diagnosis is usually made on kidney biopsy or after nephrectomy in transplant recipients, who often present with fever of unknown origin and deteriorating graft function.

Prophylactic treatment with isoniazid or rifampin for patients at high risk (Mantoux skin test reaction of >10 mm) has decreased the development of active tuberculosis. An important sequela of this treatment is the induction of cytochrome P450 enzymes by the antituberculous drugs, which results in a severe drop in the circulating therapeutic levels of calcineurin inhibitors and, consequently, severe acute rejection. Therefore, increasing the dose of calcineurin inhibitors and frequent monitoring of their circulating levels are mandatory in such cases.^{431,472}

Viral hepatitis, as discussed previously, is also common in the ME, especially among patients undergoing dialysis. Eligible patients with HBV and HCV infections are offered transplantation in many ME kidney transplantation centers after appropriate presurgical workup and management.^{406,472,474}

The most common types of neoplasia among kidney transplant recipients are skin malignancies, lymphoproliferative disorders, and Kaposi sarcoma. The incidence of some uncommon tumors in the general population (e.g., Kaposi sarcoma) can be 400- to 500-fold higher among kidney transplant recipients. Kaposi sarcoma is most often seen in

transplant recipients of Mediterranean, Jewish, and Arabic descent. Its reported incidence is 0.5% in most Western industrialized countries and as high as 5.3% in Saudi Arabia.^{472,475–478}

Clinical Relevance

Kaposi Sarcoma Postransplant

Kaposi sarcoma is more common in patients from the ME, potentially linked to greater rates of human herpesvirus 8 seropositivity than in Europe and North America.

The preponderance of cases of Kaposi sarcoma in certain ethnic groups appears to be linked to the geographic distribution of human herpesvirus 8 infection, inasmuch as more than 80% of transplant recipients with Kaposi sarcoma are seropositive for human herpesvirus 8 before undergoing transplantation.⁴⁷⁸

SUMMARY

The ME has a strategic global location. Its fascinating geography is combined with rich national histories, cultures, and resources. In addition, the countries of the ME differ in their economics, political systems, culture, and bioecology, all of which eventually translate into disparities in their health care systems, as well as in disease epidemiologic features, causes, management, and outcomes. The burden of kidney diseases is high in all ME countries and is compounded by many risk factors in the region, although data from developing countries are possibly underestimates. Practicing nephrologists in the ME are familiar with the genetic background, social habits, and culture of their patients. However, this is not so for many individuals of ME origin who now live in non-ME countries. Nephrologists in those countries need to take these considerations into account when diagnosing or treating the kidney diseases and comorbid conditions of their patients of ME origin. Therefore, the emerging epidemic of diabetes mellitus in many ME countries is pertinent, and the prevailing genetic kidney diseases in the various population groups who live in the ME, have been reviewed in this chapter.

Moreover, the ME has experience in management of mass disasters and consequent kidney complications, which can be implemented in regions outside the ME. However, several issues still need to be addressed, and hurdles must be overcome to improve the overall management of kidney diseases in the ME. Well-conducted epidemiologic cohort studies are urgently needed, and regional and national registries must be established as sources of transparent and accurate data. The results of the studies should provide accurate estimates of the public health burden of AKI, CKD, and ESKD, their risk factors, and comorbid conditions. This information should also improve the quality of therapy provided to patients. In addition, these efforts should also focus on the special needs of refugees in countries where human-engendered and natural disasters have occurred.

The entire international nephrology community agrees that improving existing diagnostic methods and establishing preventive strategies for the detection and treatment of kidney

diseases at the earliest possible stage is of utmost importance, especially in countries with limited resources or health care expenditures. Therefore, to achieve these goals, ME countries need to invest in the following: (1) training qualified nephrologists and medical personnel; (2) public education and campaigns for lifestyle modification to combat obesity and diabetes mellitus and to explain proper use of over-the-counter drugs; (3) premarital genetic counseling to decrease the burden of genetic diseases; (4) vigorous treatment of comorbid conditions, which include infectious diseases, and strategies for preventing dehydration in tropical ME countries; and (5) use of low-cost, generic medications.^{52,76,86,469,479}

Because transplantation is the most cost-effective treatment for ESKD, ME countries also need to create a societal environment that motivates and encourages its population to support organ donation for transplantation in patients with ESKD. Encouraging organ donation through deceased donors, living related donors, and paired-exchange kidney transplantation programs, as well as organ sharing among the MESOT member states, will enable ME countries to reach national sufficiency for organ transplantation. The establishment of such programs will then help reduce the long waiting lists in each ME country and combat unethical commercial kidney transplantations.^{76,86,96,427}

ACKNOWLEDGMENT

The authors are deeply grateful to Dr. Dvora Rubinger who contributed to the previous versions of this chapter.

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

KEY REFERENCES

- Held CC, Cummings JT. *Middle East Patterns: Places, Peoples, and Politics*. 6th ed. Boulder, CO: Westview Press; 2014.
- Mokdad AH, Forouzanfar MH, Daoud F, et al. Health in times of uncertainty in the eastern Mediterranean region, 1990-2013: a systematic analysis for the Global Burden of Disease Study 2013. *Lancet Glob Health*. 2016;4:e704-e713.
- Van Biesen W, Vanholder R, Erandez T, et al. Caring for migrants and refugees with end-stage kidney disease in Europe. *Am J Kidney Dis*. 2018;71:701-709.
- Van Biesen W, Vanholder R, Vanderhaegen B, et al. Renal replacement therapy for refugees with end-stage kidney disease: an international survey of the nephrological community. *Kidney Int Suppl*. 2016;6:35-41.
- Gibney RT, Sever MS, Vanholder RC. Disaster nephrology: crush injury and beyond. *Kidney Int*. 2014;85:1049-1057.
- Hoste EA, Bagshaw SM, Bellomo R, et al. Epidemiology of acute kidney injury in critically ill patients: the multinational AKI-EPI study. *Intensive Care Med*. 2015;41:1411-1423.
- Barsoum RS. Malarial acute renal failure. *J Am Soc Nephrol*. 2000;11:2147-2154.
- Mehta RL, Burdmann EA, Cerda J, et al. Recognition and management of acute kidney injury in the International Society of Nephrology 0by25 Global Snapshot: a multinational cross-sectional study. *Lancet*. 2016;387:2017-2025.
- Hatamizadeh P, Najafi I, Vanholder R, et al. Epidemiologic aspects of the Bam earthquake in Iran: the nephrologic perspective. *Am J Kidney Dis*. 2006;47:428-438.
- Kantarci G, Vanholder R, Tuglular S, et al. Acute renal failure due to crush syndrome during Marmara earthquake. *Am J Kidney Dis*. 2002;40:682-689.
- Rifai AO, Murad LB, Sekkarie MA, et al. Continuous venovenous hemofiltration using a stand-alone blood pump for acute kidney injury in field hospitals in Syria. *Kidney Int*. 2015;87:254-261.
- Süleymanlar G, Utaş C, Arinsoy T, et al. A population-based survey of Chronic RENal Disease In Turkey—the CREDIT study. *Nephrol Dial Transplant*. 2011;26:1862-1871.

- Benghanem Gharbi M, Elseviers M, Zamd M, et al. Chronic kidney disease, hypertension, diabetes, and obesity in the adult population of Morocco: how to avoid “over”- and “under”-diagnosis of CKD. *Kidney Int*. 2016;89:1363-1371.
- Karnib HH, Gharavi AG, Aftimos G, et al. A 5-year survey of biopsy proven kidney diseases in Lebanon: significant variation in prevalence of primary glomerular diseases by age, population structure and consanguinity. *Nephrol Dial Transplant*. 2010;25:3962-3969.
- Bello AK, Levin A, Tonelli M, et al. Assessment of global kidney health care status. *JAMA*. 2017;317:1864-1881.
- Vivante A, Afek A, Frenkel-Nir Y, et al. Persistent asymptomatic isolated microscopic hematuria in Israeli adolescents and young adults and risk for end-stage renal disease. *JAMA*. 2011;306:729-736.
- Calderon-Margalit R, Golan E, Twig G, et al. History of childhood kidney disease and risk of adult end-stage renal disease. *N Engl J Med*. 2018;378:428-438.
- Gouda Z, Mashaal G, Bello AK, et al. Egypt Information, Prevention, and Treatment of Chronic Kidney Disease (EGIPT-CKD) programme: prevalence and risk factors for microalbuminuria among the relatives of patients with CKD in Egypt. *Saudi J Kidney Dis Transpl*. 2011;22:1055-1063.
- Alsuwaida AO, Farag YM, Al Sayyari AA, et al. Epidemiology of chronic kidney disease in the Kingdom of Saudi Arabia (SEEK-Saudi investigators) - a pilot study. *Saudi J Kidney Dis Transpl*. 2010;21:1066-1072.
- Tohidi M, Hasheminia M, Mohebi R, et al. Incidence of chronic kidney disease and its risk factors, results of over 10 year follow up in an Iranian cohort. *PLoS ONE*. 2012;7:e45304.
- Yüzbashian E, Asghari G, Mirmiran P, et al. Adherence to low-sodium Dietary Approaches to Stop Hypertension-style diet may decrease the risk of incident chronic kidney disease among high-risk patients: a secondary prevention in prospective cohort study. *Nephrol Dial Transplant*. 2018;33:1159-1168.
- Najafi I, Alatab S, Atabak S, et al. Seventeen years' experience of peritoneal dialysis in Iran: first official report of the Iranian Peritoneal Dialysis Registry. *Perit Dial Int*. 2014;34:636-642.
- El Matri A. History of nephrology in Mediterranean Arab countries. *J Nephrol*. 2013;26:170-174.
- AlSahow A, AlRukhaimi M, Al Wakeel J, et al. Demographics and key clinical characteristics of hemodialysis patients from the Gulf Cooperation Council countries enrolled in the dialysis outcomes and practice patterns study phase 5 (2012-2015). *Saudi J Kidney Dis Transpl*. 2016;27:S12-S23.
- Mahdavi-Mazdeh M. The Iranian model of living renal transplantation. *Kidney Int*. 2012;82:627-634.
- Süleymanlar G, Serdengeçti K, Altıparmak MR, et al. Trends in renal replacement therapy in Turkey, 1996-2008. *Am J Kidney Dis*. 2011;57:456-465.
- Utaş C. The development of PD in Turkey. *Perit Dial Int*. 2008;28:217-219.
- Kamrar KZ, Eliezer G, Tamy S, et al. Survival Disparities within American and Israeli Dialysis Populations: Learning from Similarities and Distinctions across Race and Ethnicity. *Semin Dial*. 2010;23:586-594.
- Counil É, Cherni N, Kharrat M, et al. Trends of incident dialysis patients in Tunisia between 1992 and 2001. *Am J Kidney Dis*. 2008;51:463-470.
- Alashek WA, McIntyre CW, Taal MW. Determinants of survival in patients receiving dialysis in Libya. *Hemodial Int*. 2013;17:249-255.
- Frishberg Y, Ben-Neriah Z, Suvanto M, et al. Misleading findings of homozygosity mapping resulting from three novel mutations in NPHS1 encoding nephrin in a highly inbred community. *Genet Med*. 2007;9:180-184.
- Yousri NA, Fakhro KA, Robay A, et al. Whole-exome sequencing identifies common and rare variant metabolic QTLs in a Middle Eastern population. *Nat Commun*. 2018;9:333.
- Ben-Shalom E, Frishberg Y. Primary hyperoxalurias: diagnosis and treatment. *Pediatr Nephrol*. 2015;30:1781-1791.
- Belostotsky R, Seboun E, Idelson GH, et al. Mutations in DHAPSL are responsible for primary hyperoxaluria type III. *Am J Hum Genet*. 2010;87:392-399.
- Zelikovic I, Szargel R, Hawash A, et al. A novel mutation in the chloride channel gene, CLCNKB, as a cause of Gitelman and Bartter syndromes. *Kidney Int*. 2003;63:24-32.
- Magen D, Sprecher E, Zelikovic I, et al. A novel missense mutation in SLC5A2 encoding SGLT2 underlies autosomal-recessive renal glucosuria and aminoaciduria. *Kidney Int*. 2005;67:34-41.

245. Magen D, Berger L, Coady MJ, et al. A loss-of-function mutation in NaPi-IIa and renal Fanconi's syndrome. *N Engl J Med*. 2010;362:1102–1109.
263. Manukyan G, Aminov R. Update on pyrin functions and mechanisms of familial Mediterranean fever. *Front Microbiol*. 2016;7:456.
275. Ben-Chetrit E, Backenroth R. Amyloidosis induced, end stage renal disease in patients with familial Mediterranean fever is highly associated with point mutations in the MEFV gene. *Ann Rheum Dis*. 2001;60:146–149.
356. van der Hilst J, Moutschen M, Messiaen PE, et al. Efficacy of anti-IL-1 treatment in familial Mediterranean fever: a systematic review of the literature. *Biologics*. 2016;10:75–80.
364. Green H, Lichtenberg S, Rahamimov R, et al. Familial Mediterranean fever is associated with increased mortality after kidney transplantation—a 19 Years' single center experience. *Transplantation*. 2017;101:2621–2626.
371. Horvat LD, Shariff SZ, Garg AX. Global trends in the rates of living kidney donation. *Kidney Int*. 2009;75:1088–1098.
401. Lavanchy D. The global burden of hepatitis C. *Liver Int*. 2009;29(suppl 1):74–81.
406. Harfouche M, Chemaitelly H, Mahmud S, et al. Epidemiology of hepatitis C virus among hemodialysis patients in the Middle East and North Africa: systematic syntheses, meta-analyses, and meta-regressions. *Epidemiol Infect*. 2017;145:3243–3263.
418. Goodkin DA, Bieber B, Jadoul M, et al. Mortality, Hospitalization, and Quality of Life among Patients with Hepatitis C Infection on Hemodialysis. *Clin J Am Soc Nephrol*. 2017;12:287–297.
427. Shaheen FA. Organ donation in the Middle East countries. *Ethn Dis*. 2009;19:S1-16-17.
446. Lavee J, Ashkenazi T, Stoler A, et al. Preliminary marked increase in the national organ donation rate in Israel following implementation of a new organ transplantation law. *Am J Transplant*. 2013;13:780–785.
457. Fallahzadeh MK, Jafari L, Roozbeh J, et al. Comparison of health status and quality of life of related versus paid unrelated living kidney donors. *Am J Transplant*. 2013;13:3210–3214.
462. Danovitch GM, Chapman J, Capron AM, et al. Organ trafficking and transplant tourism: the role of global professional ethical standards—the 2008 Declaration of Istanbul. *Transplantation*. 2013;95:1306–1312.
479. Levin A, Tonelli M, Bonventre J, et al. Global kidney health 2017 and beyond: a roadmap for closing gaps in care, research, and policy. *Lancet*. 2017;390:1888–1917.

REFERENCES

- Held CC, Cummings JT. *Middle East Patterns: Places, Peoples, and Politics*. 6th ed. Boulder, CO: Westview Press; 2014.
- Saleh SS, Alameddine MS, Natafqi NM, et al. The path towards universal health coverage in the Arab uprising countries Tunisia, Egypt, Libya, and Yemen. *Lancet*. 2014;383:368–381.
- World Bank: *Countries and economies*. <http://data.worldbank.org/country>.
- World Bank: *How we classify countries*. <http://data.worldbank.org/about/country-classifications>.
- Barber RM, Fullman N, Sorensen RJD, et al. Healthcare Access and Quality Index based on mortality from causes amenable to personal health care in 195 countries and territories, 1990–2015: a novel analysis from the Global Burden of Disease Study 2015. *Lancet*. 2017;390:231–266.
- Mokdad AH, Forouzanfar MH, Daoud F, et al. Health in times of uncertainty in the eastern Mediterranean region, 1990–2013: a systematic analysis for the Global Burden of Disease Study 2013. *Lancet Glob Health*. 2016;4:e704–e713.
- Barsoum RS. Burden of chronic kidney disease: North Africa. *Kidney Int Suppl* (2011). 2013;3:164–166.
- Chao F, You D, Pedersen J, et al. National and regional under-5 mortality rate by economic status for low-income and middle-income countries: a systematic assessment. *Lancet Glob Health*. 2018;6:e535–e547.
- Van Biesen W, Vanholder R, Hernandez T, et al. Caring for migrants and refugees with end-stage kidney disease in Europe. *Am J Kidney Dis*. 2018;71:701–709.
- Perini W, Snijder MB, Peters RJG, et al. Ethnic disparities in estimated cardiovascular disease risk in Amsterdam, the Netherlands. *Neth Heart J*. 2018;26:252–262.
- Muhsen K, Green MS, Soskolne V, et al. Inequalities in non-communicable diseases between the major population groups in Israel: achievements and challenges. *Lancet*. 2017;389:2531–2541.
- Tinghög P, Malm A, Arwidson C, et al. Prevalence of mental ill health, traumas and postmigration stress among refugees from Syria resettled in Sweden after 2011: a population-based survey. *BMJ Open*. 2017;7.
- Belqacem S. Health issues in the Arab American community. Commentary: the growing risk factors for noncommunicable diseases in the Arab world. *Ethn Dis*. 2007;17:S3–S53–S52.
- Shah SM, Ali R, Loney T, et al. Prevalence of diabetes among migrant women and duration of residence in the United Arab Emirates: a cross sectional study. *PLoS ONE*. 2017;12:e0169949.
- Asfour LW, Stanley ZD, Weitzman M, et al. Uncovering risky behaviors of expatriate teenagers in the United Arab Emirates: a survey of tobacco use, nutrition and physical activity habits. *BMC Public Health*. 2015;15:944.
- AlFaleh HF, Al Shamiri MQ, Ullah A, et al. Disparities in health care delivery and hospital outcomes between non-Saudis and Saudi nationals presenting with acute coronary syndromes in Saudi Arabia. *PLoS ONE*. 2015;10:e0124012.
- Graetz V, Rechel B, Groot W, et al. Utilization of health care services by migrants in Europe—a systematic literature review. *Br Med Bull*. 2017;121:5–18.
- Hussin AH. Adjustment factors to per capita health-care indicators in countries with expatriate male-majority populations. *East Mediterr Health J*. 2014;20:690–697.
- Chernin G, Gal-Oz A, Schwartz IF, et al. Care of undocumented-undocumented immigrants in a large urban dialysis unit. *BMC Nephrol*. 2012;13:112.
- Devi S. Yemeni health under relentless pressure. *Lancet*. 2018;391:646.
- Butler D. What the numbers say about refugees. *Nature*. 2017;543:22–23.
- Sekkarie MA, Zanolli AR, Rifai AO, et al. The Syrian conflict: assessment of the ESRD system and response to hemodialysis needs during a humanitarian and medical crisis. *Kidney Int*. 2015;87:262–265.
- Mateen FJ, Carone M, Al-Saedy H, et al. Medical conditions among Iraqi refugees in Jordan: data from the United Nations Refugee Assistance Information System. *Bull World Health Organ*. 2012;90:444–451.
- The L. Syria suffers as the world watches. *Lancet*. 2017;389:1075.
- Fouad FM, Sparrow A, Tarakji A, et al. Health workers and the weaponisation of health care in Syria: a preliminary inquiry for The Lancet-American University of Beirut Commission on Syria. *Lancet*. 2017.
- Zimmerman C, Kiss L, Hossain M. Migration and health: a framework for 21st century policy-making. *PLoS Med*. 2011;8:e1001034.
- Van Biesen W, Vanholder R, Vanderhaeghe B, et al. Renal replacement therapy for refugees with end-stage kidney disease: an international survey of the nephrological community. *Kidney Int Suppl*. 2016;6:35–41.
- Gibney RT, Sever MS, Vanholder RC. Disaster nephrology: crush injury and beyond. *Kidney Int*. 2014;85:1049–1057.
- Musani A, Shaikh IA. The humanitarian consequences and actions in the Eastern Mediterranean Region over the last 60 years—a health perspective. *East Mediterr Health J*. 2008;14(suppl):S150–S156.
- Lameire NH, Bagga A, Cruz D, et al. Acute kidney injury: an increasing global concern. *Lancet*. 2013;382:170–179.
- Hoste EA, Bagshaw SM, Bellomo R, et al. Epidemiology of acute kidney injury in critically ill patients: the multinational AKI-EPI study. *Intensive Care Med*. 2015;41:1411–1423.
- Negi S, Koreeda D, Kobayashi S, et al. Acute kidney injury: Epidemiology, outcomes, complications, and therapeutic strategies. *Semin Dial*. 2018.
- Hadidy S, Asfari R, Shammaa MZ, et al. Acute renal failure among a Syrian population. Incidence, cause, treatment and outcome. *Int Urol Nephrol*. 1989;21:455–461.
- Al-Azzam SI, Al-Husein BA, Abu-Dahoud EY, et al. Etiologies of acute renal failure in a sample of hospitalized Jordanian patients. *Ren Fail*. 2008;30:373–376.
- Al-Homrany M. Epidemiology of acute renal failure in hospitalized patients: experience from southern Saudi Arabia. *East Mediterr Health J*. 2003;9:1061–1067.
- Shema L, Ore L, Geron R, et al. Hospital-acquired acute kidney injury in Israel. *Isr Med Assoc J*. 2009;11:269–274.
- Balushi F, Khan S, Riyami D, et al. Acute kidney injury in a teaching hospital in Oman. *Saudi J Kidney Dis Transpl*. 2011;22:825–828.
- Kavaz A, Ozcarar ZB, Kendirli T, et al. Acute kidney injury in a paediatric intensive care unit: comparison of the pRIFLE and AKIN criteria. *Acta Paediatr*. 2012;101:e126–e129.
- Oweis AO, Alshelleh SA. Incidence and outcomes of acute kidney injury in octogenarians in Jordan. *BMC Res Notes*. 2018;11:279.
- Arabi YM, Arifi AA, Balkhy HH, et al. Clinical course and outcomes of critically ill patients with middle East respiratory syndrome coronavirus infection. *Ann Intern Med*. 2014;160.
- Fowler RA, Lapinsky SE, Hallett D, et al. Critically ill patients with severe acute respiratory syndrome. *JAMA*. 2003;290:367–373.
- KDIGO clinical practice guideline for acute kidney injury. *Kidney Int Suppl*. 2012;2:1–138.
- Duzova A, Bakkaloglu A, Kalyoncu M, et al. Cause and outcome of acute kidney injury in children. *Pediatr Nephrol*. 2010;25:1453–1461.
- Said R. Acute renal failure in Jordan. *Saudi J Kidney Dis Transpl*. 1998;9:301–305.
- Al-Malki H, Sadek M, Rashed A, et al. Acute renal failure in the State of Qatar: presentation and outcome. *Transplant Proc*. 2009;41:1530–1532.
- Al Rohani M, Aljawshaei H, Aduolimi E. Acute renal failure in Yemeni patients. *Saudi J Kidney Dis Transpl*. 2011;22:829–833.
- Barsoum RS. Malarial acute renal failure. *J Am Soc Nephrol*. 2000;11:2147–2154.
- Barsoum RS. Parasitic kidney disease: milestones in the evolution of our knowledge. *Am J Kidney Dis*. 2013;61:501–513.
- Sitprija V. Altered fluid, electrolyte and mineral status in tropical disease, with an emphasis on malaria and leptospirosis. *Nat Clin Pract Nephrol*. 2008;4:91–101.
- Bernieh B, Al Hakim M, Boobes Y, et al. Pattern of acute renal failure in a tertiary hospital in the United Arab Emirates. *Transplant Proc*. 2004;36:1780–1783.
- Alsultan MA. The renal recovery of critically ill patients with acute renal failure requiring dialysis. *Saudi J Kidney Dis Transpl*. 2013;24:1175–1179.
- Mehta RL, Burdmann EA, Cerda J, et al. Recognition and management of acute kidney injury in the International Society of Nephrology 0by25 Global Snapshot: a multinational cross-sectional study. *Lancet*. 2016;387:2017–2025.
- Mehta RL, Cerda J, Burdmann EA, et al. International Society of Nephrology's 0by25 initiative for acute kidney injury (zero preventable deaths by 2025): a human rights case for nephrology. *Lancet*. 2015;385:2616–2643.

54. Raimann JG, Riella MC, Levin NW. International Society of Nephrology's 0by25 initiative (zero preventable deaths from acute kidney injury by 2025): focus on diagnosis of acute kidney injury in low-income countries. *Clin Kidney J*. 2018;11:12–19.
55. Hatamizadeh P, Najafi I, Vanholder R, et al. Epidemiologic aspects of the Bam earthquake in Iran: the nephrologic perspective. *Am J Kidney Dis*. 2006;47:428–438.
56. Kantarci G, Vanholder R, Tuglular S, et al. Acute renal failure due to crush syndrome during Marmara earthquake. *Am J Kidney Dis*. 2002;40:682–689.
57. Better OS, Rubinstein I, Reis DN. Muscle crush compartment syndrome: fulminant local edema with threatening systemic effects. *Kidney Int*. 2003;63:1155–1157.
58. Sever MS, Vanholder R. Recommendation for the management of crush victims in mass disasters. *Nephrol Dial Transplant*. 2012;27(suppl 1):i1–i67.
59. Rifai AO, Murad LB, Sekkarie MA, et al. Continuous venovenous hemofiltration using a stand-alone blood pump for acute kidney injury in field hospitals in Syria. *Kidney Int*. 2015;87:254–261.
60. Iraj N, Saeed S, Mostafa H, et al. Prophylactic fluid therapy in crushed victims of Bam earthquake. *Am J Emerg Med*. 2011;29:738–742.
61. Correa-Rotter R, Naicker S, Katz JJ, et al. Demographic and epidemiologic transition in the developing world: role of albuminuria in the early diagnosis and prevention of renal and cardiovascular disease. *Kidney Int*. 2004;Suppl:S32–S37.
62. Farag YM, Kari JA, Singh AK. Chronic kidney disease in the Arab world: a call for action. *Nephron Clin Pract*. 2012;121:c120–c123.
63. Hijazi S. Major health issues in Jordan. *Ethn Dis*. 2005;15:S1-77-82.
64. Khogali M. Health and disease in a changing Arab world 2000/2025/2050: global, environmental, and climate change and emerging diseases. *Ethn Dis*. 2005;15:S1-74-75.
65. International Diabetes Federation: *IDF diabetes atlas*, 8th ed, Brussels, Belgium, 2017, International Diabetes Federation. <http://www.diabetesatlas.org>. 2018.
66. Alashek WA, McIntyre CW, Taal MW. Epidemiology and a cause of dialysis-treated end-stage kidney disease in Libya. *BMC Nephrol*. 2012;13:33.
67. Al-Azzam SI, Abu-Dahoud EY, El-Khatib HA, et al. Etiologies of chronic renal failure in Jordanian population. *J Nephrol*. 2007;20:336–339.
68. Alhyas L, McKay A, Majeed A. Prevalence of type 2 diabetes in the States of the co-operation council for the Arab States of the Gulf: a systematic review. *PLoS ONE*. 2012;7:e40948.
69. Bennet L, Groop L, Lindblad U, et al. Ethnicity is an independent risk indicator when estimating diabetes risk with FINDRISC scores: a cross sectional study comparing immigrants from the Middle East and native Swedes. *Prim Care Diabetes*. 2014.
70. Farag YM, Al Wakeel JS. Diabetic nephropathy in the Arab Gulf countries. *Nephron Clin Pract*. 2011;119:c317–c322, discussion c322–c323.
71. Jaber LA, Brown MB, Hammad A, et al. The prevalence of the metabolic syndrome among arab americans. *Diabetes Care*. 2004;27:234–238.
72. Kridli SA, Herman WH, Brown M, et al. The epidemiology of diabetes and its risk factors among Chaldean Americans. *Ethn Dis*. 2005;15:S1-30-31.
73. Safarinejad MR. The epidemiology of adult chronic kidney disease in a population-based study in Iran: prevalence and associated risk factors. *J Nephrol*. 2009;22:99–108.
74. Suleiman K, Ghattas B, Makhoul C. The health status of the Palestinian Arab community in Israel in relation to the Jewish community of Israel. *Ethn Dis*. 2005;15:S1–S83.
75. Süleymanlar G, Utaş C, Arinsoy T, et al. A population-based survey of Chronic REnal Disease In Turkey—the CREDIT study. *Nephrol Dial Transplant*. 2011;26:1862–1871.
76. Bello AK, Levin A, Tonelli M, et al. *Global kidney health atlas: A report by the International Society of Nephrology on the current state of organization and structures for kidney care across the globe*. Brussels, Belgium, 2017, International Society of Nephrology.
77. Benghanem Gharbi M, Elseviers M, Zamd M, et al. Chronic kidney disease, hypertension, diabetes, and obesity in the adult population of Morocco: how to avoid “over”- and “under”-diagnosis of CKD. *Kidney Int*. 2016;89:1363–1371.
78. Al-Eisa AA, Samhan M, Naseef M. End-stage renal disease in Kuwaiti children: an 8-year experience. *Transplant Proc*. 2004;36:1788–1791.
79. Barbari A, Stephan A, Masri M, et al. Consanguinity-associated kidney diseases in Lebanon: an epidemiological study. *Mol Immunol*. 2003;39:1109–1114.
80. Bener A, Hussain R, Teebi AS. Consanguineous marriages and their effects on common adult diseases: studies from an endogamous population. *Med Princ Pract*. 2007;16:262–267.
81. Finer G, Shalev H, Landau D. Genetic kidney diseases in the pediatric population of southern Israel. *Pediatr Nephrol*. 2006;21:910–916.
82. Karnib HH, Gharavi AG, Aftimos G, et al. A 5-year survey of biopsy proven kidney diseases in Lebanon: significant variation in prevalence of primary glomerular diseases by age, population structure and consanguinity. *Nephrol Dial Transplant*. 2010;25:3962–3969.
83. Ozen S. Renal amyloidosis in familial Mediterranean fever. *Kidney Int*. 2004;65:1118–1127.
84. Zlotogora J, Shalev SA. Marriage patterns and reproductive decision-making in the inhabitants of a single Muslim village during a 50-year period. *Hum Hered*. 2014;77:10–15.
85. Taal MW. Screening for chronic kidney disease: preventing harm or harming the healthy? *PLoS Med*. 2012;9:e1001345.
86. Bello AK, Levin A, Tonelli M, et al. Assessment of global kidney health care status. *JAMA*. 2017;317:1864–1881.
87. Vivante A, Golan E, Tzur D, et al. Body mass index in 1.2 million adolescents and risk for end-stage renal disease. *Arch Intern Med*. 2012;172:1644–1650.
88. Vivante A, Afek A, Frenkel-Nir Y, et al. Persistent asymptomatic isolated microscopic hematuria in Israeli adolescents and young adults and risk for end-stage renal disease. *JAMA*. 2011;306:729–736.
89. Calderon-Margalit R, Golan E, Twig G, et al. History of childhood kidney disease and risk of adult end-stage renal disease. *N Engl J Med*. 2018;378:428–438.
90. Obrador GT, Mahdavi-Mazdeh M, Collins AJ. Establishing the Global Kidney Disease Prevention Network (KDPN): a position statement from the National Kidney Foundation. *Am J Kidney Dis*. 2011;57:361–370.
91. Perico N, Bravo RF, De Leon FR, et al. Screening for chronic kidney disease in emerging countries: feasibility and hurdles. *Nephrol Dial Transplant*. 2009;24:1355–1358.
92. Perico N, Remuzzi G. Prevention programs for chronic kidney disease in low-income countries. *Intern Emerg Med*. 2016;11:385–389.
93. *International Society of Nephrology Global Outreach Clinical Research and Prevention—Awarded Projects*. http://www.theisn.org/images/current_and_past_projects_march_2014.pdf.
94. Gouda Z, Mashaal G, Bello AK, et al. Egypt Information, Prevention, and Treatment of Chronic Kidney Disease (EGIPT-CKD) programme: prevalence and risk factors for microalbuminuria among the relatives of patients with CKD in Egypt. *Saudi J Kidney Dis Transpl*. 2011;22:1055–1063.
95. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl*. 2013;3:1–150.
96. Shaheen FA, Souqiyeh MZ. Kidney health in the Middle East. *Clin Nephrol*. 2010;74(suppl 1):S85–S88.
97. Alsuwaidia AO, Farag YM, Al Sayyari AA, et al. Epidemiology of chronic kidney disease in the Kingdom of Saudi Arabia (SEEK-Saudi investigators) - a pilot study. *Saudi J Kidney Dis Transpl*. 2010;21:1066–1072.
98. El-Reshaid W, El-Reshaid K, Kapoor M, et al. Chronic renal disease in Kuwaiti nationals: a prospective study during the past 4 years. *Ren Fail*. 2005;27:227–233.
99. Mahdavi-Mazdeh M, Saeed Hashemi Nazri S, Hajghasemi E, et al. Screening for decreased renal function in taxi drivers in Tehran, Iran. *Ren Fail*. 2010;32:62–68.
100. Hosseinpahan F, Kasraei F, Nassiri AA, et al. High prevalence of chronic kidney disease in Iran: a large population-based study. *BMC Public Health*. 2009;9:44.
101. Tohidi M, Hasheminia M, Mohebi R, et al. Incidence of chronic kidney disease and its risk factors, results of over 10 year follow up in an Iranian cohort. *PLoS ONE*. 2012;7:e45304.
102. Asghari G, Farhadnejad H, Mirmiran P, et al. Adherence to the Mediterranean diet is associated with reduced risk of incident chronic kidney diseases among Tehranian adults. *Hypertens Res*. 2017;40:96–102.
103. Yuzbashian E, Asghari G, Mirmiran P, et al. Adherence to low-sodium Dietary Approaches to Stop Hypertension-style diet may decrease the risk of incident chronic kidney disease among high-risk patients: a secondary prevention in prospective cohort study. *Nephrol Dial Transplant*. 2018;33:1159–1168.

104. Asghari G, Yuzbashian E, Mirmiran P, et al. The association between Dietary Approaches to Stop Hypertension and incidence of chronic kidney disease in adults: the Tehran Lipid and Glucose Study. *Nephrol Dial Transplant*. 2017;32:ii224–ii230.
105. Thomas B, Matsushita K, Abate KH, et al. Global cardiovascular and renal outcomes of reduced GFR. *J Am Soc Nephrol*. 2017;28:2167–2179.
106. Aghighi M, Mahdavi-Mazdeh M, Zamyadi M, et al. Changing epidemiology of end-stage renal disease in last 10 years in Iran. *Iran J Kidney Dis*. 2009;3:192–196.
107. Najafi I, Alatab S, Atabak S, et al. Seventeen Years' experience of peritoneal dialysis in Iran: first official report of the Iranian Peritoneal Dialysis Registry. *Perit Dial Int*. 2014;34:636–642.
108. El Matri A. History of nephrology in Mediterranean Arab countries. *J Nephrol*. 2013;26:170–174.
109. Al Arrayed S. Renal replacement therapy in bahrain. *Saudi J Kidney Dis Transpl*. 1998;9:457–458.
110. Soliman AR, Fathy A, Roshd D. The growing burden of end-stage renal disease in Egypt. *Ren Fail*. 2012;34:425–428.
111. AlSahow A, AlRukhaimi M, Al Wakeel J, et al. Demographics and key clinical characteristics of hemodialysis patients from the Gulf Cooperation Council countries enrolled in the dialysis outcomes and practice patterns study phase 5 (2012-2015). *Saudi J Kidney Dis Transpl*. 2016;27:S12–S23.
112. SCOT Data. Dialysis in the Kingdom of Saudi Arabia. *Saudi J Kidney Dis Transpl*. 2017;28:949–957.
113. Batieha A, Abdallah S, Maghaireh M, et al. Epidemiology and cost of haemodialysis in Jordan. *East Mediterr Health J*. 2007;13: 654–663.
114. Khader MI, Snouber S, Alkhatib A, et al. Prevalence of patients with end-stage renal disease on dialysis in the West Bank, Palestine. *Saudi J Kidney Dis Transpl*. 2013;24:832–837.
115. Younis M, Jabr S, Al-Khatib A, et al. A cost analysis of kidney replacement therapy options in Palestine. *Inquiry*. 2015;52.
116. Ereke E, Suleymanlar G, Serdengecti K. Nephrology, dialysis and transplantation in Turkey. *Nephrol Dial Transplant*. 2002;17:2087–2093.
117. Erlik D, Better OS, Shramek A, et al. Cadaveric renal transplantation in man: first report from Israel. *Isr J Med Sci*. 1967;3:88–92.
118. Abomelha MS. Renal failure and transplantation activity in the Arab world. Arab Society of Nephrology and Renal Transplantation. *Nephrol Dial Transplant*. 1996;11:28–29.
119. Saeed B, Derani R, Hajibrahim M, et al. Organ failure in Syria: initiating a national deceased donation program. *Saudi J Kidney Dis Transpl*. 2007;18:270–276.
120. Mahdavi-Mazdeh M. The Iranian model of living renal transplantation. *Kidney Int*. 2012;82:627–634.
121. Al Sayyari AA. The history of renal transplantation in the Arab world: a view from Saudi Arabia. *Am J Kidney Dis*. 2008;51:1033–1046.
122. Saeed B. Pediatric versus adult kidney transplantation activity in Arab countries. *Saudi J Kidney Dis Transpl*. 2013;24:1031–1038.
123. Shaheen FA, Salmeen NM, Al-Muzairy IA, et al. Renal transplantation in the Gulf Co-operation Council countries. *Saudi J Kidney Dis Transpl*. 1995;6:5–8.
124. Shigidi MM, Ramachandiran G, Rashed AH, et al. Demographic data and hemodialysis population dynamics in Qatar: a five year survey. *Saudi J Kidney Dis Transpl*. 2009;20:493–500.
125. Najafi I. Peritoneal dialysis in Iran and the Middle East. *Perit Dial Int*. 2009;29(suppl 2):S217–S221.
126. Masri M, Haberal M. Solid-organ transplant activity in MESOT countries. *Exp Clin Transplant*. 2013;11:93–98.
127. *Renal replacement therapy in Israel, 1990-2015. Israel Ministry of Health and the Israel Society for Nephrology and Hypertension [ICDC Publication No. 367, January 2018, in Hebrew]*. https://www.health.gov.il/PublicationsFiles/RRT1990_2015.pdf.
128. *Turkish Society of Nephrology: Registry of the nephrology, dialysis and transplantation in Turkey: registry 2016*. http://www.nefroloji.org.tr/folders/file/2016_REGISTRY.pdf.
129. *The Hashemite Kingdom of Jordan Ministry of Health, Non-Communicable Diseases Directorate: National Registry of End Stage Renal Disease (ESRD) annual report 2016*. <http://www.moh.gov.jo/Echobusv3.0/SystemAssets/debfff26b-c6df-4244-b7f6-4244cdf58733.pdf>.
130. *The 2012 report of the Lebanese kidney registry*. <http://kidneyregistrylb.com/national-registry-2/annual-report-2012/>.
131. Kramer A, Pippias M, Noordzij M, et al. The European Renal Association - European Dialysis and Transplant Association (ERA-EDTA) registry annual report 2015: a summary. *Clin Kidney J*. 2018;11:108–122.
132. Saran R, Robinson B, Abbott KC, et al. US Renal Data System 2017 annual data report: epidemiology of kidney disease in the United States. *Am J Kidney Dis*. 2018;71:A7.
133. Süleymanlar G, Serdengeçti K, Altıparmak MR, et al. Trends in renal replacement therapy in Turkey, 1996-2008. *Am J Kidney Dis*. 2011;57:456–465.
134. Utaş C. The development of PD in Turkey. *Perit Dial Int*. 2008;28: 217–219.
135. Kamyar KZ, Eliezer G, Tamy S, et al. Survival Disparities within American and Israeli Dialysis Populations: Learning from Similarities and Distinctions across Race and Ethnicity. *Semin Dial*. 2010;23: 586–594.
136. Counil É, Cherni N, Kharat M, et al. Trends of incident dialysis patients in Tunisia between 1992 and 2001. *Am J Kidney Dis*. 2008; 51:463–470.
137. Bouguerra R, Alberti H, Salem LB, et al. The global diabetes pandemic: the Tunisian experience. *Eur J Clin Nutr*. 2007;61:160–165.
138. Alashek WA, McIntyre CW, Taal MW. Provision and quality of dialysis services in Libya. *Hemodial Int*. 2011;15:444–452.
139. Alashek WA, McIntyre CW, Taal MW. Hepatitis B and C infection in haemodialysis patients in Libya: prevalence, incidence and risk factors. *BMC Infect Dis*. 2012;12:265.
140. Alashek WA, McIntyre CW, Taal MW. Vascular access in patients receiving hemodialysis in Libya. *J Vasc Access*. 2012;13:468–474.
141. Alashek WA, McIntyre CW, Taal MW. Determinants of survival in patients receiving dialysis in Libya. *Hemodial Int*. 2013;17:249–255.
142. Health indicators for renal replacement therapy in Saudi Arabia. *Saudi J Kidney Dis Transpl*. 2018;29:225–231.
143. Haghighi AN, Broumand B, D'Amico M, et al. The epidemiology of end-stage renal disease in Iran in an international perspective. *Nephrol Dial Transplant*. 2002;17:28–32.
144. Mahdavi-Mazdeh M, Zamyadi M, Nafar M. Assessment of management and treatment responses in haemodialysis patients from Tehran province, Iran. *Nephrol Dial Transplant*. 2008;23:288–293.
145. Goodkin DA, Young EW, Kurokawa K, et al. Mortality among hemodialysis patients in Europe, Japan, and the United States: case-mix effects. *Am J Kidney Dis*. 2004;44:16–21.
146. *Renal replacement therapy in Israel, 1990-2010. Israel Ministry of Health and the Israel Society for Nephrology and Hypertension [ICDC Publication No. 350, December 2012, in Hebrew]*. http://www.isnh.org.il/RRT1990_2010.pdf.
147. Badheeb AM. Causes of Chronic Renal Failure in Hemodialysis Unit: a single center experience in Yemen. *Saudi J Kidney Dis Transpl*. 2006;17:66–69.
148. Barsoum RS. End-stage renal disease in the developing world. *Artif Organs*. 2002;26:735–736.
149. Abboud O. Incidence, prevalence, and treatment of end-stage renal disease in the Middle East. *Ethn Dis*. 2006;16:S2-2-4.
150. Ghods AJ, Nasrollahzadeh D, Kazemine M. Afghan refugees in Iran model renal transplantation program: ethical considerations. *Transplant Proc*. 2005;37:565–567.
151. Barsoum RS. End-stage renal disease in North Africa. *Kidney Int*. 2003;Suppl:S111–S114.
152. Mortada WI, Sobh MA, El-Defrawy MM, et al. Study of lead exposure from automobile exhaust as a risk for nephrotoxicity among traffic policemen. *Am J Nephrol*. 2001;21:274–279.
153. Tuglular S, Yalcinkaya F, Paydas S, et al. A retrospective analysis for acause and clinical findings of 287 secondary amyloidosis cases in Turkey. *Nephrol Dial Transplant*. 2002;17:2003–2005.
154. Utaş C. Patient and technique survival on CAPD in Turkey. *Perit Dial Int*. 2001;21:602–606.
155. Abumwais JQ. Cause of chronic renal failure in Jenin district, Palestine. *Saudi J Kidney Dis Transpl*. 2012;23:158–161.
156. Teebi AS, Teebi SA. Genetic diversity among the Arabs. *Community Genet*. 2005;8:21–26.
157. Zlotogora J. Molecular basis of autosomal recessive diseases among the Palestinian Arabs. *Am J Med Genet*. 2002;109:176–182.
158. Raz AE, Atar M, Rodnay M, et al. Between acculturation and ambivalence: knowledge of genetics and attitudes towards genetic testing in a consanguineous bedouin community. *Community Genet*. 2003;6:88–95.
159. Cohen T, Vardi-Saliternik R, Friedlander Y. Consanguinity, intracommunity and intercommunity marriages in a population sample of Israeli Jews. *Ann Hum Biol*. 2004;31:38–48.
160. Mattoo TK. Genetically transmitted renal diseases in children: a Saudi perspective. *Saudi J Kidney Dis Transpl*. 1998;9:105–109.

161. Al-Eisa A, Naseef M, Al-Hamad N, et al. Chronic renal failure in Kuwaiti children: an eight-year experience. *Pediatr Nephrol*. 2005;20:1781–1785.
162. Jaoud IC, Elalaoui SC, Sbiti A, et al. Consanguineous marriages in Morocco and the consequence for the incidence of autosomal recessive disorders. *J Biosoc Sci*. 2009;41:575–581.
163. Hassan MO, Jaju D, Albarwani S, et al. Non-dipping blood pressure in the metabolic syndrome among Arabs of the Oman family study. *Obesity (Silver Spring)*. 2007;15:2445–2453.
164. Vogel F, Motulsky AG. *Human Genetics: Problems and Approaches*. New York: Springer-Verlag; 1997.
165. Frishberg Y, Ben-Neriah Z, Suvanto M, et al. Misleading findings of homozygosity mapping resulting from three novel mutations in NPHS1 encoding nephrin in a highly inbred community. *Genet Med*. 2007;9:180–184.
166. Saeed MB. The major causes of chronic renal insufficiency in Syrian children: a one-year, single-center experience. *Saudi J Kidney Dis Transpl*. 2005;16:84–88.
167. Yousri NA, Fakhro KA, Robay A, et al. Whole-exome sequencing identifies common and rare variant metabolic QTLs in a Middle Eastern population. *Nat Commun*. 2018;9:333.
168. Bourquia A. Autosomal dominant polycystic kidney disease (ADPKD). in Morocco. Multicenter study about 308 families. *Nephrologie*. 2002;23:93–96.
169. Kazancioglu R, Ecder T, Altintepe L, et al. Demographic and clinical characteristics of patients with autosomal dominant polycystic kidney disease: a multicenter experience. *Nephron Clin Pract*. 2011;117:c270–c275.
170. Roozbeh J, Razmkon AR, Jalaiean H, et al. Outcome of kidney transplantation in patients with polycystic kidney disease: a single center study. *Saudi J Kidney Dis Transpl*. 2008;19:72–75.
171. Ward CJ, Yuan D, Maszyk TV, et al. Cellular and subcellular localization of the ARPKD protein; fibrocystin is expressed on primary cilia. *Hum Mol Genet*. 2003;12:2703–2710.
172. Mutation Database Autosomal Recessive Polycystic Kidney Disease (ARPKD/PKHD1). Department of Human Genetics, RWTH Aachen University, Germany. <http://www.humgen.rwth-aachen.de/index.php?page=database>.
173. Bergmann C, Senderek J, Sedlacek B, et al. Spectrum of mutations in the gene for autosomal recessive polycystic kidney disease (ARPKD/PKHD1). *J Am Soc Nephrol*. 2003;14:76–89.
174. Bergmann C, Senderek J, Windelen E, et al. Clinical consequences of PKHD1 mutations in 164 patients with autosomal-recessive polycystic kidney disease (ARPKD). *Kidney Int*. 2005;67:829–848.
175. Al-Belushi M, Al Ibrahim A, Ahmed M, et al. A review of Meckel-Gruber syndrome—incidence and outcome in the state of Qatar. *J Matern Fetal Neonatal Med*. 2016;29:2013–2016.
176. Aalimi U, Spiegel E, Chervinsky I, et al. High incidence and broad genetic variability of Meckel-Gruber syndrome in the Arab population residing in north-east Israel. *Harefuah*. 2015;154:632–636, 676.
177. Landau D, Shalev H. Childhood genetic renal diseases in southern Israel. *Harefuah*. 2010;149:180–185, 193.
178. Soliman NA, Hildebrandt F, Otto EA, et al. Clinical characterization and NPHP1 mutations in nephronophthisis and associated ciliopathies: a single center experience. *Saudi J Kidney Dis Transpl*. 2012;23:1090–1098.
179. Tory K, Rousset-Rouviere C, Gubler MC, et al. Mutations of NPHP2 and NPHP3 in infantile nephronophthisis. *Kidney Int*. 2009;75:839–847.
180. Dehghani M, Mojarad M, Ghayoor Karimiani E, et al. A common ancestral Asn242Ser mutation in TMEM67 identified in multiple Iranian families with Joubert syndrome. *Public Health Genomics*. 2017;20:188–193.
181. Karnib HH, Sanna-Cherchi S, Zalloua PA, et al. Characterization of a large Lebanese family segregating IgA nephropathy. *Nephrol Dial Transplant*. 2007;22:772–777.
182. Boute N, Gribouval O, Roselli S, et al. NPHS2, encoding the glomerular protein podocin, is mutated in autosomal recessive steroid-resistant nephrotic syndrome. *Nat Genet*. 2000;24:349–354.
183. Frishberg Y, Rinat C, Megged O, et al. Mutations in NPHS2 encoding podocin are a prevalent cause of steroid-resistant nephrotic syndrome among Israeli-Arab children. *J Am Soc Nephrol*. 2002;13:400–405.
184. Frishberg Y, Feinstein S, Rinat C, et al. The heart of children with steroid-resistant nephrotic syndrome: is it all podocin? *J Am Soc Nephrol*. 2006;17:227–231.
185. Berdeli A, Mir S, Yavascan O, et al. NPHS2 (podocin) mutations in Turkish children with idiopathic nephrotic syndrome. *Pediatr Nephrol*. 2007;22:2031–2040.
186. Özçakar ZB, Cengiz FB, Çakar N, et al. Analysis of NPHS2 mutations in Turkish steroid-resistant nephrotic syndrome patients. *Pediatr Nephrol*. 2006;21:1093–1096.
187. Basiratnia M, Yavarian M, Torabinezhad S, et al. NPHS2 gene in steroid-resistant nephrotic syndrome: prevalence, clinical course, and mutational spectrum in South-West Iranian children. *Iran J Kidney Dis*. 2013;7:357–362.
188. Landau D, Oved T, Geiger D, et al. Familial steroid-sensitive nephrotic syndrome in Southern Israel: clinical and genetic observations. *Pediatr Nephrol*. 2007;22:661–669.
189. Al-Hamed MH, Al-Sabban E, Al-Mojalli H, et al. A molecular genetic analysis of childhood nephrotic syndrome in a cohort of Saudi Arabian families. *J Hum Genet*. 2013;58:480–489.
190. Trautmann A, Bodria M, Ozaltin F, et al. Spectrum of steroid-resistant and congenital nephrotic syndrome in children: the PodoNet registry cohort. *Clin J Am Soc Nephrol*. 2015;10:592–600.
191. Kari JA, El-Desoky SM, Gari M, et al. Steroid-resistant nephrotic syndrome: impact of genetic testing. *Ann Saudi Med*. 2013;33:533–538.
192. Mir S, Yavascan O, Berdeli A, et al. TRPC6 gene variants in Turkish children with steroid-resistant nephrotic syndrome. *Nephrol Dial Transplant*. 2012;27:205–209.
193. Braun DA, Rao J, Mollet G, et al. Mutations in KEOPS-complex genes cause nephrotic syndrome with primary microcephaly. *Nat Genet*. 2017;49:1529–1538.
194. Ashraf S, Gee HY, Woerner S, et al. ADCK4 mutations promote steroid-resistant nephrotic syndrome through CoQ10 biosynthesis disruption. *J Clin Invest*. 2013;123:5179–5189.
195. Atmaca M, Gulhan B, Korkmaz E, et al. Follow-up results of patients with ADCK4 mutations and the efficacy of CoQ10 treatment. *Pediatr Nephrol*. 2017;32:1369–1375.
196. Wang L, Ellis MJ, Gomez JA, et al. Mechanisms of the proteinuria induced by Rho GTPases. *Kidney Int*. 2012;81:1075–1085.
197. Gee HY, Saisawat P, Ashraf S, et al. ARHGDI mutations cause nephrotic syndrome via defective RHO GTPase signaling. *J Clin Invest*. 2013;123:3243–3253.
198. Beck BB, Hoyer-Kuhn H, Gobel H, et al. Hyperoxaluria and systemic oxalosis: an update on current therapy and future directions. *Expert Opin Investig Drugs*. 2013;22:117–129.
199. Hoppe B. An update on primary hyperoxaluria. *Nat Rev Nephrol*. 2012;8:467–475.
200. Ben-Shalom E, Frishberg Y. Primary hyperoxalurias: diagnosis and treatment. *Pediatr Nephrol*. 2015;30:1781–1791.
201. Belostotsky R, Seboun E, Idelson GH, et al. Mutations in DHAPSL are responsible for primary hyperoxaluria type III. *Am J Hum Genet*. 2010;87:392–399.
202. Belostotsky R, Pitt JJ, Frishberg Y. Primary hyperoxaluria type III—a model for studying perturbations in glyoxylate metabolism. *J Mol Med*. 2012;90:1497–1504.
203. Frishberg Y, Rinat C, Shalata A, et al. Intra-familial clinical heterogeneity: absence of genotype-phenotype correlation in primary hyperoxaluria type 1 in Israel. *Am J Nephrol*. 2005;25:269–275.
204. Levin-Iaina N, Dinour D, Romero L, et al. Late diagnosis of primary hyperoxaluria type 2 in the adult: effect of a novel mutation in GRHPR gene on enzymatic activity and molecular modeling. *J Urol*. 2009;181:2146–2151.
205. Rinat C, Wanders RJ, Drukker A, et al. Primary hyperoxaluria type I: a model for multiple mutations in a monogenic disease within a distinct ethnic group. *J Am Soc Nephrol*. 1999;10:2352–2358.
206. Soliman NA, Nabhan MM, Abdelrahman SM, et al. Clinical spectrum of primary hyperoxaluria type 1: Experience of a tertiary center. *Nephrol Ther*. 2017;13:176–182.
207. Harambat J, Fargue S, Acquaviva C, et al. Genotype-phenotype correlation in primary hyperoxaluria type 1: the p.Gly170Arg AGXT mutation is associated with a better outcome. *Kidney Int*. 2010;77:443–449.
208. Abu-Ghanem Y, Kleinmann N, Winkler HZ, et al. Nephrolithiasis in Israel: epidemiological characteristics of return patients in a tertiary care center. *Isr Med Assoc J*. 2016;18:725–728.
209. Almardini RI, Alfarah MG, Salaita GM. The clinical pattern of primary hyperoxaluria in pediatric patient at Queen Rania Abdulla Children Hospital. *Arab J Nephrol Transplant*. 2014;7:119–123.

210. Elmaci AM, Ece A, Akin F. Pediatric urolithiasis: metabolic risk factors and follow-up results in a Turkish region with endemic stone disease. *Urolithiasis*. 2014;42:421–426.
211. Al-Eisa AA, Al-Hunayyan A, Gupta R. Pediatric urolithiasis in Kuwait. *Int Urol Nephrol*. 2002;33:3–6.
212. Sanjad SA, Al-Abbad A, Al-Sabban E. Primary hyperoxaluria type 1: An underestimated cause of nephrocalcinosis and chronic renal failure in Saudi Arabian children. *Ann Saudi Med*. 1999;19:4–7.
213. Aydogdu SD, Kirel B, Coskun T, et al. Prevalence of cystinuria among elementary schoolchildren in Eskisehir, Turkey. *Scand J Urol Nephrol*. 2009;43:138–141.
214. Pras E, Raben N, Golomb E, et al. Mutations in the SLC3A1 transporter gene in cystinuria. *Am J Hum Genet*. 1995;56:1297–1303.
215. Al-Riyami AA, Suleiman AJ, Afifi M, et al. A community-based study of common hereditary blood disorders in Oman. *East Mediterr Health J*. 2001;7:1004–1011.
216. Tanzer F, Ozgur A, Bardakci F. Type I cystinuria and its genetic basis in a population of Turkish school children. *Int J Urol*. 2007;14:914–917.
217. Colombo R. Dating the origin of the V170M mutation causing non-type I cystinuria in Libyan Jews by linkage disequilibrium and physical mapping of the SLC7A9 gene. *Genomics*. 2000;69:131–134.
218. Sidi R, Levy-Nissenbaum E, Kreiss I, et al. Clinical manifestations in Israeli cystinuria patients and molecular assessment of carrier rates in Libyan Jewish controls. *Isr Med Assoc J*. 2003;5:439–442.
219. Lotan D, Yoskovitz G, Bisceglia L, et al. A combined approach to the molecular analysis of cystinuria: from urinalysis to sequencing via genotyping. *Isr Med Assoc J*. 2007;9:513–516.
220. Aldahmesh MA, Abu-Safieh L, Khan AO, et al. Allelic heterogeneity in inbred populations: the Saudi experience with Alstrom syndrome as an illustrative example. *Am J Med Genet A*. 2009;149A:662–665.
221. Friedlaender MM, Kopolovic J, Rubinger D, et al. Renal biopsy in Fabry's disease eight years after successful renal transplantation. *Clin Nephrol*. 1987;27:206–211.
222. Raas-Rothschild A, Friedlaender MM, Pizov G, et al. The kidney in Fabry disease. *J Pediatr*. 2005;146:148.
223. Raas-Rothschild A. Pediatrics: implementing the promise of early intervention for Fabry disease. *Clin Ther*. 2007;29(supplA):S6.
224. El-Reshaid W, El-Reshaid K, Kapoor MM, et al. Glomerulopathy in Kuwait: the spectrum over the past 7 years. *Ren Fail*. 2003;25:619–630.
225. Ozkara HA, Topcu M. Sphingolipidoses in Turkey. *Brain Dev*. 2004;26:363–366.
226. Okur I, Ezgu F, Biberoglu G, et al. Screening for Fabry disease in patients undergoing dialysis for chronic renal failure in Turkey: identification of new case with novel mutation. *Gene*. 2013;527:42–47.
227. Kabalan SN, Abbas S, Tawil L. A search for Fabry disease among male end-stage renal disease patients in Lebanon and a review of the literature. *J Med Liban*. 2013;61:144–147.
228. Abdel-al YK, Badawi MH, Yaesh SA, et al. Bartter's syndrome in Arabic children: review of 13 cases. *Pediatr Int*. 1999;41:299–303.
229. Zelikovic I, Szargel R, Hawash A, et al. A novel mutation in the chloride channel gene, CLCNKB, as a cause of Gitelman and Bartter syndromes. *Kidney Int*. 2003;63:24–32.
230. Vivante A, Mark-Danieli M, Davidovits M, et al. Renal hypodysplasia associates with a WNT4 variant that causes aberrant canonical WNT signaling. *J Am Soc Nephrol*. 2013;24:550–558.
231. Vivante A, Mann N, Yonath H, et al. A dominant mutation in nuclear receptor interacting protein 1 causes urinary tract malformations via dysregulation of retinoic acid signaling. *J Am Soc Nephrol*. 2017;28:2364–2376.
232. Aldahmesh MA, Humeidan A, Almojalli HA, et al. Characterization of CTNS mutations in Arab patients with cystinosis. *Ophthalmic Genet*. 2009;30:185–189.
233. Bahat H, Dinour D, Ganon L, et al. Non-urate transporter 1-related renal hypouricemia and acute renal failure in an Israeli-Arab family. *Pediatr Nephrol*. 2009;24:999–1003.
234. Besbas N, Ozaltin F, Jeck N, et al. CLCN5 mutation (R347X) associated with hypokalaemic metabolic alkalosis in a Turkish child: an unusual presentation of Dent's disease. *Nephrol Dial Transplant*. 2005;20:1476–1479.
235. Dinour D, Chang MH, Satoh J, et al. A novel missense mutation in the sodium bicarbonate cotransporter (NBCe1/SLC4A4) causes proximal tubular acidosis and glaucoma through ion transport defects. *J Biol Chem*. 2004;279:52238–52246.
236. Dinour D, Davidovitz M, Levin-Iaina N, et al. Truncating mutations in the chloride/proton ClC-5 antiporter gene in Seven Jewish Israeli families with Dent's 1 disease. *Nephron Clin Pract*. 2009;112:c262–c267.
237. Dinour D, Gray NK, Campbell S, et al. Homozygous SLC2A9 mutations cause severe renal hypouricemia. *J Am Soc Nephrol*. 2010;21:64–72.
238. Jahanzad I, Amouei S, Attaranzadeh A. Familial lecithin-cholesterol acyltransferase deficiency. *Arch Iran Med*. 2009;12:179–181.
239. Magen D, Sprecher E, Zelikovic I, et al. A novel missense mutation in SLC5A2 encoding SGLT2 underlies autosomal-recessive renal glucosuria and aminoaciduria. *Kidney Int*. 2005;67:34–41.
240. Al-Haggag M. Cystinosis as a lysosomal storage disease with multiple mutant alleles: Phenotypic-genotypic correlations. *World J Nephrol*. 2013;2:94–102.
241. Al-Haggag M, Bakr A, Tajima T, et al. Familial hypomagnesemia with hypercalciuria and nephrocalcinosis: unusual clinical associations and novel claudin16 mutation in an Egyptian family. *Clin Exp Nephrol*. 2009;13:288–294.
242. Dinour D, Bahn A, Ganon L, et al. URAT1 mutations cause renal hypouricemia type 1 in Iraqi Jews. *Nephrol Dial Transplant*. 2011;26:2175–2181.
243. Dinour D, Gray NK, Ganon L, et al. Two novel homozygous SLC2A9 mutations cause renal hypouricemia type 2. *Nephrol Dial Transplant*. 2012;27:1035–1041.
244. Frishberg Y, Dinour D, Belostotsky R, et al. Dent's disease manifesting as focal glomerulosclerosis: Is it the tip of the iceberg? *Pediatr Nephrol*. 2009;24:2369–2373.
245. Magen D, Berger L, Coady MJ, et al. A loss-of-function mutation in NaPi-IIa and renal Fanconi's syndrome. *N Engl J Med*. 2010;362:1102–1109.
246. Mannstadt M, Magen D, Segawa H, et al. Fanconi-Bickel syndrome and autosomal recessive proximal tubulopathy with hypercalciuria (ARPTH) are allelic variants caused by GLUT2 mutations. *J Clin Endocrinol Metab*. 2012;97:E1978–E1986.
247. Ozgul RK, Satman I, Collin GB, et al. Molecular analysis and long-term clinical evaluation of three siblings with Alstrom syndrome. *Clin Genet*. 2007;72:351–356.
248. Soliman NA, Elmonem MA, van den Heuvel L, et al. Mutational Spectrum of the CTNS Gene in Egyptian Patients with Nephropathic Cystinosis. *JIMD Rep*. 2014.
249. Topaloglu R, Vilboux T, Coskun T, et al. Genetic basis of cystinosis in Turkish patients: a single-center experience. *Pediatr Nephrol*. 2012;27:115–121.
250. Vivante A, Lotan D, Pode-Shakked N, et al. Familial autosomal recessive renal tubular acidosis: importance of early diagnosis. *Nephron Physiol*. 2011;119:p31–p39.
251. Jaradat S, Al-Rababah B, Hazza I, et al. Molecular analysis of the CTNS gene in Jordanian families with nephropathic cystinosis. *Nefrologia*. 2015;35:547–553.
252. Aleem A. Renal abnormalities in patients with sickle cell disease: a single center report from Saudi Arabia. *Saudi J Kidney Dis Transpl*. 2008;19:194–199.
253. Steensma DP, Hoyer JD, Fairbanks VF. Hereditary red blood cell disorders in middle eastern patients. *Mayo Clin Proc*. 2001;76:285–293.
254. Eliakim M, Levy M, Ehrenfeld M. *Recurrent Polyserositis (Familial Mediterranean Fever)*. Amsterdam: Elsevier-North Holland Biomedical Press; 1981.
255. Consortium FF. A candidate gene for familial Mediterranean fever. *Nat Genet*. 1997;17:25–31.
256. Consortium IF. Ancient missense mutations in a new member of the RoRet gene family are likely to cause familial Mediterranean fever. The International FMF Consortium. *Cell*. 1997;90:797–807.
257. *Infefers: the registry of hereditary auto-inflammatory disorders mutations*. <http://fmf.igh.cnrs.fr/ISSAID/infefers/>. Accessed July 16, 2018.
258. Matzner Y, Abedat S, Shapiro E, et al. Expression of the familial Mediterranean fever gene and activity of the C5a inhibitor in human primary fibroblast cultures. *Blood*. 2000;96:727–731.
259. Centola M, Wood G, Frucht DM, et al. The gene for familial Mediterranean fever, MEFV, is expressed in early leukocyte development and is regulated in response to inflammatory mediators. *Blood*. 2000;95:3223–3231.
260. Stehlik C, Reed JC. The PYRIN connection: novel players in innate immunity and inflammation. *J Exp Med*. 2004;200:551–558.
261. Xu H, Yang J, Gao W, et al. Innate immune sensing of bacterial modifications of Rho GTPases by the PIRIN inflammasome. *Nature*. 2014;513:237–241.

262. Gao W, Yang J, Liu W, et al. Site-specific phosphorylation and microtubule dynamics control Pyrin inflammasome activation. *Proc Natl Acad Sci USA*. 2016;113:E4857–E4866.
263. Manukyan G, Aminov R. Update on pyrin functions and mechanisms of familial Mediterranean fever. *Front Microbiol*. 2016;7:456.
264. Bakkaloglu A. Familial Mediterranean fever. *Pediatr Nephrol*. 2003;18:853–859.
265. Touitou I. Standardized testing for mutations in familial Mediterranean fever. *Clin Chem*. 2003;49:1781–1782.
266. Kumpfel T, Gerdes LA, Wacker T, et al. Familial Mediterranean fever-associated mutation pyrin E148Q as a potential risk factor for multiple sclerosis. *Mult Scler*. 2012;18:1229–1238.
267. el Garf A, Salah S, Iskander I, et al. MEFV mutations in Egyptian patients suffering from familial Mediterranean fever: analysis of 12 gene mutations. *Rheumatol Int*. 2010;30:1293–1298.
268. Kasifoglu T, Bilge SY, Sari I, et al. Amyloidosis and its related factors in Turkish patients with familial Mediterranean fever: a multicentre study. *Rheumatology (Oxford)*. 2014;53:741–745.
269. Mohammadnejad L, Farajnia S. Mediterranean Fever gene analysis in the azeri turk population with familial mediterranean Fever: evidence for new mutations associated with disease. *Cell J*. 2013;15:152–159.
270. Ozdemir O, Sezgin I, Kurtulgan HK, et al. Prevalence of known mutations in the MEFV gene in a population screening with high rate of carriers. *Mol Biol Rep*. 2011;38:3195–3200.
271. Ait-Idir D, Khilan A, Djerdjouri B, et al. Spectrum of mutations and carrier frequency of familial Mediterranean fever gene in the Algerian population. *Rheumatology (Oxford)*. 2011;50:2306–2310.
272. Al-Alami JR, Tayeh MK, Najib DA, et al. Familial Mediterranean fever mutation frequencies and carrier rates among a mixed Arabic population. *Saudi Med J*. 2003;24:1055–1059.
273. Ayesh SK, Nassar SM, Al-Sharef WA, et al. Genetic screening of familial Mediterranean fever mutations in the Palestinian population. *Saudi Med J*. 2005;26:732–737.
274. Belmahi L, Seifiani A, Fouveau C, et al. Prevalence and distribution of MEFV mutations among Arabs from the Maghreb patients suffering from familial Mediterranean fever. *C R Biol*. 2006;329:71–74.
275. Ben-Chetrit E, Backenroth R. Amyloidosis induced, end stage renal disease in patients with familial Mediterranean fever is highly associated with point mutations in the MEFV gene. *Ann Rheum Dis*. 2001;60:146–149.
276. Ben-Chetrit E, Urieli-Shoval S, Calko S, et al. Molecular diagnosis of FMF: lessons from a study of 446 unrelated individuals. *Clin Exp Rheumatol*. 2002;20:S25–S29.
277. Bidari A, Ghavidel-Parsa B, Najmabadi H, et al. Common MEFV mutation analysis in 36 Iranian patients with familial Mediterranean fever: clinical and demographic significance. *Mod Rheumatol*. 2010;20:566–572.
278. Brik R, Shinawi M, Kepten I, et al. Familial Mediterranean fever: clinical and genetic characterization in a mixed pediatric population of Jewish and Arab patients. *Pediatrics*. 1999;103:e70.
279. Caglayan AO, Demiryilmaz F, Ozyazgan I, et al. MEFV gene compound heterozygous mutations in familial Mediterranean fever phenotype: a retrospective clinical and molecular study. *Nephrol Dial Transplant*. 2010;25:2520–2523.
280. El Gezery DA, Abou-Zeid AA, Hashad DI, et al. MEFV gene mutations in Egyptian patients with familial Mediterranean fever. *Genet Test Mol Biomarkers*. 2010;14:263–268.
281. Chaabouni HB, Ksantini M, M'Rad R, et al. MEFV mutations in Tunisian patients suffering from familial Mediterranean fever. *Semin Arthritis Rheum*. 2007;36:397–401.
282. Esmaceli M, Bonyadi M, Rafeey M, et al. Common MEFV mutation analysis in Iranian Azeri Turkish patients with familial Mediterranean fever. *Semin Arthritis Rheum*. 2008;37:334–338.
283. Langevitz P, Zemer D, Livneh A, et al. Protracted febrile myalgia in patients with familial Mediterranean fever. *J Rheumatol*. 1994;21:1708–1709.
284. Livneh A, Langevitz P, Zemer D, et al. Criteria for the diagnosis of familial Mediterranean fever. *Arthritis Rheum*. 1997;40:1879–1885.
285. Majeed HA, El-Khateeb M, El-Shanti H, et al. The spectrum of familial Mediterranean fever gene mutations in Arabs: report of a large series. *Semin Arthritis Rheum*. 2005;34:813–818.
286. Matti H, Joma M, Al-Cheikh S, et al. Familial Mediterranean fever in the Syrian population: gene mutation frequencies, carrier rates and phenotype-genotype correlation. *Eur J Med Genet*. 2006;49:481–486.
287. Medlej-Hashim M, Rawashdeh M, Chouery E, et al. Genetic screening of fourteen mutations in Jordanian familial Mediterranean fever patients. *Hum Mutat*. 2000;15:384.
288. Sabbagh AS, Ghasham M, Abdel Khalek R, et al. MEFV gene mutations spectrum among Lebanese patients referred for Familial Mediterranean Fever work-up: experience of a major tertiary care center. *Mol Biol Rep*. 2008;35:447–451.
289. Settin A, El-Baz R, Abd Rasool M, et al. Clinical and molecular diagnosis of Familial Mediterranean Fever in Egyptian children. *J Gastrointest Liver Dis*. 2007;16:141–145.
290. Sharkia R, Mahajnah M, Zalan A, et al. Comparative screening of FMF mutations in various communities of the Israeli society. *Eur J Med Genet*. 2013;56:351–355.
291. Shinawi M, Brik R, Berant M, et al. Familial Mediterranean fever: high gene frequency and heterogeneous disease among an Israeli-Arab population. *J Rheumatol*. 2000;27:1492–1495.
292. Solak M, Yildiz H, Koken R, et al. Analysis of familial Mediterranean fever gene mutations in 202 patients with familial Mediterranean fever. *Genet Test*. 2008;12:341–344.
293. Tunca M, Akar S, Onen F, et al. Familial Mediterranean fever (FMF) in Turkey: results of a nationwide multicenter study. *Medicine (Baltimore)*. 2005;84:1–11.
294. Zaks N, Shinar Y, Padeh S, et al. Analysis of the three most common MEFV mutations in 412 patients with familial Mediterranean fever. *Isr Med Assoc J*. 2003;5:585–588.
295. Koshy R, Sivasdas A, Scaria V. Genetic epidemiology of familial Mediterranean fever through integrative analysis of whole genome and exome sequences from Middle East and North Africa. *Clin Genet*. 2018;93:92–102.
296. Dode C, Pecheux C, Cazeneuve C, et al. Mutations in the MEFV gene in a large series of patients with a clinical diagnosis of familial Mediterranean fever. *Am J Med Genet*. 2000;92:241–246.
297. Sugiura T, Kawaguchi Y, Fujikawa S, et al. Familial Mediterranean fever in three Japanese patients, and a comparison of the frequency of MEFV gene mutations in Japanese and Mediterranean populations. *Mod Rheumatol*. 2008;18:57–59.
298. Chae JJ, Aksentijevich I, Kastner DL. Advances in the understanding of familial Mediterranean fever and possibilities for targeted therapy. *Br J Haematol*. 2009;146:467–478.
299. El-Shanti H, Majeed HA, El-Khateeb M. Familial mediterranean fever in Arabs. *Lancet*. 2006;367:1016–1024.
300. Guz G, Kanbay M, Ozturk MA. Current perspectives on familial Mediterranean fever. *Curr Opin Infect Dis*. 2009;22:309–315.
301. Park H, Bourla AB, Kastner DL, et al. Lighting the fires within: the cell biology of autoinflammatory diseases. *Nat Rev Immunol*. 2012;12:570–580.
302. Hashkes PJ, Spalding SJ, Giannini EH, et al. Riloncept for colchicine-resistant or -intolerant familial Mediterranean fever: a randomized trial. *Ann Intern Med*. 2012;157:533–541.
303. Richards N, Schaner P, Diaz A, et al. Interaction between pyrin and the apoptotic speck protein (ASC) modulates ASC-induced apoptosis. *J Biol Chem*. 2001;276:39320–39329.
304. Gumucio DL, Diaz A, Schaner P, et al. Fire and ICE: the role of pyrin domain-containing proteins in inflammation and apoptosis. *Clin Exp Rheumatol*. 2002;20:S45–S53.
305. Perfetto L, Gherardini PF, Davey NE, et al. Exploring the diversity of SPRY/B30.2-mediated interactions. *Trends Biochem Sci*. 2013;38:38–46.
306. Kimura T, Jain A, Choi SW, et al. TRIM-mediated precision autophagy targets cytoplasmic regulators of innate immunity. *J Cell Biol*. 2015;210:973–989.
307. Weinert C, Morger D, Djekic A, et al. Crystal structure of TRIM20 C-terminal coiled-coil/B30.2 fragment: implications for the recognition of higher order oligomers. *Sci Rep*. 2015;5:10819.
308. Hoffman HM, Mueller JL, Broide DH, et al. Mutation of a new gene encoding a putative pyrin-like protein causes familial cold autoinflammatory syndrome and Muckle-Wells syndrome. *Nat Genet*. 2001;29:301–305.
309. Stoffels M, Szperl A, Simon A, et al. MEFV mutations affecting pyrin amino acid 577 cause autosomal dominant autoinflammatory disease. *Ann Rheum Dis*. 2014;73:455–461.
310. Majeed HA, El-Shanti H, Al-Khateeb MS, et al. Genotype/phenotype correlations in Arab patients with familial Mediterranean fever. *Semin Arthritis Rheum*. 2002;31:371–376.
311. Gershoni-Baruch R, Brik R, Zacks N, et al. The contribution of genotypes at the MEFV and SAA1 loci to amyloidosis and disease

- severity in patients with familial Mediterranean fever. *Arthritis Rheum.* 2003;48:1149–1155.
312. D'Cruz AA, Babon JJ, Norton RS, et al. Structure and function of the SPRY/B30.2 domain proteins involved in innate immunity. *Protein Sci.* 2013;22:1–10.
 313. Kone-Paut I, Hentgen V, Guillaume-Czitrom S, et al. The clinical spectrum of 94 patients carrying a single mutated MEFV allele. *Rheumatology (Oxford)*. 2009;48:840–842.
 314. Sohar E, Gafni J, Pras M, et al. Familial Mediterranean fever. A survey of 470 cases and review of the literature. *Am J Med.* 1967;43:227–253.
 315. Rocken C, Shakespeare A. Pathology, diagnosis and pathogenesis of AA amyloidosis. *Virchows Arch.* 2002;440:111–122.
 316. Shohat M, Magal N, Shohat T, et al. Phenotype-genotype correlation in familial Mediterranean fever: evidence for an association between Met694Val and amyloidosis. *Eur J Hum Genet.* 1999;7:287–292.
 317. Camus D, Shinar Y, Aamar S, et al. 'Silent' carriage of two familial Mediterranean fever gene mutations in large families with only a single identified patient. *Clin Genet.* 2012;82:288–291.
 318. Touitou I, Sarkisian T, Medlej-Hashim M, et al. Country as the primary risk factor for renal amyloidosis in familial Mediterranean fever. *Arthritis Rheum.* 2007;56:1706–1712.
 319. Ben-Zvi I, Brandt B, Berkun Y, et al. The relative contribution of environmental and genetic factors to phenotypic variation in familial Mediterranean fever (FMF). *Gene.* 2012;491:260–263.
 320. Samuels J, Aksentijevich I, Torosyan Y, et al. Familial Mediterranean fever at the millennium. Clinical spectrum, ancient mutations, and a survey of 100 American referrals to the National Institutes of Health. *Medicine (Baltimore)*. 1998;77:268–297.
 321. Deveci K, Gokakin AK, Senel S, et al. Cystatin C in serum as an early marker of renal involvement in Familial Mediterranean Fever patients. *Eur Rev Med Pharmacol Sci.* 2013;17:253–260.
 322. Twig G, Livneh A, Vivante A, et al. Mortality risk factors associated with familial Mediterranean fever among a cohort of 1.25 million adolescents. *Ann Rheum Dis.* 2014;73:704–709.
 323. El-Shanti HI. Familial mediterranean Fever and renal disease. *Saudi J Kidney Dis Transpl.* 2003;14:378–385.
 324. Obici L, Perfetti V, Palladini G, et al. Clinical aspects of systemic amyloid diseases. *Biochim Biophys Acta.* 2005;1753:11–22.
 325. Dember LM. Amyloidosis-associated kidney disease. *J Am Soc Nephrol.* 2006;17:3458–3471.
 326. Libbey CA, Skinner M, Cohen AS. Use of abdominal fat tissue aspirate in the diagnosis of systemic amyloidosis. *Arch Intern Med.* 1983;143:1549–1552.
 327. Sungur C, Sungur A, Ruacan S, et al. Diagnostic value of bone marrow biopsy in patients with renal disease secondary to familial Mediterranean fever. *Kidney Int.* 1993;44:834–836.
 328. Eliakim M, Rachmilewitz M, Rosenmann E, et al. Renal manifestations in recurrent polyserositis (familial Mediterranean fever). *Isr J Med Sci.* 1970;6:228–245.
 329. Gok F, Sari E, Erdogan O, et al. Familial Mediterranean fever and IgA nephropathy: case report and review of the literature. *Clin Nephrol.* 2008;70:62–64.
 330. Said R, Hamzeh Y, Said S, et al. Spectrum of renal involvement in familial Mediterranean fever. *Kidney Int.* 1992;41:414–419.
 331. Sozeri B, Mir S, Ertan P, et al. Rapidly progressive glomerulonephritis in a child with Henoch-Schonlein Vasculitis and familial Mediterranean fever. *Pediatr Rheumatol Online J.* 2009;7:8.
 332. Yalcinkaya F, Tumer N. Glomerular lesions other than amyloidosis in patients with familial Mediterranean fever. *Nephrol Dial Transplant.* 1999;14:21–23.
 333. Goldfinger SE. Colchicine for familial Mediterranean fever. *N Engl J Med.* 1972;287:1302.
 334. Zemer D, Revach M, Pras M, et al. A controlled trial of colchicine in preventing attacks of familial mediterranean fever. *N Engl J Med.* 1974;291:932–934.
 335. Mansfield E, Chae JJ, Komarow HD, et al. The familial Mediterranean fever protein, pyrin, associates with microtubules and colocalizes with actin filaments. *Blood.* 2001;98:851–859.
 336. Kiraz S, Ertenli I, Arici M, et al. Effects of colchicine on inflammatory cytokines and selectins in familial Mediterranean fever. *Clin Exp Rheumatol.* 1998;16:721–724.
 337. Chae JJ, Wood G, Richard K, et al. The familial Mediterranean fever protein, pyrin, is cleaved by caspase-1 and activates NF-kappaB through its N-terminal fragment. *Blood.* 2008;112:1794–1803.
 338. Manukyan G, Petrek M, Tomankova T, et al. Colchicine modulates expression of pro-inflammatory genes in neutrophils from patients with familial Mediterranean fever and healthy subjects. *J Biol Regul Homeost Agents.* 2013;27:329–336.
 339. Taskiran EZ, Cetinkaya A, Balci-Peynircioglu B, et al. The effect of colchicine on pyrin and pyrin interacting proteins. *J Cell Biochem.* 2012;113:3536–3546.
 340. Nuki G. Colchicine: its mechanism of action and efficacy in crystal-induced inflammation. *Curr Rheumatol Rep.* 2008;10:218–227.
 341. Lidar M, Yonath H, Shechter N, et al. Incomplete response to colchicine in M694V homozygote FMF patients. *Autoimmun Rev.* 2012;12:72–76.
 342. Ben-Zvi I, Krichely-Vachdi T, Feld O, et al. Colchicine-free remission in familial Mediterranean fever: featuring a unique subset of the disease—a case control study. *Orphanet J Rare Dis.* 2014;9:3.
 343. Tufan A, Babaoglu MO, Akdogan A, et al. Association of drug transporter gene ABCB1 (MDR1) 3435C to T polymorphism with colchicine response in familial Mediterranean fever. *J Rheumatol.* 2007;34:1540–1544.
 344. Hoffman HM, Rosengren S, Boyle DL, et al. Prevention of cold-associated acute inflammation in familial cold autoinflammatory syndrome by interleukin-1 receptor antagonist. *Lancet.* 2004;364:1779–1785.
 345. Seyahi E, Ozdogan H, Celik S, et al. Treatment options in colchicine resistant familial Mediterranean fever patients: thalidomide and etanercept as adjunctive agents. *Clin Exp Rheumatol.* 2006;24:S99–S103.
 346. Ter Haar N, Lachmann H, Ozen S, et al. Treatment of autoinflammatory diseases: results from the Eurofer Registry and a literature review. *Ann Rheum Dis.* 2013;72:678–685.
 347. Tunca M, Tankurt E, Akbaylor Akpinar H, et al. The efficacy of interferon alpha on colchicine-resistant familial Mediterranean fever attacks: a pilot study. *Br J Rheumatol.* 1997;36:1005–1008.
 348. Tunca M, Akar S, Soyuturk M, et al. The effect of interferon alpha administration on acute attacks of familial Mediterranean fever: a double-blind, placebo-controlled trial. *Clin Exp Rheumatol.* 2004;22:S37–S40.
 349. Tweezer-Zaks N, Rabinovich E, Lidar M, et al. Interferon-alpha as a treatment modality for colchicine-resistant familial Mediterranean fever. *J Rheumatol.* 2008;35:1362–1365.
 350. Mor A, Pillinger MH, Kishimoto M, et al. Familial Mediterranean fever successfully treated with etanercept. *J Clin Rheumatol.* 2007;13:38–40.
 351. Ozgocmen S, Ozcakar L, Ardicoglu O, et al. Familial Mediterranean fever responds well to infliximab: single case experience. *Clin Rheumatol.* 2006;25:83–87.
 352. Belkhir R, Moulouguet-Doleris L, Hachulla E, et al. Treatment of familial Mediterranean fever with anakinra. *Ann Intern Med.* 2007;146:825–826.
 353. Moser C, Pohl G, Haslinger I, et al. Successful treatment of familial Mediterranean fever with Anakinra and outcome after renal transplantation. *Nephrol Dial Transplant.* 2009;24:676–678.
 354. Stankovic Stojanovic K, Delmas Y, Torres PU, et al. Dramatic beneficial effect of interleukin-1 inhibitor treatment in patients with familial Mediterranean fever complicated with amyloidosis and renal failure. *Nephrol Dial Transplant.* 2012;27:1898–1901.
 355. Lachmann HJ, Kone-Paut I, Kuemmerle-Deschner JB, et al. Use of canakinumab in the cryopyrin-associated periodic syndrome. *N Engl J Med.* 2009;360:2416–2425.
 356. van der Hilst J, Moutschen M, Messiaen PE, et al. Efficacy of anti-IL-1 treatment in familial Mediterranean fever: a systematic review of the literature. *Biologics.* 2016;10:75–80.
 357. Hoffman HM, Throne ML, Amar NJ, et al. Efficacy and safety of rilonacept (interleukin-1 Trap) in patients with cryopyrin-associated periodic syndromes: results from two sequential placebo-controlled studies. *Arthritis Rheum.* 2008;58:2443–2452.
 358. Bergesio F, Ciciani AM, Manganaro M, et al. Renal involvement in systemic amyloidosis: an Italian collaborative study on survival and renal outcome. *Nephrol Dial Transplant.* 2008;23:941–951.
 359. Altıparmak MR, Pamuk ON, Ataman R, et al. Continuous ambulatory peritoneal dialysis in familial Mediterranean fever amyloidosis patients with end-stage renal failure: a single-centre experience from Turkey. *Nephron Clin Pract.* 2004;98:c119–c123.
 360. Sahin S, Sahin GM, Ergin H, et al. The effect of dialytic modalities on clinical outcomes in ESRD patients with familial Mediterranean fever. *Ren Fail.* 2007;29:315–319.
 361. Celik A, Saglam F, Dolek D, et al. Outcome of kidney transplantation for renal amyloidosis: a single-center experience. *Transplant Proc.* 2006;38:435–439.

362. Keven K, Sengul S, Kutlay S, et al. Long-term outcome of renal transplantation in patients with familial Mediterranean fever amyloidosis: a single-center experience. *Transplant Proc.* 2004;36:2632–2634.
363. Sherif AM, Refaie AF, Sobh MA, et al. Long-term outcome of live donor kidney transplantation for renal amyloidosis. *Am J Kidney Dis.* 2003;42:370–375.
364. Green H, Lichtenberg S, Rahamimov R, et al. Familial Mediterranean fever is associated with increased mortality after kidney transplantation—a 19 years' single center experience. *Transplantation.* 2017;101:2621–2626.
365. Heering P, Hetzel R, Grabensee B, et al. Renal transplantation in secondary systemic amyloidosis. *Clin Transplant.* 1998;12:159–164.
366. Ozdemir BH, Ozdemir FN, Sezer S, et al. Among therapy modalities of end-stage renal disease, renal transplantation improves survival in patients with amyloidosis. *Transplant Proc.* 2006;38:432–434.
367. Maes BD, Dalle I, Geboes K, et al. Erosive enterocolitis in mycophenolate mofetil-treated renal-transplant recipients with persistent afebrile diarrhea. *Transplantation.* 2003;75:665–672.
368. Al-Rohani M. Renal failure in Yemen. *Transplant Proc.* 2004;36:1777–1779.
369. Al-Saedy AJ, Al-Kahichy HR. The current status of hemodialysis in Baghdad. *Saudi J Kidney Dis Transpl.* 2011;22:362–367.
370. Ersoy FF. The TULIP story. *Perit Dial Int.* 2008;28:220–223.
371. Horvat LD, Shariff SZ, Garg AX. Global trends in the rates of living kidney donation. *Kidney Int.* 2009;75:1088–1098.
372. Sheiban AK, Yehia A, Mohamed YA, et al. Renal replacement therapy in Yemen. *Saudi J Kidney Dis Transpl.* 1996;7:313–314.
373. Karah N, Kurhani MN, Fadloun E, et al. Dialysis therapy for Syrian refugees in Lebanon: a call for action. *Bull World Health Organ.* 2018;96:370.
374. Shigidi MM, Fituri OM, Chandy SK, et al. Peritoneal dialysis, an expanding mode of renal replacement therapy in Qatar. *Saudi J Kidney Dis Transpl.* 2011;22:587–593.
375. Elzorkany KMA. Current state of continuous ambulatory peritoneal dialysis in Egypt. *Saudi J Kidney Dis Transpl.* 2017;28:1369–1374.
376. Akpolat T. CAPD: a control strategy to prevent spread of HCV infection in end-stage renal disease. *Perit Dial Int.* 2001;21:77–79.
377. Akpolat T, Dilek M, Yavuz M, et al. Low seroconversion rates in CAPD patients compared with hemodialysis patients: potential advantages for transplant candidates. *Perit Dial Int.* 2002;22:520–523.
378. Al-wakeel J, Al-Ghonaim M, Al-Suwaida A, et al. Peritoneal membrane characteristics in patients on peritoneal dialysis. *Saudi J Kidney Dis Transpl.* 2011;22:49–53.
379. Ahmed AM, Allam MF, Habil ES, et al. Compliance with haemodialysis practice guidelines in Egypt. *East Mediterr Health J.* 2013;19:4–9.
380. Al-Naseri SK, Mahdi ZM, Hashim MF. Quality of water in hemodialysis centers in Baghdad, Iraq. *Hemodial Int.* 2013;17:517–522.
381. Barsoum RS. Overview: end-stage renal disease in the developing world. *Artif Organs.* 2002;26:737–746.
382. Amini M, Aghighi M, Masoudkabar F, et al. Hemodialysis adequacy and treatment in Iranian patients: a national multicenter study. *Iran J Kidney Dis.* 2011;5:103–109.
383. Hamadneh SA, Nueirat SA, Qadoomi J, et al. Vascular access mortality and hospitalization among hemodialysis patients in Palestine. *Saudi J Kidney Dis Transpl.* 2018;29:120–126.
384. Abdelsalam MS, Rashwan M, Althaf MM, et al. Comparison of survival between dialysis patients with incident high-flux hemodialysis versus on-line hemodiafiltration: a single center experience in Saudi Arabia. *Saudi J Kidney Dis Transpl.* 2018;29:107–113.
385. Romano-Zelekha O, Golan E, Ifrah A, et al. Differences in quality of life between Jewish and Arab patients on hemodialysis. *Qual Life Res.* 2017;26:3343–3352.
386. Abouchacra S, Obaidli A, Al-Ghamdi SM, et al. Gulf Cooperation Council-dialysis outcomes and practice patterns study: An overview of anemia management trends at the regional and country specific levels in the Gulf Cooperation Council countries. *Saudi J Kidney Dis Transpl.* 2016;27:51–61.
387. Al Salmi I, AlRukhaimi M, AlSahow A, et al. Mineral bone disorder and its management among hemodialysis patients in the Gulf Cooperation Council: Initial findings from the dialysis outcomes and practice patterns study (2012-2015). *Saudi J Kidney Dis Transpl.* 2016;27:62–80.
388. Al-Ali FS, Bieber BA, Pisoni RL, et al. Nutritional status and outcomes in hemodialysis patients from the Gulf Cooperation Council countries enrolled in the dialysis outcome and practice patterns study phase 5 (2012-2015). *Saudi J Kidney Dis Transpl.* 2016;27(suppl 1):S31–S41.
389. AlYousef A, AlGhareeb S, Al Wakeel J, et al. Hemodialysis delivery, dialysis dose achievement, and vascular access types in hemodialysis patients from the Gulf Cooperation Council countries enrolled in the dialysis outcomes and practice patterns study phase 5 (2012-2015). *Saudi J Kidney Dis Transpl.* 2016;27(suppl 1):S42–S50.
390. Shaheen FA, Al Wakeel J, Al-Ghamdi SM, et al. Cardiovascular and cerebrovascular comorbidities in hemodialysis patients from the Gulf Cooperation Council countries enrolled in the dialysis outcome and practice pattern study phase 5 (2012-2015). *Saudi J Kidney Dis Transpl.* 2016;27(suppl):S24–S30.
391. Barsoum RS. Chronic kidney disease in the developing world. *N Engl J Med.* 2006;354:997–999.
392. Bahadi A, El Farouki MR, Zajjari Y, et al. Initiating hemodialysis in Morocco: Impact of late referral. *Nephrol Ther.* 2017;13:525–531.
393. Dialysis in the Kingdom of Saudi Arabia. *Saudi J Kidney Dis Transpl.* 2013;24:853–861.
394. Alavian SM, Bagheri-Lankarani K, Mahdavi-Mazdeh M, et al. Hepatitis B and C in dialysis units in Iran: changing the epidemiology. *Hemodial Int.* 2008;12:378–382.
395. Afifi A. Renal osteodystrophy in developing countries. *Artif Organs.* 2002;26:767–769.
396. Schrooten I, Elseviers MM, Lamberts LV, et al. Increased serum strontium levels in dialysis patients: an epidemiological survey. *Kidney Int.* 1999;56:1886–1892.
397. Afifi A, El-Sayed H, El-Setouhi M, et al. Hyperphosphatemia among end-stage renal disease patients in developing countries: a forgotten issue? *Hemodial Int.* 2005;9:409–415.
398. Souqiyeh MZ, Shaheen FA. Survey of attitudes of physicians toward the current evaluation and treatment of chronic kidney disease-mineral and bone disorder (CKD-MBD). *Saudi J Kidney Dis Transpl.* 2010;21:93–101.
399. Al Rukhaimi M, Al Sahow A, Boobes Y, et al. Adaptation and implementation of the “Kidney Disease: Improving Global Outcomes (KDIGO)” guidelines for evaluation and management of mineral and bone disorders in chronic kidney disease for practice in the Middle East countries. *Saudi J Kidney Dis Transpl.* 2014;25:133–148.
400. Messina JP, Humphreys I, Flaxman A, et al. Global distribution and prevalence of hepatitis C virus genotypes. *Hepatology.* 2015;61:77–87.
401. Lavanchy D. The global burden of hepatitis C. *Liver Int.* 2009;29(suppl 1):74–81.
402. Sunbul M. Hepatitis B virus genotypes: global distribution and clinical importance. *World J Gastroenterol.* 2014;20:5427–5434.
403. Te HS, Jensen DM. Epidemiology of hepatitis B and C viruses: a global overview. *Clin Liver Dis.* 2010;14:1–21, vii.
404. Mahmud S, Al-Kanaani Z, Chemaitelly H, et al. Hepatitis C virus genotypes in the Middle East and North Africa: Distribution, diversity, and patterns. *J Med Virol.* 2018;90:131–141.
405. Harfouche M, Chemaitelly H, Kouyoumjian SP, et al. Hepatitis C virus viremic rate in the Middle East and North Africa: Systematic synthesis, meta-analyses, and meta-regressions. *PLoS ONE.* 2017;12:e0187177.
406. Harfouche M, Chemaitelly H, Mahmud S, et al. Epidemiology of hepatitis C virus among hemodialysis patients in the Middle East and North Africa: systematic syntheses, meta-analyses, and meta-regressions. *Epidemiol Infect.* 2017;145:3243–3263.
407. Fehr T, Ambuhl PM. Chronic hepatitis virus infections in patients on renal replacement therapy. *Nephrol Dial Transplant.* 2004;19:1049–1053.
408. Köhler H. The prevalence of hepatitis C in different countries of the ERA/EDTA area. *Nephrol Dial Transplant.* 1995;10:468–469.
409. Rahnavardi M, Hosseini Moghaddam SM, Alavian SM. Hepatitis C in hemodialysis patients: current global magnitude, natural history, diagnostic difficulties, and preventive measures. *Am J Nephrol.* 2008;28:628–640.
410. Barsoum RS, William EA, Khalil SS. Hepatitis C and kidney disease: a narrative review. *J Adv Res.* 2017;8:113–130.
411. Liang TJ, Ghany MG. Current and future therapies for hepatitis C virus infection. *N Engl J Med.* 2013;368:1907–1917.
412. Chung RT, Baumert TF. Curing chronic hepatitis C—the arc of a medical triumph. *N Engl J Med.* 2014;370:1576–1578.
413. Frank C, Mohamed MK, Strickland GT, et al. The role of parenteral antischistosomal therapy in the spread of hepatitis C virus in Egypt. *Lancet.* 2000;355:887–891.

414. Progress toward prevention and control of hepatitis C virus infection—Egypt, 2001–2012. *MMWR Morb Mortal Wkly Rep.* 2012;61:545–549.
415. Mohamoud YA, Mumtaz GR, Riome S, et al. The epidemiology of hepatitis C virus in Egypt: a systematic review and data synthesis. *BMC Infect Dis.* 2013;13:288.
416. Al Balwi MA. Prevalence of mixed hepatitis C virus (HCV) genotypes among recently diagnosed dialysis patients with HCV infection. *Saudi J Kidney Dis Transpl.* 2011;22:712–716.
417. Karkar A. Hepatitis C in dialysis units: the Saudi experience. *Hemodial Int.* 2007;11:354–367.
418. Goodkin DA, Bieber B, Jadoul M, et al. Mortality, Hospitalization, and Quality of Life among Patients with Hepatitis C Infection on Hemodialysis. *Clin J Am Soc Nephrol.* 2017;12:287–297.
419. Sayiner AA, Zeytinoglu A, Ozkahya M, et al. HCV infection in haemodialysis and CAPD patients. *Nephrol Dial Transplant.* 1999;14:256–257.
420. Turkish Society of Nephrology: Registry of the nephrology, dialysis and transplantation in Turkey: registry 2008. <http://tsn.org.tr/folders/file/registry%20kitap.pdf>. Accessed January 14, 2014.
421. Abou Rached A, El Khoury L, El Imad T, et al. Incidence and prevalence of hepatitis B and hepatitis C viruses in hemodialysis patients in Lebanon. *World J Nephrol.* 2016;5:101–107.
422. El-kader YE-OA, Elmanama AA, Ayeshe BM. Prevalence and risk factors of hepatitis B and C viruses among haemodialysis patients in Gaza strip, Palestine. *Virol J.* 2010;7:210.
423. Burdick RA, Bragg-Gresham JL, Woods JD, et al. Patterns of hepatitis B prevalence and seroconversion in hemodialysis units from three continents: the DOPPS. *Kidney Int.* 2003;63:2222–2229.
424. Fissell RB, Bragg-Gresham JL, Woods JD, et al. Patterns of hepatitis C prevalence and seroconversion in hemodialysis units from three continents: the DOPPS. *Kidney Int.* 2004;65:2335–2342.
425. Forzley M. Advancing the health of Arab Americans: key points to obtaining resources and establishing programs focused on special populations. *Ethn Dis.* 2005;15:S1-90-92.
426. Fogazzi GB, Castelnovo C. Maintenance dialysis in patients from developing countries: the experience of an Italian center. *J Nephrol.* 2004;17:552–558.
427. Shaheen FA. Organ donation in the Middle East countries. *Ethn Dis.* 2009;19:S1-16-17.
428. Paris W, Nour B. Organ transplantation in Egypt. *Prog Transplant.* 2010;20:274–278.
429. Mohsin N, Militala E, Budruddin M, et al. Attitude of the Omani population toward organ transplantation. *Transplant Proc.* 2010;42:4305–4308.
430. Mohsin NB, Metry A. Reasons of preclusion of living-related donor renal transplants in Oman. *Exp Clin Transplant.* 2010;8:303–306.
431. Rizvi SA, Naqvi SA, Hussain Z, et al. Renal transplantation in developing countries. *Kidney Int Suppl.* 2003;63:S96–S100.
432. Abderrahim E, Zammouri A, Bacha MM, et al. Thirty years of experience at the first Tunisian kidney transplant center. *Exp Clin Transplant.* 2017;15:84–89.
433. Muralidharan A, White S. The need for kidney transplantation in low- and middle-income countries in 2012: an epidemiological perspective. *Transplantation.* 2015;99:476–481.
434. Einollahi B. Iranian experience with the non-related renal transplantation. *Saudi J Kidney Dis Transpl.* 2004;15:421–428.
435. Ehtuish EF, Abouna GM, Shebani AH, et al. Kidney transplantation in Libya: a North African and Middle Eastern perspective. *Exp Clin Transplant.* 2006;4:425–428.
436. Einollahi B, Taheri S. Renal transplantation practice in Iran and the Middle East: report from Iran and a review of the literature. *Ann Transplant.* 2008;13:5–14.
437. El-Nono IH, Al-Ba'adani TH, Ghilan AM, et al. Adult-to-adult living related donor renal transplantation in Yemen: the first experience. *Saudi J Kidney Dis Transpl.* 2007;18:265–269.
438. El-Nono IH, Telha KA, Al-Alimy GM, et al. Challenges in renal transplantation in Yemen. *Ann Transplant.* 2015;20:92–96.
439. Masri MA, Haberal MA, Shaheen FA, et al. Middle East Society for Organ Transplantation (MESOT) transplant registry. *Exp Clin Transplant.* 2004;2:217–220.
440. Albar M. Organ transplantation: a Sunni Islamic perspective. *Saudi J Kidney Dis Transpl.* 2012;23:817–822.
441. Hamdy S. Not quite dead: why Egyptian doctors refuse the diagnosis of death by neurological criteria. *Theor Med Bioeth.* 2013;34:147–160.
442. Faraj W, Fakhri H, Mukherji D, et al. Organ donation after cardiac death in the Middle East. *Transplant Proc.* 2010;42:713–715.
443. Cohen J, Rahamimov R, Hoffman A, et al. Derivation and Implementation of a Protocol in Israel for Organ Donation after Cardio-Circulatory Death. *Isr Med Assoc J.* 2017;19:566–569.
444. Bulka RR. Jewish perspective on organ transplantation. *Transplant Proc.* 1990;22:945–946.
445. Ulshafer TR. A Catholic perspective on religion and organ transplantation. *Del Med J.* 1988;60:505–507.
446. Lavee J, Ashkenazi T, Stoler A, et al. Preliminary marked increase in the national organ donation rate in Israel following implementation of a new organ transplantation law. *Am J Transplant.* 2013;13:780–785.
447. Heidary Rouchi A, Mahdavi-Mazdeh M. Human tissue processing and transplantation in MESOT states: how to promote? *Int J Organ Transplant Med.* 2016;7:102–105.
448. SCOT Data. Organ donation and transplantation in the Kingdom of Saudi Arabia 2016. *Saudi J Kidney Dis Transpl.* 2017;28:1209–1214.
449. Nöl L. The data on transplantation in Saudi Arabia are complex and require additional framework for interpretation. *Kidney Int.* 2009;76:914, author reply 914–915.
450. Saeed B. The effect of the Syrian crisis on organ transplantation in Syria. *Exp Clin Transplant.* 2015;13:206–208.
451. Haghighi AN, Ghahramani N. Living unrelated kidney donor transplantation in Iran. *Nat Clin Pract Nephrol.* 2006;2:E1.
452. Ghods AJ. Renal transplantation in Iran. *Nephrol Dial Transplant.* 2002;17:222–228.
453. Ghods AJ, Savaj S. Iranian model of paid and regulated living-unrelated kidney donation. *Clin J Am Soc Nephrol.* 2006;1:1136–1145.
454. Simforoosh N, Basiri A, Fattahi MR, et al. Living unrelated versus living related kidney transplantation: 20 years' experience with 2155 cases. *Transplant Proc.* 2006;38:422–425.
455. USRDS: the United States Renal Data System. *Am J Kidney Dis.* 2003;42:1–230.
456. Harmon W, Delmonico F. Payment for kidneys: a government-regulated system is not ethically achievable. *Clin J Am Soc Nephrol.* 2006;1:1146–1147.
457. Fallahzadeh MK, Jafari L, Roozbeh J, et al. Comparison of health status and quality of life of related versus paid unrelated living kidney donors. *Am J Transplant.* 2013;13:3210–3214.
458. Khatami SM, Mahdavi-Mazdeh M. Perfect is the enemy of good: the Iranian system of paid donation. *Am J Transplant.* 2014;14:1222–1223.
459. Randhawa G. Policy perspectives: international survey of nephrologists' perceptions of and attitudes towards rewards and compensation for kidney donation. *Nephrol Dial Transplant.* 2013;28:1343–1345.
460. Ghahramani N, Karpavicius Z, Ghahramani M, et al. International survey of nephrologists' perceptions and attitudes about rewards and compensations for kidney donation. *Nephrol Dial Transplant.* 2013;28:1610–1621.
461. Shimazono Y. The state of the international organ trade: a provisional picture based on integration of available information. *Bull World Health Organ.* 2007;85:955–962.
462. Danovitch GM, Chapman J, Capron AM, et al. Organ trafficking and transplant tourism: the role of global professional ethical standards—the 2008 Declaration of Istanbul. *Transplantation.* 2013;95:1306–1312.
463. Delmonico FL, Dew MA. Living donor kidney transplantation in a global environment. *Kidney Int.* 2007;71:608–614.
464. Sever MS, Kazancioglu R, Yildiz A, et al. Outcome of living unrelated (commercial) renal transplantation. *Kidney Int.* 2001;60:1477–1483.
465. The declaration of Istanbul on organ trafficking and transplant tourism. *Clin J Am Soc Nephrol.* 2008;3:1227–1231.
466. The Declaration of Istanbul on organ trafficking and transplant tourism. International Summit on Transplant Tourism and Organ Trafficking. *Kidney Int.* 2008;74:854–859.
467. Efrat A. The politics of combating the organ trade: lessons from the Israeli and Pakistani experience. *Am J Transplant.* 2013;13:1650–1654.
468. Moszynski P. Organ traffickers are targeting refugees in Egypt, warns charity. *BMJ.* 2011;343:d8254.
469. Shaheen FA, Al-Khader AA. Preventive strategies of renal failure in the Arab world. *Kidney Int.* 2005;Suppl:S37–S40.
470. Sen N, Turunc T, Karatasli M, et al. Tuberculosis in patients with end-stage renal disease undergoing dialysis in an endemic region of Turkey. *Transplant Proc.* 2008;40:81–84.
471. Naqvi A, Rizvi A, Hussain Z, et al. Developing world perspective of posttransplant tuberculosis: morbidity, mortality, and cost implications. *Transplant Proc.* 2001;33:1787–1788.
472. Al-Khader AA, Shaheen FA. Posttransplant complications encountered in renal transplantation in the Middle East. *Transplant Proc.* 2004;36:180–183.

473. Abdelrahman M, Sinha AK, Karkar A. Tuberculosis in end-stage renal disease patients on hemodialysis. *Hemodial Int*. 2006;10:360–364.
474. Mahmoud IM, Sobh MA, El-Habashi AF, et al. Interferon therapy in hemodialysis patients with chronic hepatitis C: study of tolerance, efficacy and post-transplantation course. *Nephron Clin Pract*. 2005;100:c133–c139.
475. Samhan M, Al-Mousawi M, Donia F, et al. Malignancy in renal recipients. *Transplant Proc*. 2005;37:3068–3070.
476. Raeisi D, Payandeh M, Madani SH, et al. Kaposi's sarcoma after kidney transplantation: a 21-years experience. *Int J Hematol Oncol Stem Cell Res*. 2013;7:29–33.
477. Harzallah K, Abderrahim E, Chareffedine K, et al. Cancers after renal transplantation: multicenter experience. *Saudi J Kidney Dis Transpl*. 2008;19:825–830.
478. Euvrard S, Kinitakis J, Claudy A. Skin cancers after organ transplantation. *N Engl J Med*. 2003;348:1681–1691.
479. Levin A, Tonelli M, Bonventre J, et al. Global kidney health 2017 and beyond: a roadmap for closing gaps in care, research, and policy. *Lancet*. 2017;390:1888–1917.

BOARD REVIEW QUESTIONS

1. Which kidney disease is likely to be prevalent in the Middle East?
 - a. Fabry disease
 - b. Alport syndrome due to *COL4A5* mutations
 - c. Congenital nephrotic syndrome of the Finnish type
 - d. Dent disease

Answer: c

Rationale: Congenital nephrotic syndrome of the Finnish type is a recessively inherited podocytopathy. Middle Eastern communities are characterized by a high rate of consanguinity and large families, resulting in rapid population growth. This setting increases the carrier rate of recessively inherited diseases and consequently the prevalence of affected individuals, but has no bearing on X-linked inherited disorders (options a, b, and d).

2. Which statement describes correctly primary hyperoxaluria (PH) in the Middle East (ME)?
 - a. There is a 10-fold increase in the prevalence of PH2 in the ME compared with Europe.
 - b. There is a founder mutation in *HOGA1* leading to PH3 among Ashkenazi Jews.
 - c. The common phenotype of PH1 is of a first kidney stone during the sixth decade.
 - d. Most PH1 patients respond to pyridoxine and maintain preserved kidney function.

Answer: b

Rationale: PH1 may serve as a disease model for recessively inherited diseases in the ME. It is prevalent and quite often present with end-stage kidney disease (ESKD) during the first years of life. PH3 was described in Israel, but subsequently identified in other communities around the world. There is a founder mutation among Ashkenazi Jews.

3. Which therapy is most appropriate to prevent acute kidney injury (AKI) in victims of crush syndrome during and after extrication?
 - a. Control of bleeding and administration of any type of intravenous fluid
 - b. Early fluid resuscitation with intravenous 5% dextrose, 1 L/hour containing 20 mmol/L of potassium
 - c. Early fluid resuscitation with intravenous 5% dextrose, 1 L/hour
 - d. Early fluid resuscitation with intravenous 0.9% NaCl, 1 L/hour, limited to 3-6 L/day if close monitoring is absent
 - e. Early fluid resuscitation with intravenous 0.9% NaCl, 1 L/hour containing 20 mmol/L of potassium

Answer: d

Rationale: Hypovolemia and rhabdomyolysis are major risk factors for AKI following crush injury. Muscle tissue edema, bleeding, high ambient temperature, time spent under rubbles, and duration of extrication may all cause hypovolemia. Injured muscles, especially after reperfusion, release myoglobin and carry risks for hyperkalemia, hyperphosphatemia, hypocalcemia, hyperuricemia, AKI, and death. Therefore, all health professionals at disaster scenes should be aware of the importance of immediate restoration of intravascular volume with isotonic saline, which was proven to prevent AKI. All potassium-containing fluids should be avoided. When proper monitoring is impossible, fluid therapy should be individualized, and volume limited to 3-6 L/day to prevent life-threatening hypervolemia.